1.1 Introduction

Management Information System (MIS) is a study of people, technology, organizations, and the relationships among them in a broader sense.

However, in precise terms MIS is a software system that focuses on the management of information technology to provide efficiency and effectiveness or strategy decision making.

The term is often used in the academic study of businesses and has connections with other areas, such as information systems, information technology, informatics, e-commerce and computer science.

In this unit, we will discuss the basic concepts in Management Information System.

Explanation:

- MIS serves as a bridge connecting three core pillars: People, Technology, and Organizations.
- It is not just about computers; it is about how these components interact to create value.
- The primary goal is to support decision making and improve business efficiency.
- It connects various academic fields like computer science and business administration.

INTERACTIVE CONCEPT DIAGRAM (HOVER OVER CIRCLES)



Real-world Example:

Consider a **University Portal**. The *People* are students and faculty. The *Technology* is the website server and database. The *Organization* is the university administration. Together, they form the MIS that allows students to register for classes online.

1.2 Basic Concepts

Management Information System is an accumulation of 3 different terms:

1. Management

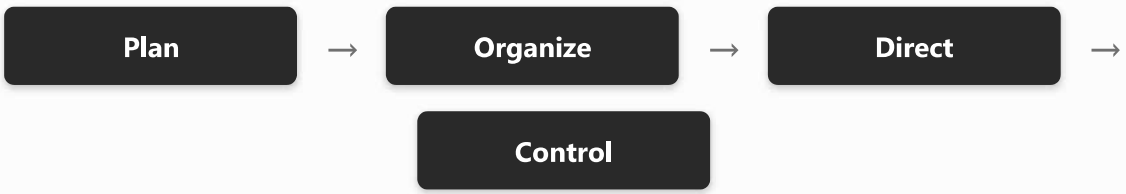
> We can define management in many ways like, "Manage Man Tactfully" or Management is an art of getting things done by others.

> However, for the purpose of Management Information System, management comprises the process and activity that a manager does in the operation of their organization, i.e., to plan, organize, direct and control operations.

Explanation:

- Management is fundamentally the art of **getting things done by others**.
- In the context of MIS, we focus on the systematic processes: Planning (setting goals), Organizing (structuring), Directing (leading), and Controlling (monitoring).
- It turns human efforts into organized production.

MANAGERIAL FUNCTIONS LOOP (HOVER BOXES)





Example:
A Restaurant Manager plans the menu, organizes the kitchen staff, directs the waiters, and controls the budget without cooking the food themselves.

2. Information

> Information simply means processed data or in the layman language, data which can be converted into meaningful and useful form for a specific user.

Explanation:

- **Data** on its own is often useless (e.g., just a number like "100").

- When this data is processed, organized, or interpreted, it becomes Information (e.g., "100 items sold").
- Information adds **context** to data, making it useful for decision-making.

DATA TRANSFORMATION PROCESS



Example:
A spreadsheet column of numbers (Data) becomes a Bar Chart showing Sales Trends (Information).

3. System

System can be defined as a set of elements joined together for a common objective.

A group of interrelated or interacting elements forming a unified whole e.g., business organization as systems.

A group of interrelated components working together towards a common goal by accepting input and producing output in an organized transformation process.

Explanation:

- A system is a collection of parts that work together as a whole.
- The parts must be **interrelated**; if one part fails, the system may fail.
- Every system has a **common objective** or goal.
- It follows a logic: Input -> Transformation -> Output.

SYSTEM TRANSFORMATION MODEL



Example:
A
Digestive System
. Input: Food. Processing: Stomach acid/enzymes breaking down food. Output: Nutrients absorbed by the body.

1.3 Overview of Management Concepts



> Every business unit has some objectives of its own. These objectives can be achieved with the coordinated efforts of several personnel.

> The works of a number of persons are properly co-ordinated to achieve the objectives through the process of management.

> Management is a vital aspect of the economic life of man, which is an organised group activity.

> It is considered as the indispensable institution in the modern social organization marked by scientific thought and technological innovations.

> One or the other form of management is essential wherever human efforts are to be undertaken collectively to satisfy wants through some productive activity, occupation or profession.

> It is management that regulates man's productive activities through coordinated use of material resources.

> Without the leadership provided by management, the resources of production remain resources and never become production.

> Management is the integrating force in all organized activity.

> Management is the process of allocating and organizations input including human and economic resources by planning, organizing, directing and controlling for the purpose of producing goods or services desired by customers so that organizational objectives are accomplished.

Explanation:

- Management is **indispensable** for any collective effort to succeed.

- It acts as the **integrating force**, ensuring different individuals work towards one common goal.
- Without management, raw materials (resources) stay idle and do not turn into finished goods (production).
- It is essential for **economic life** and involves the careful allocation of both human and economic resources.

THE MANAGEMENT CYCLE



Example:
In a Car Factory, management ensures the engine parts are assembled by workers (Human Resource) using machines (Economic Resource) to produce a car (Production) that the customer wants to buy.

1.4 Functions of Management



Management has been defined as a process of getting things done through others.

This process is identified in a set of functions performed by managers to accomplish the goals.

Planning: According to Terry and Franklin, "planning is selecting information and making assumptions concerning the future to put together the activities necessary to achieve organizational objectives."

Organizing: Organizing is the classification and categorization of requisite objectives, the grouping of activities needed to accomplish objectives, the assignment of each grouping to a manager with the authority necessary to supervise it, and the provisions for coordination horizontally and vertically in the organization structure.

Directing: Direction is telling people what to accomplish and seeing that they do it to the finest of their capability.

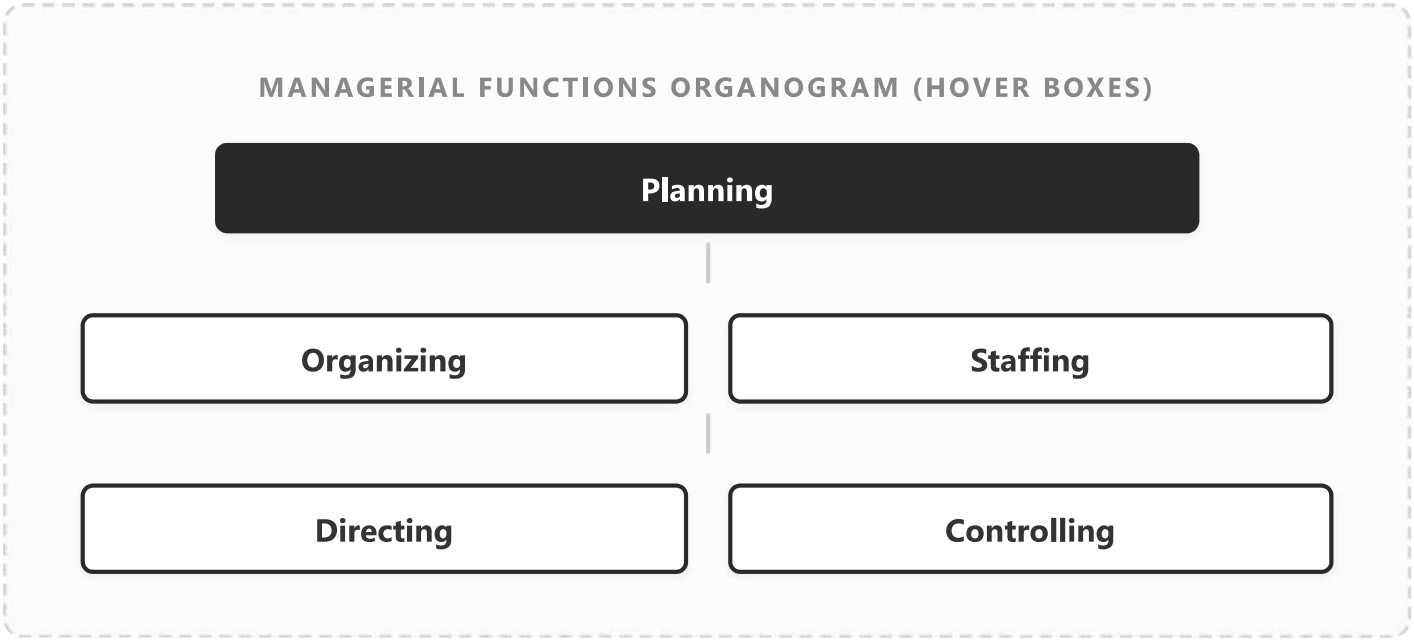
Staffing: Staffing function requires recognition of human resource needs, filling the organizational structure and keeping it filled with competent people.

Controlling: "Control is the course of action that measures present performance and guides it towards some predetermined goal."

Explanation:

- **Planning:** This is the primary function. It involves looking ahead, predicting the future, and setting a roadmap. As defined by Terry and Franklin, it is about selecting information and making assumptions to achieve objectives.
- **Organizing:** This involves creating a structure. It is the grouping of activities and assigning authority (the right to make decisions) to managers.
- **Directing:** This is the human element. It involves guiding, leading, and supervising employees to ensure they perform their best.

- **Staffing:** This involves the "people" aspect: recruitment (hiring), training, and compensation (pay).
- **Controlling:** This is the checking mechanism. It involves measuring actual performance against the plan and taking corrective action if needed.



Example:
A Project Manager
Plans
the project timeline,
Organizes
the software teams,
Staffs
the project with developers,
Directs
them on daily tasks, and finally
Controls
the budget to ensure the project is finished on time and within cost.

1.5 Levels of Management

According to the expert there are three types of level of management: (i) Top Level Management (ii) Middle Level Management (iii) Low Level or Operative Management.

Top Level Management: Consists of board of directors, managing directors or executive committee members. Objectives include setting key objectives, policies, making appointments, and reviewing work.

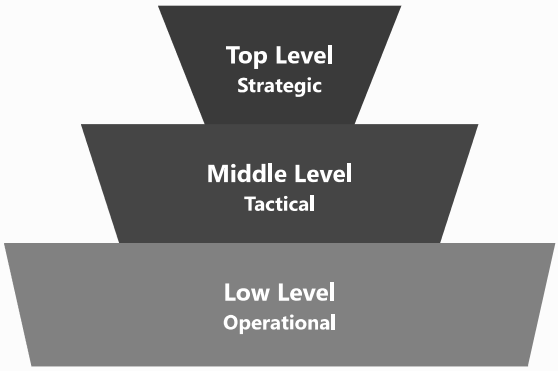
Middle Level Management: Consists of managers of various departments. Objectives include following top-level policies, motivating personnel, collecting analysis reports, mutual understanding, and recommendations.

Low Level Management: Consists of supervisors, daily workers. Functions include issuing orders, assigning jobs, arranging tools, solving worker problems, and maintaining discipline.

Explanation:

- **Top Level:** The "Brain" of the organization. They deal with long-term strategy and policies. They rarely deal with daily operational details.
- **Middle Level:** The "Bridge". They interpret top-level policies for the lower levels. They are department heads (e.g., Sales Manager, HR Manager).
- **Low Level:** The "Hands". They oversee the actual work and workers. They ensure daily tasks are completed correctly.

MANAGEMENT HIERARCHY (HOVER PYRAMID LEVELS)



Example:

In a Bank:

Top:

The Chairman (Decides to open 10 new branches).

Middle:

The Regional Manager (Plans how to staff those branches).

Low:

The Branch Supervisor (Ensures tellers are serving customers correctly today).

1.6 Concept of a System

A System is a group of interrelated components working together toward a common goal by accepting inputs and producing outputs in an organized transformation process.

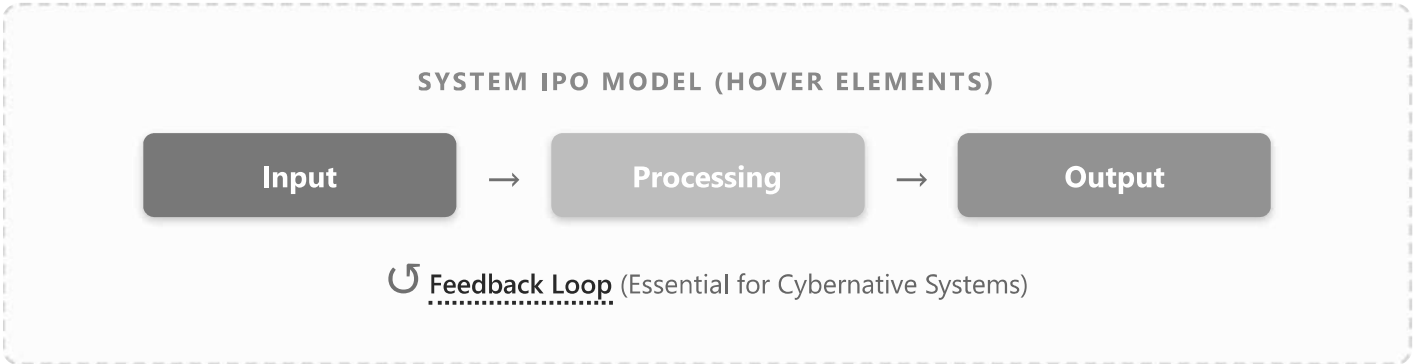
System Concepts: Technology, Application, Development, Management.

Components: Input, Processing, Output.

Types: Dynamic (interacts/controlled), Cybernative (feedback/control), Open (interacts with environment), Adoptive (changes to survive).

Explanation:

- A system follows the IPO model: **Input** is captured, **Processing** transforms it, and **Output** is delivered.
- **Dynamic System:** Components interact and are managed.
- **Cybernative System:** Uses **feedback** to self-regulate (like a thermostat).
- **Open System:** Interacts with its environment (imports/exports).
- **Adoptive System:** Changes itself to survive in changing environments.



Example:

An
Air Conditioner
is an Open & Cybernative System.
Input: Hot Air & Electricity.
Processing: Compressor cooling the air.
Output: Cold Air.
Feedback: Thermostat sensing the room temperature to adjust cooling speed.

1.7 Data and Information

By data we mean the facts or figures representing an object, place or the events occurring in the organization. It is not enough to have data... but when these data are interpreted and processed to determine its true meaning, they become useful.

By information, we mean that the data have been shaped into a meaningful form, which may be useful for human beings.

Characteristics of Good Information: Relevance, Accuracy, Completeness, Confidence in the Source, Communication to the Right Person.

Functions of Information: Reduction of Uncertainty, Aid to Monitoring and Control, As a Means of Communication, As a Memory Supplement, As Aid to Simplification.

Explanation:

- **Data** is raw and unprocessed (e.g., just numbers). It has little value on its own.
- **Information** is processed data that has context and meaning (e.g., numbers in a report with labels).
- Good information must be **Relevant**, **Accurate**, and **Complete**.
- Information acts as a **Memory Supplement** and helps reduce **Uncertainty**.

DATA TRANSFORMATION (HOVER CELLS)

Data (Raw Facts)	PROCESS	Information (Context)

Example:

Data:

"100, 200, 150". This tells you nothing.

Information:

"Sales of January: 100, February: 200, March: 150". This tells you performance trends.

1.8 Information System

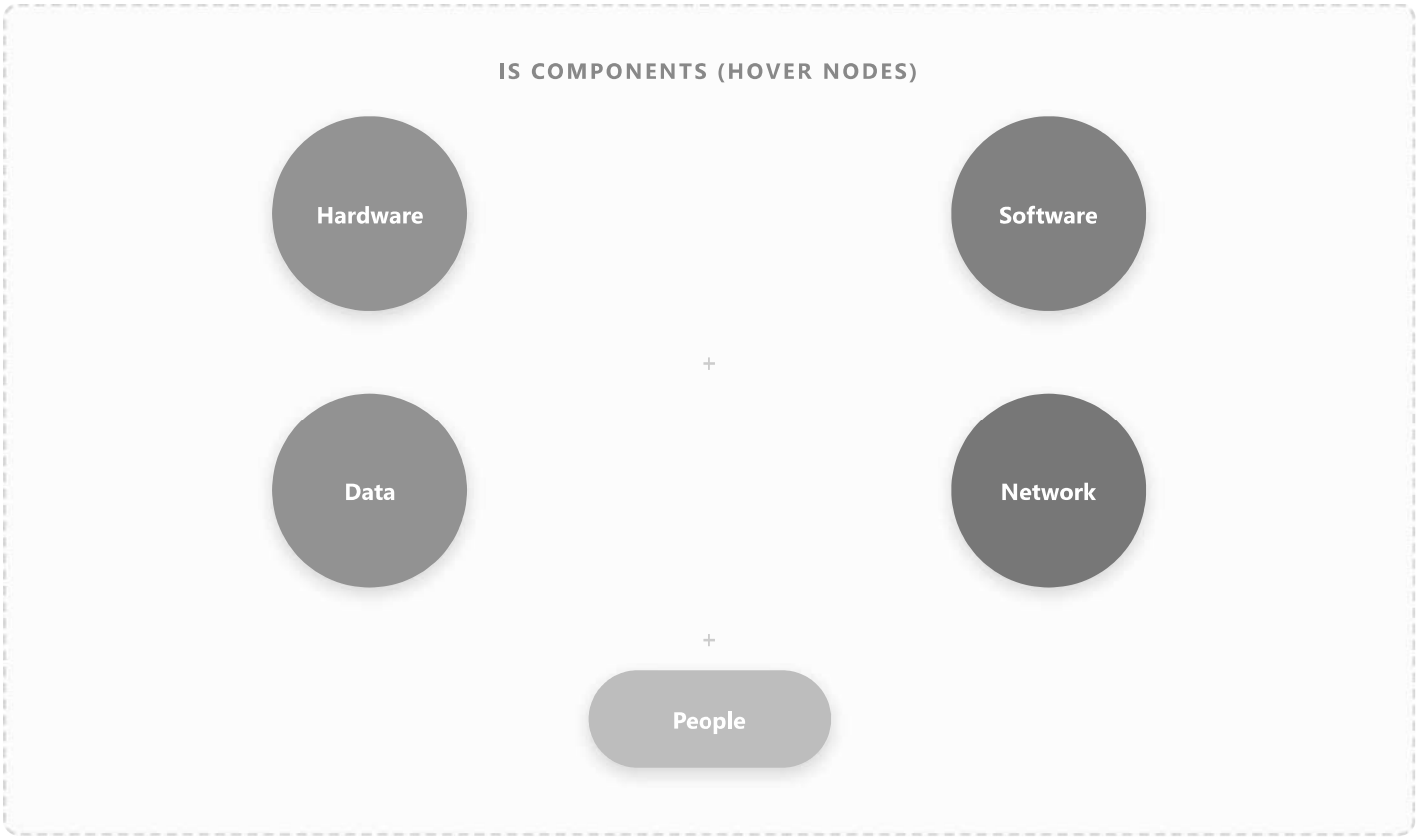
An information system can be defined as a set of interrelated components that collect (or retrieve), process, store and distribute information to support decision making, coordination and control in an organization.

Examples: Airline reservations, Bank operations, ERP integration, Logistics management.

Components: People (End-Users, IS Specialists), Data (Database, Knowledge Base), Software (Programs, Procedures), Hardware (Machines, Media), Network (Communication Media, Support).

Explanation:

- An IS is a man-made system that combines People, Hardware, Software, Data, and Network.
- **End-Users:** People who use the system (e.g., Clerks, Managers).
- **IS Specialists:** People who build/maintain it (e.g., Programmers, System Analysts).



Example:
An
ATM System
.
Hardware: The ATM Machine card reader.
Software: The banking application running on it.
Data: Your account balance.
Network: The link to the bank's central server.
People: You (the customer).

1.9 Need for Information Systems

The information system is very important for the internet technology and the traditional business concerns and is really the latest phase in the ongoing evolution of business.

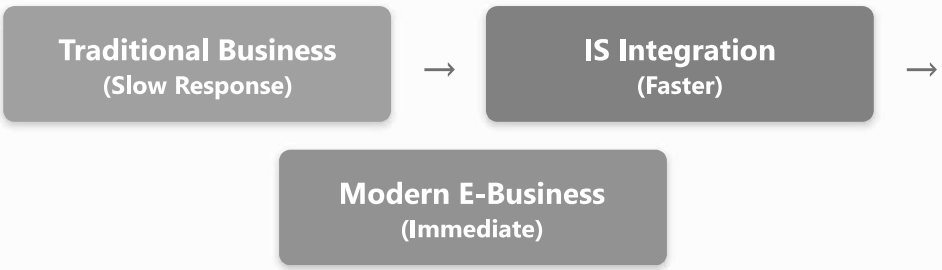
All the companies need to update their business, infrastructure and change way they work to respond more immediately to customer need.

A first step in designing and developing an MIS is to assess the information needs for decision making of management at different hierarchical levels.

Explanation:

- In the era of **Internet Technology**, traditional manual systems are too slow.
- Customers today expect **immediate responses**. IS enables this speed.
- Companies must upgrade their **infrastructure** to survive competition.
- Designing an MIS starts by understanding exactly what information the Top, Middle, and Low level managers need.

EVOLUTION OF BUSINESS RESPONSE



Example:
A local Kirana store (Traditional) takes hours to check stock manually vs. Amazon (IS Enabled) which shows you exactly how many items are left in 0.5 seconds.

1.10 Uses of Information System

Information system and technology including E-business and E-commerce technology and application has become vital component of successful business and organization.

It is a study of business administration and management.

For a manager or a business professional it is just as important to have basic understanding of information system and any other functional area in business.

Explanation:

- IS is no longer optional; it is a **vital component** of business.
- It encompasses **E-business** and **E-commerce**.
- For managers, understanding IS is as important as understanding finance or marketing. It is a core subject in **business administration**.



Example:

A Marketing Manager uses an IS to track which online ads are getting the most clicks. This helps them decide where to spend the budget next month (Strategic Planning).

1.11 Roles of Information Systems in Business

Support the Business Process: Treats inputs as a request from the customer and outputs as services to customer.

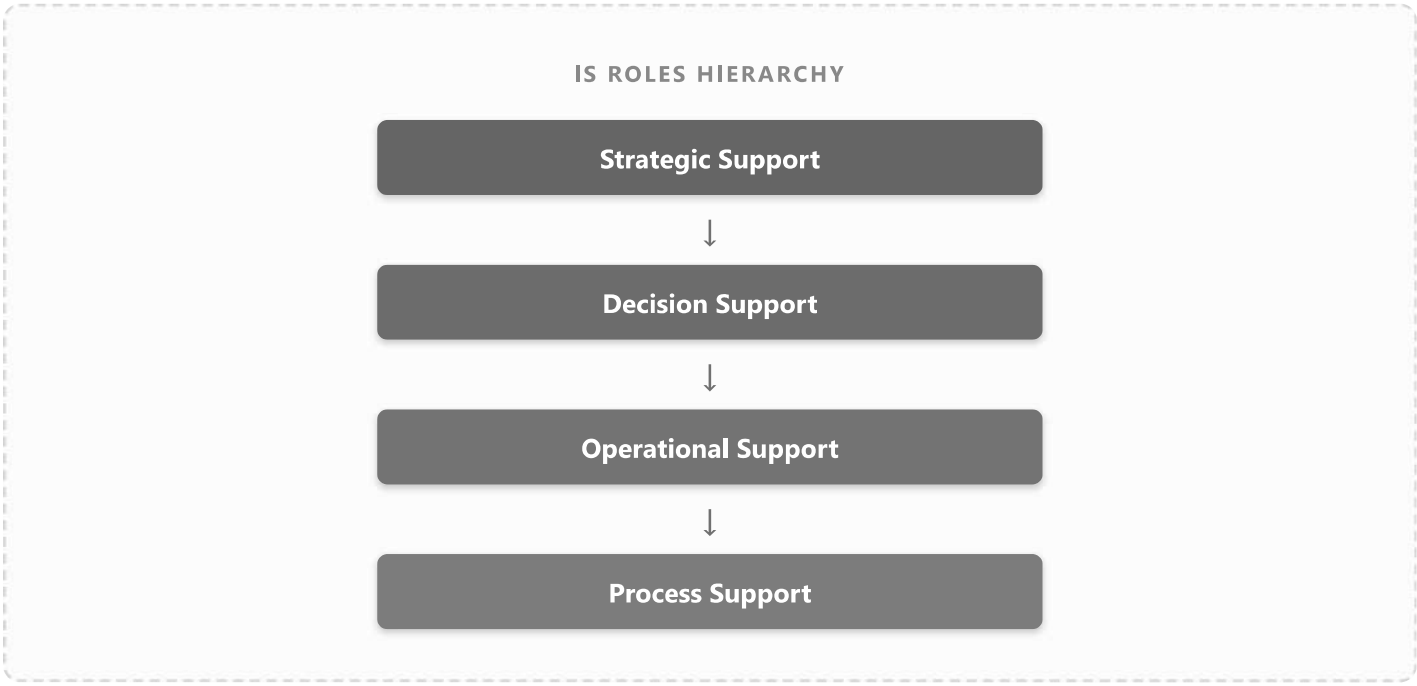
Support Operation of a Business Organization: By giving timely information, maintenance and enhancement which provides flexibility.

Support Decision Making: By employee in their daily operations and by managers to meet goals using mathematical models and IT tools.

Strategies for an Organization: Helps evolve appropriate strategies to succeed in a competitive environment.

Explanation:

- **Process Support:** Automates and streamlines daily workflows (e.g., order processing).
- **Operational Support:** Provides timely information needed to run the business day-to-day.
- **Decision Support:** Uses Mathematical models and IT tools to help managers make tough choices.
- **Strategic Support:** Helps the company find strategies to survive in the market.



Example:

Uber

uses IS for:
1. Process Support (Booking App).
2. Operational Support (Driver Tracking).
3. Decision Support (Surge Pricing Algorithms).
4. Strategic Support (Expanding to new cities based on user data).

2.6.1 Definition of MIS

Core Definition (G.B. Davis)

A Management Information System (MIS) is "an **integrated man/machine system** for providing information to hold up the operations, management and decision making functions in an organization."

The system utilizes **hardware** and **software**, manual procedures, management decision model and data base.

> An MIS need not be wholly computer based; it is however inevitable that the information deriving from the high volume of data in basic operational processes is computerized.

> What is still not so certain is whether the once off "high level" information also needed in an MIS, is best obtained using a computer.

Key Thoughts on MIS

> A management information system aims at meeting the information needs of managers, predominantly with regard to the current and past operations of the enterprise.

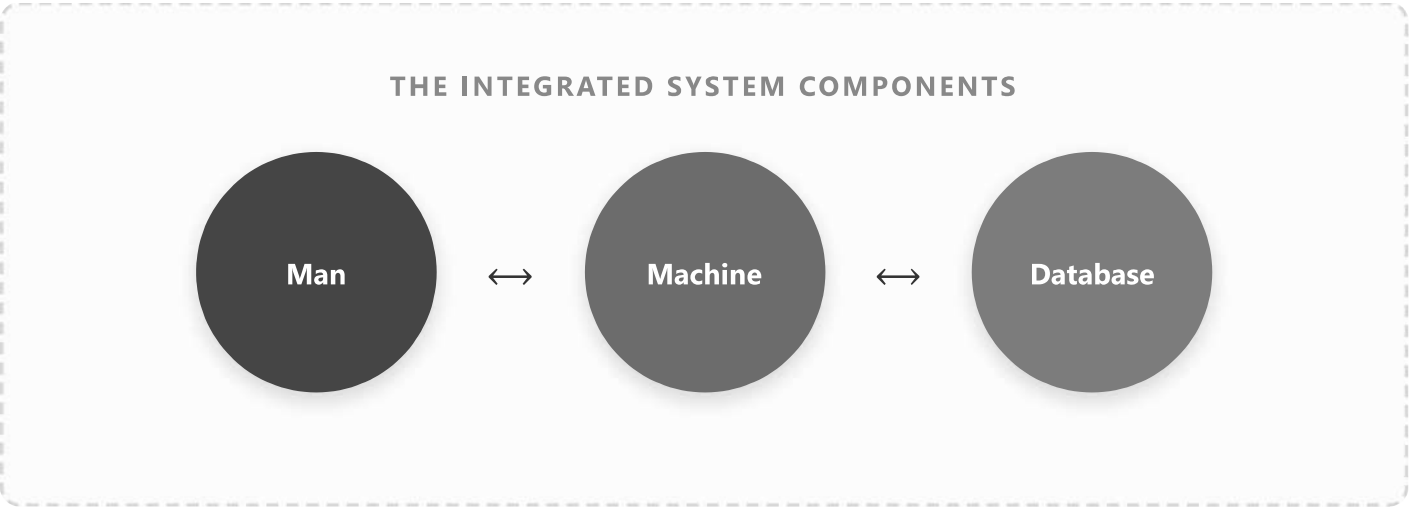
> Management information system is a system which provides precise, timely and meaningful data for management planning, analysis and control to optimize the growth of the organization.

> The MIS is a system which provides information support for decision making in the organization.

> The MIS is an integrated system of man and machine for providing the information to support the operations, the management and the decision making function in the organization.

> The MIS is a system based on the database of the organization evolved for the purpose of providing information to the people in the organization.

> The MIS is a Computer based Information System.



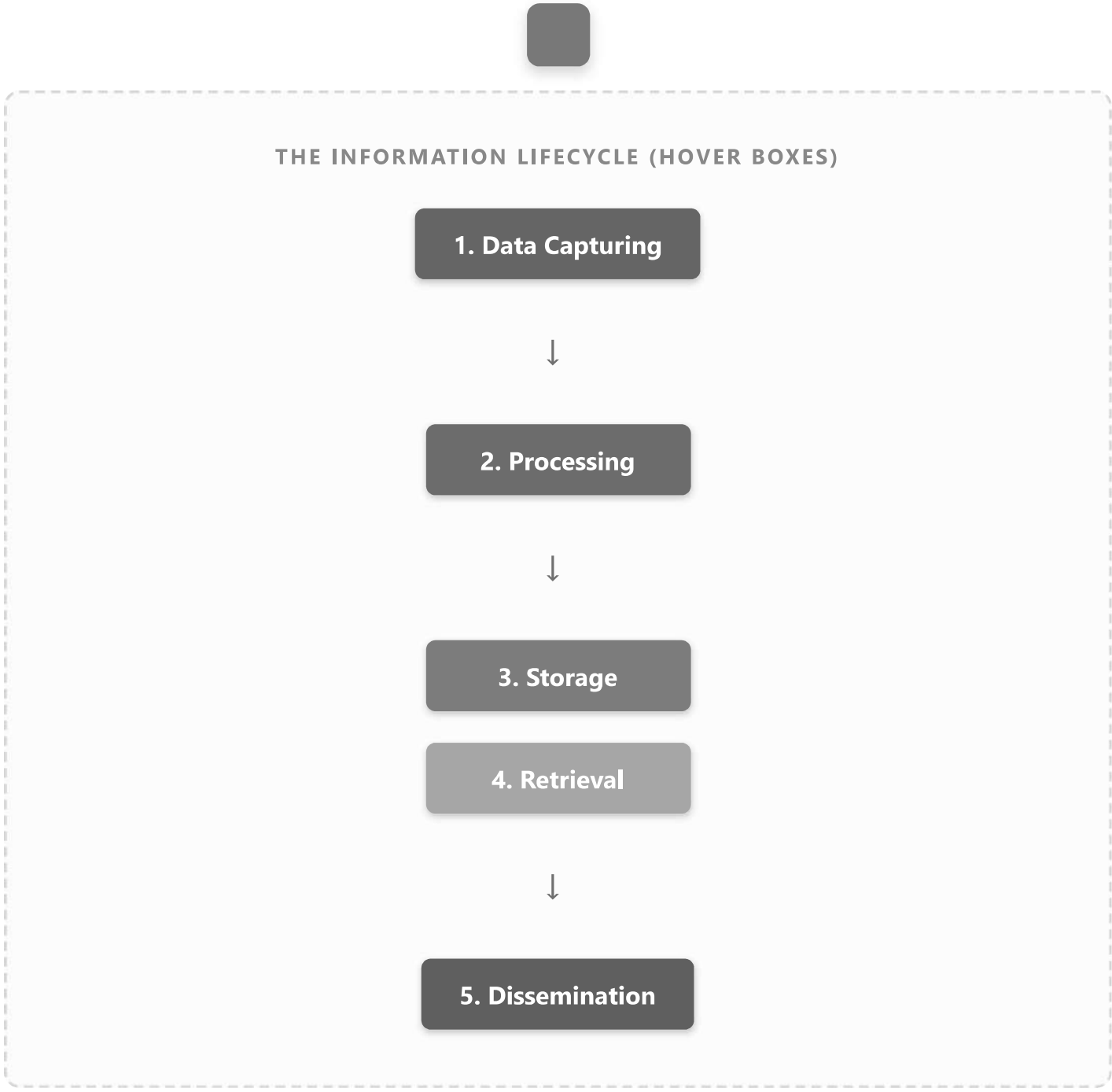
Formal System Requirements

In order to get a better grip on the activity of information processing, it is necessary to have a formal system which should take care of the following points:

- **Handling of a voluminous data.**
- **Confirmation of the validity of data and transaction.**
- **Complex processing of data and multidimensional analysis.**
- **Quick search and retrieval.**
- **Mass storage.**
- **Communication of the information system to the user on time.**
- **Fulfilling the changing needs of the information.**

The management information system uses computers and communication technology to deal with these points of supreme importance.

2.6.2 Objectives of MIS



1. Data Capturing : MIS capture data from various internal and external sources of organization. Data capturing may be manual or through computer terminals.

2. Processing of Data : The captured data is processed to convert into required information. Processing of data is done by such activities as calculating, sorting, classifying, and summarizing.

3. Storage of Information : MIS stores the processed or unprocessed data for future use. If any information is not immediately required, it is saved as an organization record, for

later use.

4. Retrieval of Information : MIS retrieves information from its stores as and when required by various users.

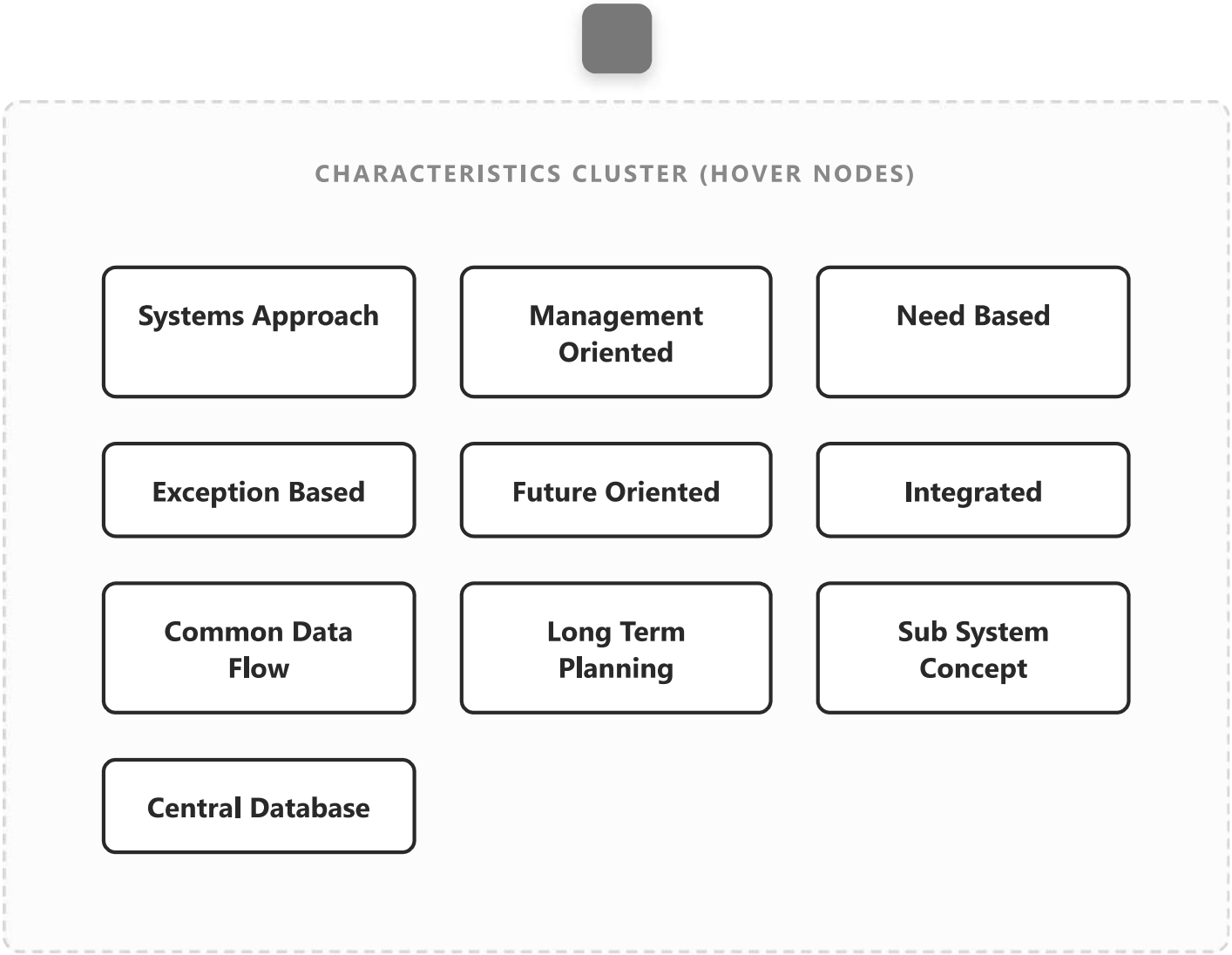
5. Dissemination of Information : Information, which is a finished product of MIS, is disseminated to the users in the organization. It is periodic or online through computer terminal.



Example:

Consider a
Sales MIS
.
Capturing: Cashier scans item.
Processing: System calculates total and deducts stock.
Storage: Transaction record saved in DB.
Retrieval: Manager asks for "Total sales of May".
Dissemination: Report displays on Manager's screen.

2.6.3 Characteristics of MIS



- **1. Systems Approach :** The information system follows a systems approach. Systems approach means taking a comprehensive view or a complete look at the interlocking sub-systems that operate within an organization.
- **2. Management Oriented :** Management oriented characteristic of MIS implies that the management actively directs the system development efforts. For planning of MIS, top-down approach should be followed. Top down approach suggests that the system development starts from the determination of management's needs and overall business objective.
- **3. Need Based :** MIS design should be as per the information needs of managers at different levels.
- **4. Exception Based :** MIS should be developed on the exception based also, which means that in an abnormal situation, there should be immediate reporting about the exceptional situation to the decision-makers at the required level.
- **5. Future Oriented :** MIS should not merely provide past of historical information; rather it should provide information, on the basis of future projections on the actions to be initiated.
- **6. Integrated :** Integration is significant because of its ability to produce more meaningful information. Integration means taking a comprehensive view or looking at the complete picture

of the interlocking subsystems that operate within the company.

- **7. Common Data Flow :** Common data flow includes avoiding duplication, combining similar functions and simplifying operations wherever possible. The development of common data flow is an economically sound and logical concept, but it must be viewed from a practical angle.
- **8. Long Term Planning :** MIS is developed over relatively long periods. A heavy element of planning should be involved.
- **9. Sub System Concept :** The MIS should be viewed as a single entity, but it must be broken down into digestible sub-systems which are more meaningful.
- **10. Central Database :** In the MIS there should be common data base for whole system.

2.6.4 Role of MIS in an Organization

The Heart & Blood Analogy:
The role of the MIS in an organization can be compared to the role of heart in the body. The information is the blood and MIS is the heart. The MIS plays exactly the same role in the organization—regulating, processing, and supplying information to all elements.

The system ensures that an appropriate data is collected from the various sources, processed, and sent further to all the needy destinations. The system is expected to fulfil the information needs of an individual, a group of individuals, the management functionaries: the managers and the top management.

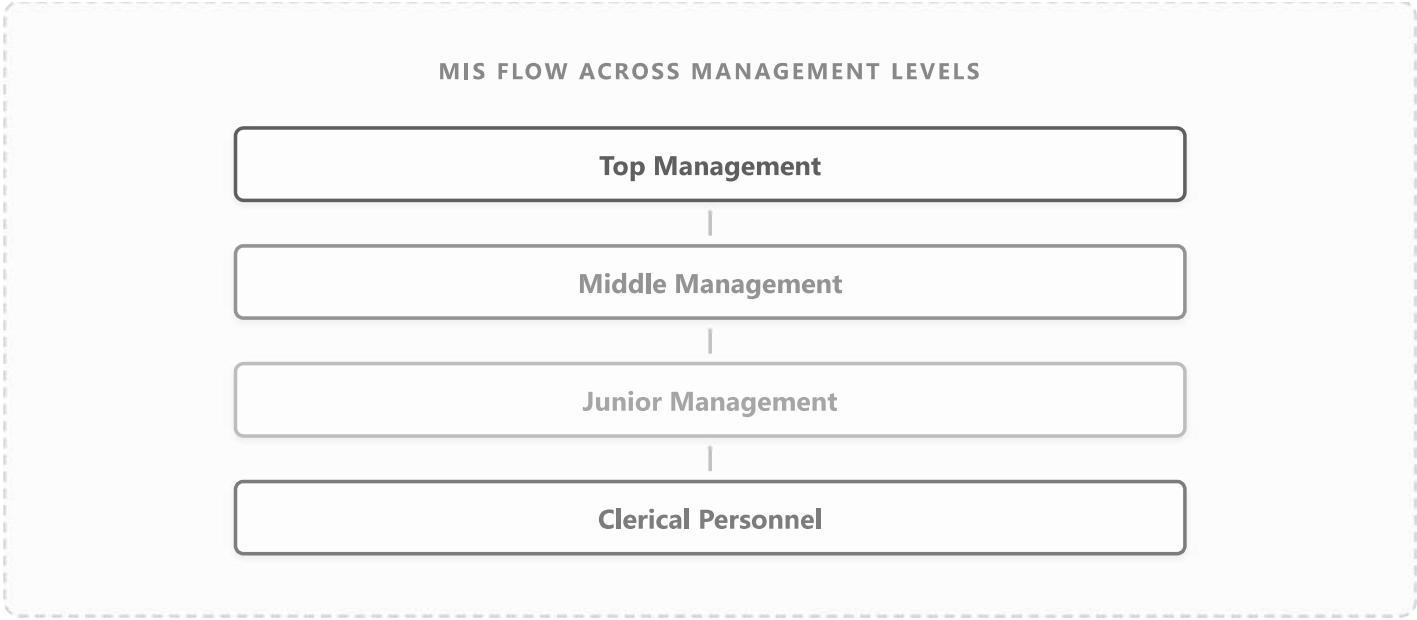
The MIS satisfies the diverse needs through a variety of systems such as Query Systems, Analysis Systems, Modelling Systems and Decision Support Systems the MIS helps in Strategic Planning, Management Control, Operational Control and Transaction Processing.

The MIS helps the clerical personnel in the transaction processing and answers their queries on the data pertaining to the transaction, the status of a particular record and references on a variety of documents.

The MIS helps the junior management personnel by providing the operational data for planning, scheduling and control, and helps them further in decision making at the operations level to correct an out of control situation.

The MIS helps the middle management in short them planning, target setting and controlling the business functions. It is supported by the use of the management tools of planning and control. The MIS helps the top management in goal setting, strategic planning and evolving the business plans and their implementation.

The MIS plays the role of information generation, communication, problem identification and helps in the process of decision making. The MIS, therefore, plays a vital role in the management, administration and operations of an organization.



3.1 Overview of MIS Classification

Understanding the Framework

Management Information Systems are not monolithic blocks; they are complex frameworks designed to address specific needs. To understand their function, we classify them into two distinct dimensions:

- **Activity Subsystems (Horizontal):** Based on the levels of management (Strategic, Tactical, Operational).
- **Functional Subsystems (Vertical):** Based on business areas (Marketing, Finance, HR, Production).

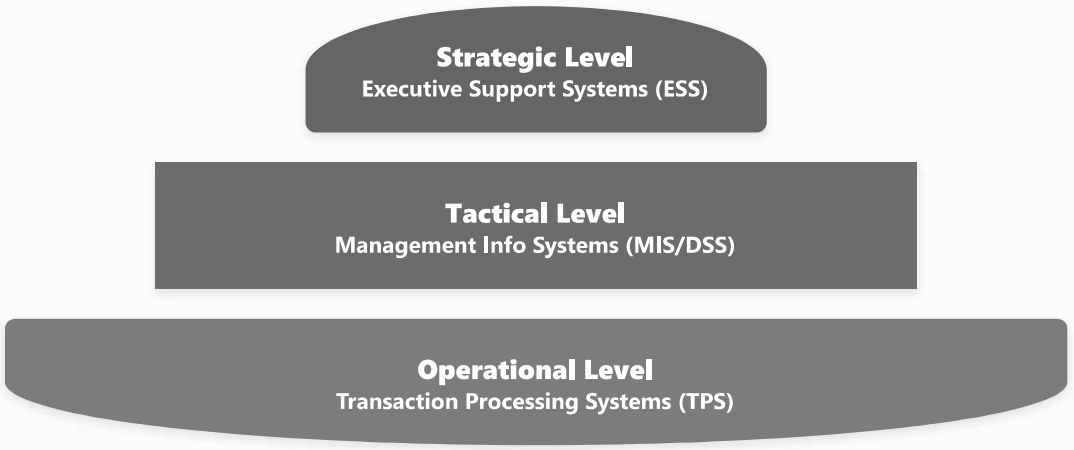


Why Classify?

Classification is essential for **System Design**. It ensures that the right information reaches the right person at the right time. Without this classification, managers would be overwhelmed by irrelevant data, and departments would operate in silos.

3.2 Activity Subsystems (Levels)

These subsystems are arranged hierarchically, corresponding to the three primary levels of management activity. Each level requires different types of information and decision support.



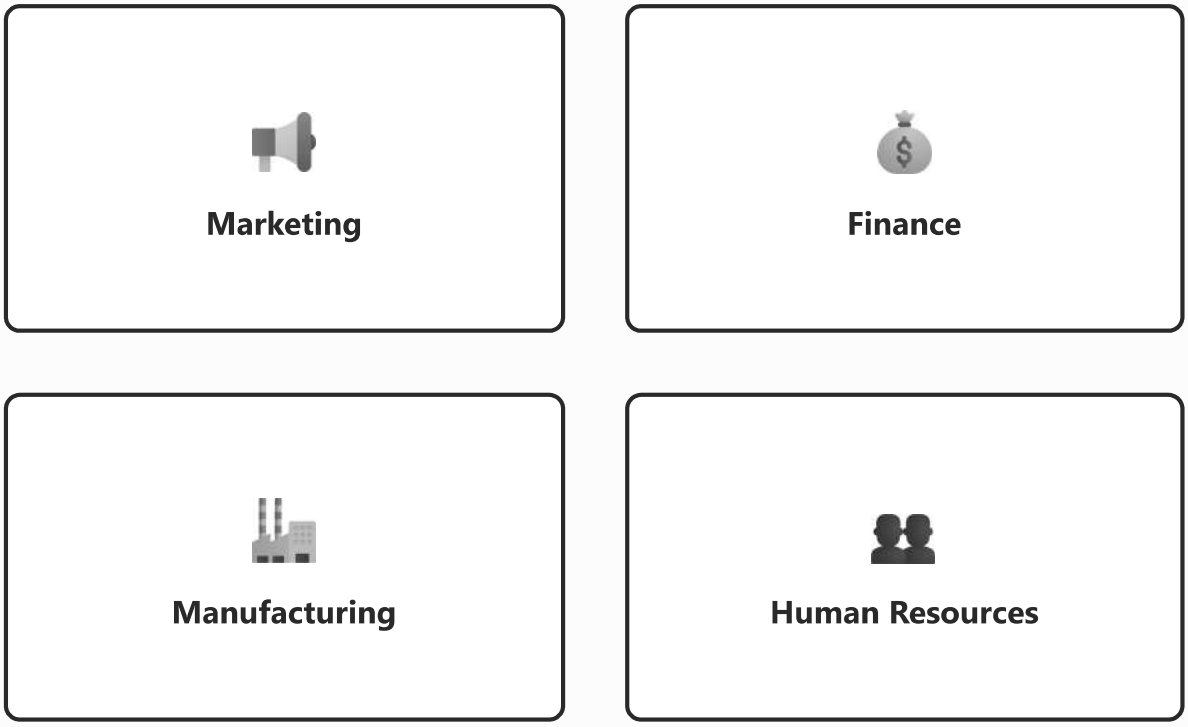
Activity Breakdown:

- **Operational:**
  "What happened today?" (e.g., Daily sales receipt).
- **Tactical:**
  "How are we doing this month?" (e.g., Monthly budget variance report).
- **Strategic:**
  "Where will we be in 5 years?" (e.g., Profit planning for new market entry).

3.3 Functional Subsystems (Business Areas)

While activity subsystems dictate *who* sees the information based on hierarchy, functional subsystems dictate *what* the information is about based on the department.

BUSINESS FUNCTIONS (HOVER FOR DETAILS)



Integration is Key

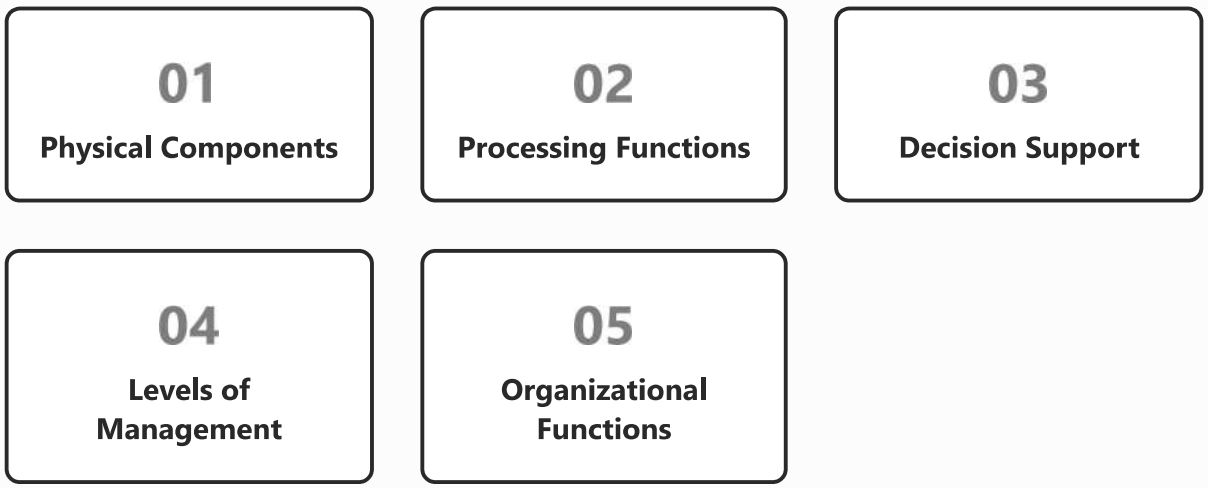
Although these subsystems are vertical, they cannot exist in isolation. For example, the **Manufacturing** subsystem needs data from the **Marketing** subsystem (sales forecasts) to plan production, and from the **Finance** subsystem (budgets) to purchase raw materials.

2.4 The Structure of MIS

What is Structure?

Structure helps determine the **shape of an entity**, which provides its basic framework. The structure of MIS is difficult to define because some entities may not have well-defined outlines and structures. However, the multiple approaches to an entity help describe the structure in a better way.

FIVE APPROACHES TO DESCRIBE MIS STRUCTURE (HOVER EACH)



Why Multiple Approaches?

No single viewpoint can fully capture the complexity of an MIS. A hardware engineer sees the structure as servers and networks; a manager sees it as reports and dashboards; a CEO sees it as strategic support. Understanding all five approaches ensures you can **communicate with every stakeholder** in the organization.

> **Analogy — A Building:**
> A building's structure can be described by its *physical materials* (brick, steel), its *functional zones* (kitchen, bedroom), its *decision spaces* (where choices are made), its *levels* (floors), and its *organizational purpose* (hospital vs. school). MIS is similarly multi-dimensional.

2.4.1 Physical Components of MIS

The physical components of the information system help us understand the structure of MIS in its most tangible form. These components work together to form the complete infrastructure.

HARDWARE DEVICE CATEGORIES (HOVER FOR EXAMPLES)

- Input Devices
- Output Devices
- Secondary Storage
- CPU
- Communication Devices

All Physical Components

Hardware: Physical data processing equipment and peripheral devices (printers, tapes, input/output devices, CPU, communication devices).

Software: Instructions and programs that direct the functioning of the hardware. Includes system software and application software.

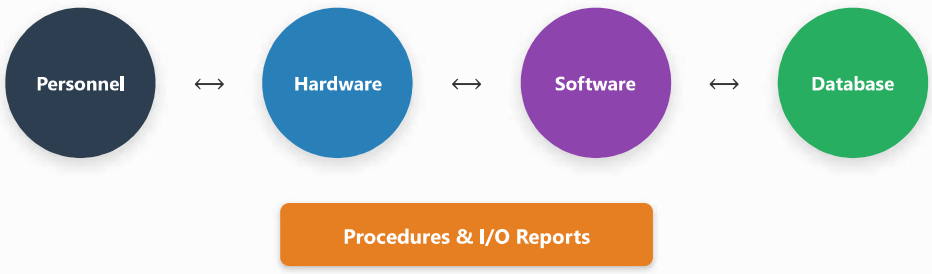
Database: Consists of all data utilized by the application software, stored in files present on the disk.

Procedures: Physical procedures such as manuals required to operate a system. These are also termed as physical elements.

Operating Personnel: Personnel such as computer programmers and system managers who execute the functions of the information systems.

Input and Output: Physical inputs and outputs from the information systems that exist in various forms, such as printouts and reports.

PHYSICAL COMPONENTS INTERCONNECTION



2.4.2 Information System Processing Functions

Beyond Just Processing Data

Information system has a central importance in any venture. This is an interdisciplinary subject and is much more than just a collection of computerized information for processing and distribution. The five core functions form a layered structure.

FIVE FUNCTIONS HIERARCHY (HOVER FOR DETAILS)



(a) Information Processing and Usability Function

This deals with the final application of the information and the way in which the information is being processed. It requires knowledge of new concepts, models, procedures, and techniques for efficient processing.

(b) Education and Learning Function

New things appear every day and the business atmosphere becomes increasingly competitive. The system must be properly understood by those who work on data and those who use information as end users (management). **Installing a system is one thing; making efficient use of it is another.** The latter is more important.

(c) Information Systems Development Function

A business is open to global competition and every organization develops its own system and methodology. An organization's image as a user of new technological products and latest methodology upgrades the market. Managers must know about emerging technology, software tools, design models, and working methodologies.

(d) Management and Control Function

This is concerned with the maintenance of the first three functions. Management invests in the information system and aims at a return on investment. Key responsibilities include:

- Acquisition/Upgradation/Extension/Amendment of hardware and software.
- Making provisions of budget and cost analysis to justify investment.
- Guidelines on statutory obligations and organizational policies.
- Contract management related to material supply and services.
- Setting priorities for tasks across departments.
- Compliance with safety standards, quality control, and performance of services.
- Control of procedures, handling user requests, complaints, amendments.
- Setting targets and monitoring system availability and working condition.
- Management of users' activities to prevent misuse of the system.

(e) Strategy and Planning Function

This includes analysis of data on the performance of the business and the system, used to formulate business strategy. Once completed, it falls under the Management and Control Function. It covers:

Production planning: Deciding the target volume of products or services.

Budgeting: Allocation of finance for producing the work volume.

Capital: Cost of investment in equipment, processing, conversion, and infrastructure.

Available manpower: Assessing and providing required skilled personnel.

Selecting technology: Evaluating vendor bids and following uniform policy for technology selection.

Skills distribution: Reviewing and inducting skills needed for coordinated task delivery.

Time constraints: Ensuring each task is performed within a defined time limit.

Quality assurance: Setting standards and assuring product/service quality before delivery.

Auditing: Internal audit and external audit (usually government-appointed).

Legal issues: Identifying statutory obligations for private and public sector compliance.

Market factors: Identifying favourable and unfavourable factors for competitive positioning.

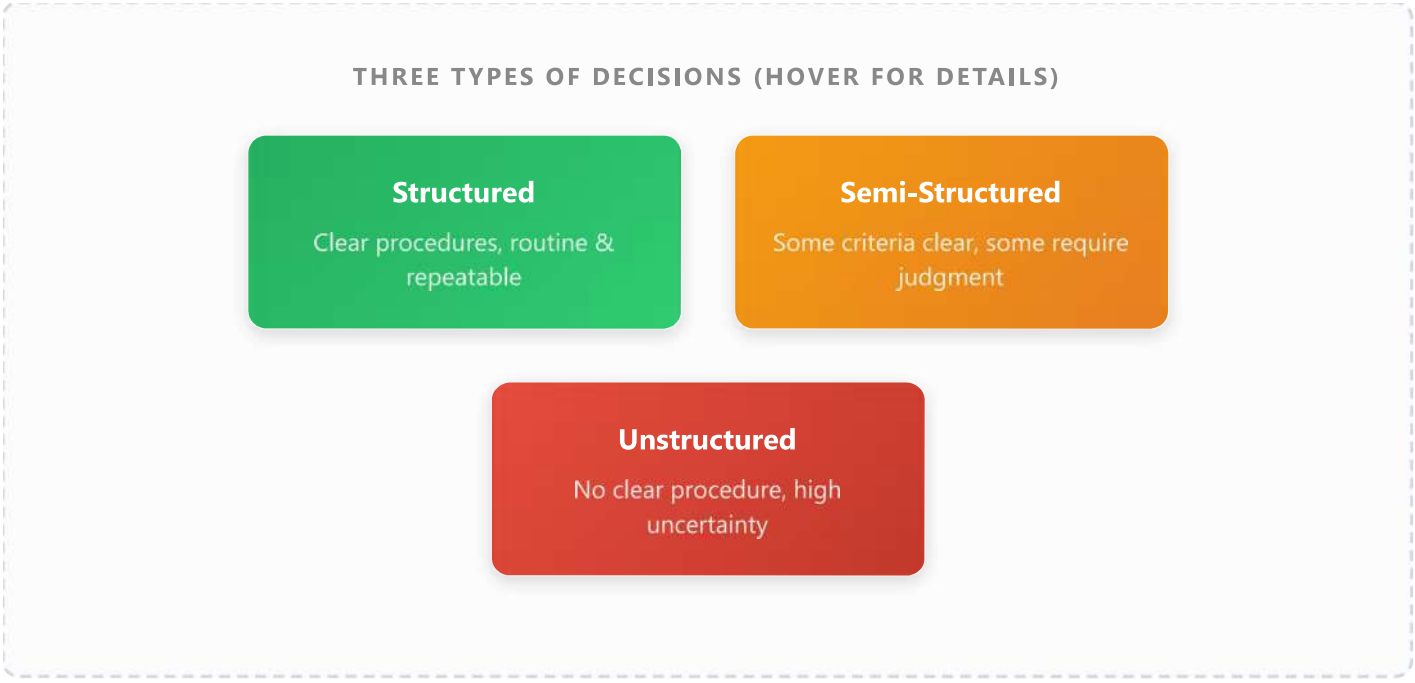
The Functional Model Summary

In a nutshell, the Information Strategy Function must ensure:

- Adequate support for corporate strategy.
- Efficient MIS with Executive Information Support.
- Assessing strategic positions relative to market and competitors.
- Support in developing existing market and product quality as well as lateral movement.
- Active participation to develop a corporate strategy.

2.4.3 Decision Support Structure

The structure of MIS also depends on how MIS supports decision-making. Decisions taken using a decision-making process may differ according to the structure of MIS provided for making decisions.



How MIS Structure Adapts

Structured Decisions: Handled by the Transaction Processing System (TPS) at the operational level. The MIS structure here is rigid, rule-based, and automated.

Semi-Structured Decisions: Handled by Decision Support Systems (DSS) and MIS at the tactical level. The structure provides reports plus analytical models — human judgment fills the gap.

Unstructured Decisions: Handled by Executive Support Systems (ESS) at the strategic level. The MIS structure here is flexible, providing external data, ad-hoc queries, and graphical dashboards.

Real-World Mapping:

- **Structured:**
  "Should we approve this employee's leave request if they have a balance?" — Automated by TPS.
- **Semi-Structured:**
  "Which of these three suppliers should we award the contract to?" — DSS provides cost/quality comparison; manager decides.
- **Unstructured:**
  "Should our company acquire a startup in AI?" — ESS provides market data, competitor analysis; CEO decides based on vision and intuition.

2.4.4 Levels of Management Activity

Management tiers use completely distinct data pools, demanding that the MIS adapt its architecture dynamically across these layers. Each level operates with a different time horizon, different data granularity, and a fundamentally different decision-making approach.

THE MANAGEMENT INFORMATION PYRAMID (HOVER EACH LEVEL)



Strategic Planning (Top Management)

Focuses on long-term goals such as entering new global markets or projecting 5-year trajectories. This tier relies on broad, macro-level market forecasts and external trends rather than internal operational data. The MIS architecture at this level must integrate data from outside the organization — competitor movements, regulatory changes, economic indicators — and present it through dashboards and ad-hoc query tools.

Management Control (Middle Management)

Focuses on resource planning, budgeting, and figuring out how to successfully execute the top-level vision. This tier requires summarized internal data — weekly/monthly reports, variance analyses, and performance against targets. The MIS structure here bridges the gap between raw operational data and strategic summaries, often employing Decision Support Systems (DSS).

Operational Control (Lower Management)

Handles **short-term, immediate daily workflows** like checking safety stock levels, tracking attendance, and overseeing inventory. This tier requires detailed, real-time, highly structured data processed by **Transaction Processing Systems (TPS)**. The MIS architecture here must be fast, reliable, and capable of handling high volumes of data entry and retrieval.

Dimension	Strategic	Tactical	Operational
Time Horizon	3 – 5+ years	6 months – 2 years	Daily / Weekly
Data Source	External + Internal	Primarily Internal	Internal (real-time)
Data Granularity	Aggregated / Summarized	Semi-detailed	Detailed / Transactional
Decision Type	Unstructured	Semi-Structured	Structured
System Used	ESS	MIS / DSS	TPS
Key Question	"Where are we heading?"	"Are we on track?"	"What happened today?"

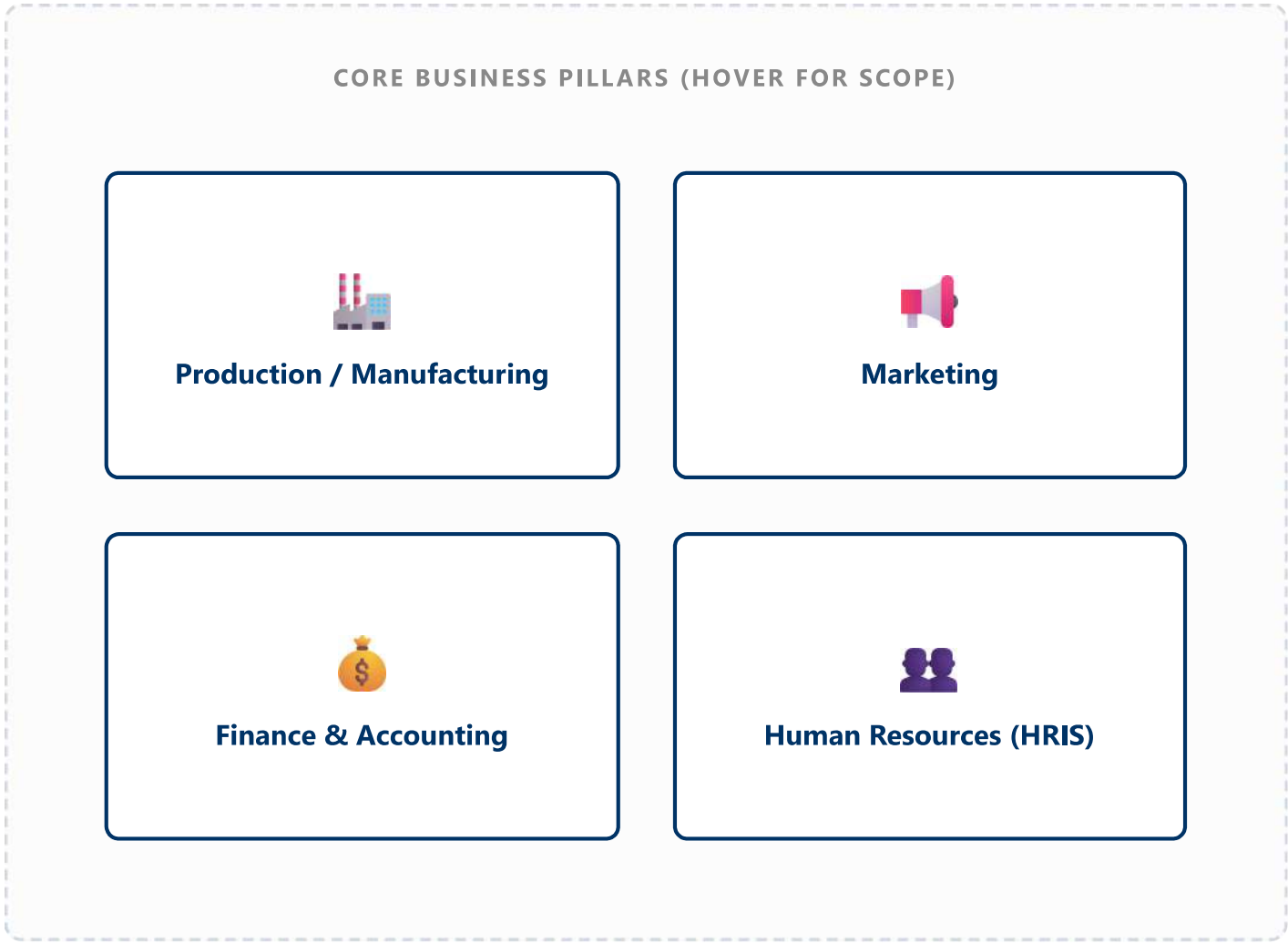


Same Data, Different Views:

A single day's sales transactions (₹50,000 from 120 customers) at the *Operational* level appear as individual receipts. At the *Tactical* level, they become a weekly sales report compared to the weekly target. At the *Strategic* level, they feed into a quarterly trend analysis compared to market growth rates.

2.4.5 Organizational Functions

Finally, an MIS structure is often departmentalized based on core business pillars. Each functional area has its own specialized subsystem that operates vertically across all three management levels — from operational transactions to strategic planning.



Production / Manufacturing Information System

Scope & Functions

This subsystem tracks the entire production lifecycle — from raw material procurement to finished goods dispatch. It ensures that manufacturing processes are efficient, cost-effective, and quality-compliant.

- **Raw Material Inventory:** Monitoring stock levels of raw materials and triggering purchase orders when safety stock is breached.
- **Batch Processing:** Scheduling and tracking production batches, including batch numbers, quantities, and timelines.

- **Machine Times:** Recording machine utilization, downtime, and maintenance schedules to optimize capacity.
- **Quality Control:** Tracking defect rates, compliance with standards, and inspection results.
- **Logistics:** Coordinating the supply chain for inbound materials and outbound distribution of finished goods.

Marketing Information System

Scope & Functions

This subsystem supports the marketing mix (Product, Price, Place, Promotion) by capturing market intelligence and tracking sales performance across geographies and segments.

- **Promotional Spends:** Tracking advertising budgets, campaign costs, and ROI on marketing initiatives.
- **Market Segments:** Analyzing customer demographics, preferences, and behavioral patterns to identify target segments.
- **Target Geographies:** Mapping sales performance by region, city, or territory to identify high-potential markets.
- **Sales Performance Tracking:** Monitoring sales by product, salesperson, period, and channel to evaluate effectiveness.
- **CRM Integration:** Managing customer relationships, loyalty programs, and feedback loops.

Finance and Accounting Information System

Scope & Functions

This subsystem manages the financial resources of the organization, ensuring legal compliance, accurate record-keeping, and effective capital allocation.

- **Revenue Tracking:** Recording and analyzing income streams from all products, services, and geographies.
- **Spending Layouts:** Monitoring expenditures across departments, projects, and cost centers.
- **Operational Cost Management:** Identifying cost drivers, tracking variable and fixed costs, and supporting cost-reduction initiatives.
- **Balance Sheet Mapping:** Maintaining assets, liabilities, and equity records for periodic financial reporting.
- **Financial Forecasting:** Projecting cash flows, revenue, and capital requirements for future periods.

Human Resource Information System (HRIS)

Scope & Functions

This subsystem manages the organization's human capital — from recruitment to retirement — ensuring that the right people are in the right roles with the right skills.

- **Payroll Structures:** Calculating salaries, deductions, bonuses, tax compliance, and disbursement schedules.
- **Hiring Funnels:** Tracking applicants through recruitment stages — sourcing, screening, interviewing, and onboarding.
- **Tracking Retirements:** Managing employee lifecycle events including retirements, resignations, and severance.
- **Performance Promotions:** Recording appraisal results, promotion histories, and performance improvement plans.
- **Employee Training:** Monitoring skill development programs, certifications, and training completion rates.

HOW FUNCTIONS SPAN MANAGEMENT LEVELS

Function	Operational Level	Tactical Level	Strategic Level
Production	Daily machine logs	Weekly capacity reports	5-year plant expansion plan
Marketing	Daily sales receipts	Monthly campaign ROI	New market entry strategy
Finance	Daily expense vouchers	Quarterly budget variance	Capital structure planning
HR	Daily attendance	Quarterly turnover report	Workforce strategy for M&A

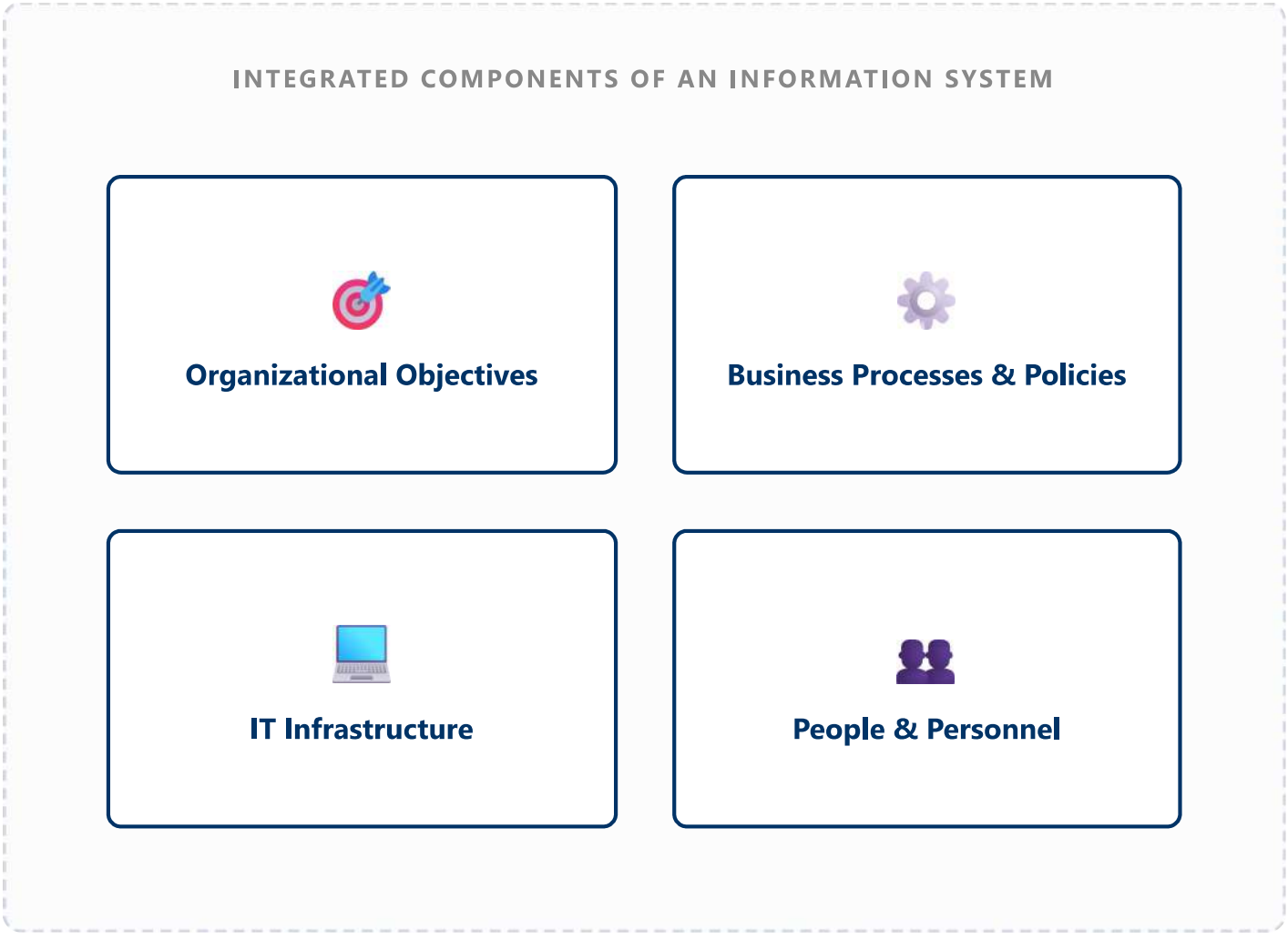
Cross-Functional Integration:

No functional subsystem operates in isolation. When Marketing launches a new campaign, the *Production* subsystem must ramp up output, the *Finance* subsystem must allocate additional budget, and the *HR* subsystem may need to hire temporary staff. The MIS structure must enable seamless data flow across these vertical pillars.

12.2 Information Systems Implementation Failures

From an organizational perspective, it has almost become impossible for any sizable business to exist without any information system. Since their advent, information systems are, many a times, projected as a silver bullet solution for deficiencies in the efficiency and effectiveness of business processes, or as a sure-shot tool for increasing them. However, what is often ignored is the fact that though information systems provide a wide array of feasible solutions to enhance efficiency and effectiveness, they also come with their own set of challenges.

The use of information systems requires the integration of multiple components—organizational objectives, business processes and policies, IT infrastructure (hardware, software, and networks), and so on. All these components, though interrelated, are managed in their own complex functions. Any issue of mismatch or compatibility in any of these components is bound to affect the implementation of the overall information system and may actually increase inefficiency, ineffectiveness, and promote any number of additional uncertainties. It is only through careful design, planning, acquisition, and implementation of information systems that organizations may benefit from them.



Failures in the implementation of information systems are not isolated occurrences and rather are common. However, the failure in the implementation of the information system is usually a failure

of the information system itself. Broadly speaking, to be labelled as a failure, the information system must meet either of the two criteria:

Criteria for IS Failure

1. **The system fails to meet the performance standards or requirements** as laid out by the management or the users of the system. This includes scenarios where systems fail to meet the laid implementation schedules, escalations in set budgets, underperformance compared to the set performance standards, etc.
2. **The results produced by the system are unacceptable or are not desirable** by the management or users of the system. This situation may arise even if the information system is implemented as per schedule and budgets sanctioned, and while conforming to the laid performance standards.



Important Distinction:
A system can be delivered on time, within budget, and meeting all technical specifications — yet still be considered a failure if the end results are not acceptable or desirable to the users and management. Delivery success does not guarantee implementation success.

12.3 Causes of Failure of Information Systems

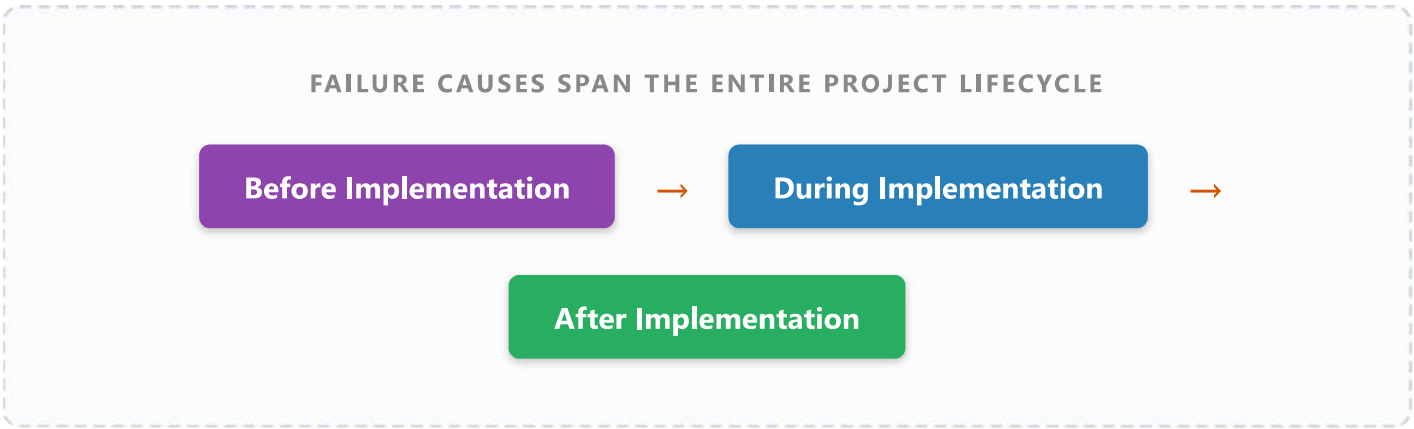
There can be multiple causes of failures in the implementation of information systems. Most of these failures have their origin in the mismanagement of various aspects related to the information system development project, or engineering principles adopted while developing the information system. Alternatively, the problems may have origins in the management of organizational change and other such practices or situations arising out of the information system development exercise.

A few of the issues that recurrently surface while analysing the causes of failure of various information system projects relate to:

- Lack of formal discussion and training
- Lack of research
- Lack of communication among various stakeholders
- Lack of leadership and their commitment towards information system implementation
- Lack of risk management and ownership of the entire development process

Spread Across Stages

These factors are spread across different stages of information system development. The following subsections discuss various reasons that may contribute to the failure of an information system **before, during, or after** implementation.



12.3.1 Lack of Research and Long-Term Commitment

The scale of implementation of information systems has increased manifolds, and with it, the associated complexities have also increased. In order to effectively handle these increased complexities, a well-planned and researched approach to the development and implementation of information systems is needed.

The information development approach in a modern context requires thorough research to develop the scale and scope of impact the proposed information system is going to have on existing processes. It is further needed to assess the readiness of the staff as well as the organization to switch over to the new system by taking into account the level of training and expertise, exposure to the proposed technology, as well as the availability and ability to implement the proposed technology and its enabling infrastructure.

It is also important to understand the impact of the proposed information system. The new system is bound to make certain changes in the existing process. An analysis of how these changes and adaptations in the processes are going to impact the entire system is also required. A thorough investigation into associated risks is necessary at the pre-implementation stage to ensure that no undue surprises crop up while implementing the information system.

Research Must Assess

- Scale and scope of impact on existing processes
- Readiness of staff and organization to switch over
- Level of training and expertise available
- Exposure to the proposed technology
- Availability and ability to implement enabling infrastructure
- How process changes will impact the entire system
- Associated risks at the pre-implementation stage

The Critical Role of Management Commitment

Management support, especially top management's commitment to develop and implement the information system, is one of the foundation stones for its successful implementation. As the development and implementation of an information system is not a short-term activity, long-term support is needed within a realistic timeframe and given objectives that the information system is supposed to achieve. Further, as the real benefits of the system's implementation may begin to be evident only after a while, patience and support at this stage are of critical importance to its success.



Foundation Stone Analogy:
Just as a building requires a strong foundation, an IS implementation project requires top management commitment as its base. Without long-term support and patience — especially during the period before benefits become evident — even well-designed systems can fail.

12.3.2 Escalation of Commitment

Escalation of commitment is a situation when an individual or organization continues to rationalize and exhibit a specific behavioral pattern even when that behavioral pattern results in outcomes that are the opposite of what was originally expected. Many a time, the information system development project goes out of hand due to this tendency of the project management to "over-commit" resources to a development exercise that is otherwise proving to be counterproductive. More related to behavioral, sociological, and psychological dimensions than traditional dimensions of management, these often go under-reported and under-estimated, making them one of the most difficult challenges faced.

A useful taxonomy to classify the factors that may promote escalation of commitment was provided by B. M. Staw and J. Ross in their work "*Knowing when to pull the plug*" published in the *Harvard Business Review* (March-April 1987). Their taxonomy classifies these factors as:

🏗️ Project Factors

These are the objective features of the project itself as perceived by management, which may include costs and benefits, complexity involved, project duration, etc. Usually, projects involving long development schedules and longer ROI periods, where setbacks are perceived as temporary problems, are more prone to escalation.

🧠 Psychological Factors

These factors arise primarily due to the failure of the parties involved to recognize negative outcomes of their course of action. They operate under a false sense of assurance that outcomes are not as negative as they look and that they will eventually succeed if they persist. These arise due to a sense of personal commitment and responsibility, or cognitive biases. Chances of escalation increase with a previous history of success and a high level of personal responsibility.

👥 Social Factors

Such factors are usually evident when competing teams are present and involved in the development and implementation of information systems. Projects are found to be more at risk of escalation in situations where there exists competitive rivalry between the decision-making group and other social groups involved as stakeholders.

🏢 Organizational Factors

These factors relate to the structure and political environment of the organization. Information system projects are more prone to failure if there is a lack of support from senior management; however, a strong involvement of top management, coupled with a sense of the project being institutionalized, may result in stronger susceptibility to escalation.

12.3.3 Lack of User Buy-In and Ownership



One of the most common reasons for failure is the recognition of its utility by its ultimate users. For any successful implementation, it is important that users are convinced that using the system will help them perform their tasks more efficiently and effectively.

It is very much possible, and frequently observed, that users may perceive the new system more as a nuisance than an enabler. However, the issue is not solely resistance to change. More often than not, the parties involved in development "assume" the utilities and benefits of the system rather than consulting actual users. Such situations result in actual users feeling pushed and forced to use the system rather than voluntarily adopting it. This leads to a failure to take complete ownership, resulting in sub-optimal utilization or overall failure.

The Core Problem: Assumption vs. Consultation

Developers often **assume** what users need instead of **consulting** them. This creates a gap between what is built and what is actually needed, leading to users feeling forced into using a system they had no voice in shaping — and consequently failing to take ownership of it.



Result of No Buy-In:

When users feel pushed and forced to use the system rather than voluntarily adopting it, they fail to take complete ownership. This results in either sub-optimal utilization of the system or its overall failure, regardless of how technically sound the system may be.

12.3.4 Information Technology Issues

One of the biggest problems is related to the choice of information technology components. Many fail to identify that **information technology (IT)** and **information systems (IS)**, though closely related, are two different concepts with different functions:

Information Technology (IT)

Relates to the products, methods, inventions, and standards used for the purpose of production and movement of information.

Information Systems (IS)

Consist of the infrastructure, applications, policies, processes, and personnel who use information technology to manage data and information resources of an organization.

Key Distinction

IT is but a **part** of information systems that facilitates the management of resources. The information system is the broader framework that encompasses IT along with people, processes, policies, and applications.

IT Strategic Planning

An important factor for success is the initial project part or strategic planning of IT. IT planning may include activities such as:

1. Recognizing organizational opportunities for using information technology;
2. Determining the resource requirements to exploit these opportunities; and
3. Developing strategies and action plans for realizing these opportunities and meeting resource needs.

To ensure suitable IT requirements match the business model and processes, organizations need to give proper consideration to the following nine planning agendas:

NINE IT PLANNING AGENDAS



Intra-organizational Political Analysis



Intra-organizational Market Analysis



Business Strategy Analysis



Business Market Analysis



Technology Analysis



Organizational Learning Analysis



Organizational Culture



IT Infrastructure Analysis



IT Risk-Taking Analysis

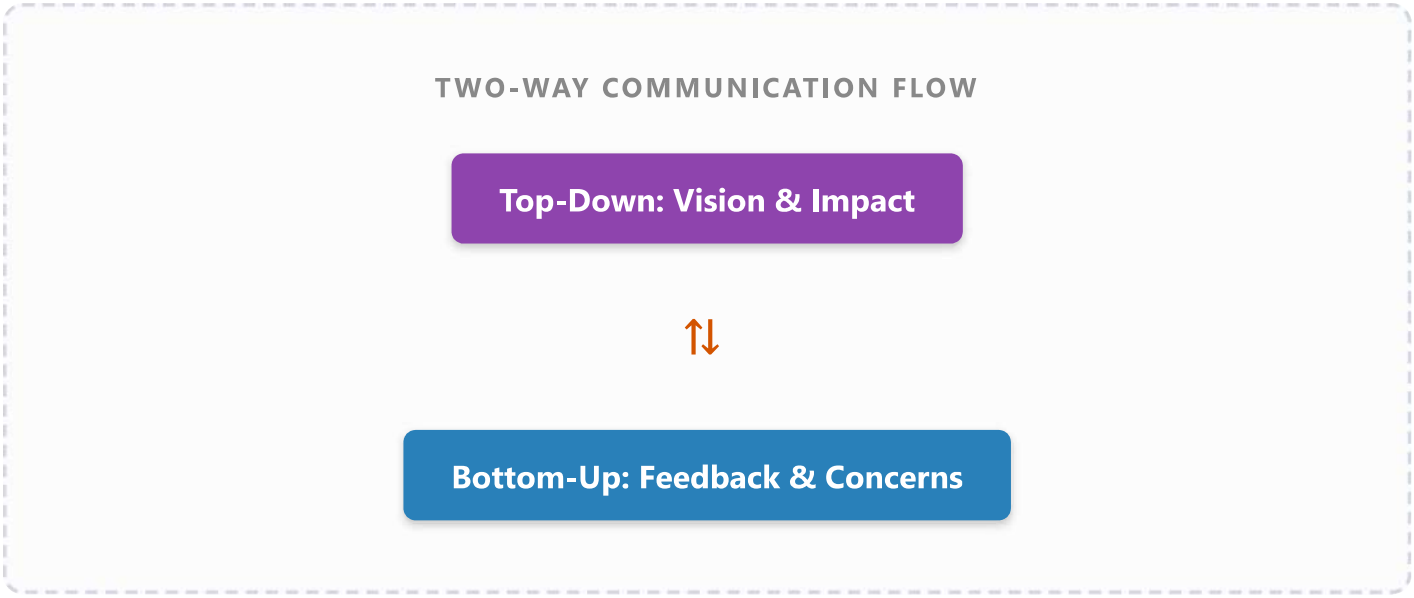


Remember:

It is important that the information system is seen as a **business solution**, rather than just an IT solution, to ensure it delivers what is expected.

12.3.5 Poor Communication

Effective communication among all stakeholders is essential. It is imperative that the vision behind the development of the system and its possible impact on organizational workflows and processes is communicated to all relevant levels. Further, it is equally important that feedback from the staff and actual users reaches those in command of planning and development to avoid unintended consequences arising from the changes.



Why Communication Fails

Communication breakdowns typically occur in two directions: (1) when the vision and impact are not clearly communicated downward to affected staff, and (2) when feedback from users and staff does not reach the planners and developers. Both gaps lead to unintended consequences and implementation failures.

12.3.6 Lack of Leadership

Planning, development, and subsequent implementation require inputs from across functional areas. Successful implementation entails allocating resources, monitoring outcomes, and controlling/coordinating people and work environments where resources often compete with other organizational functions.

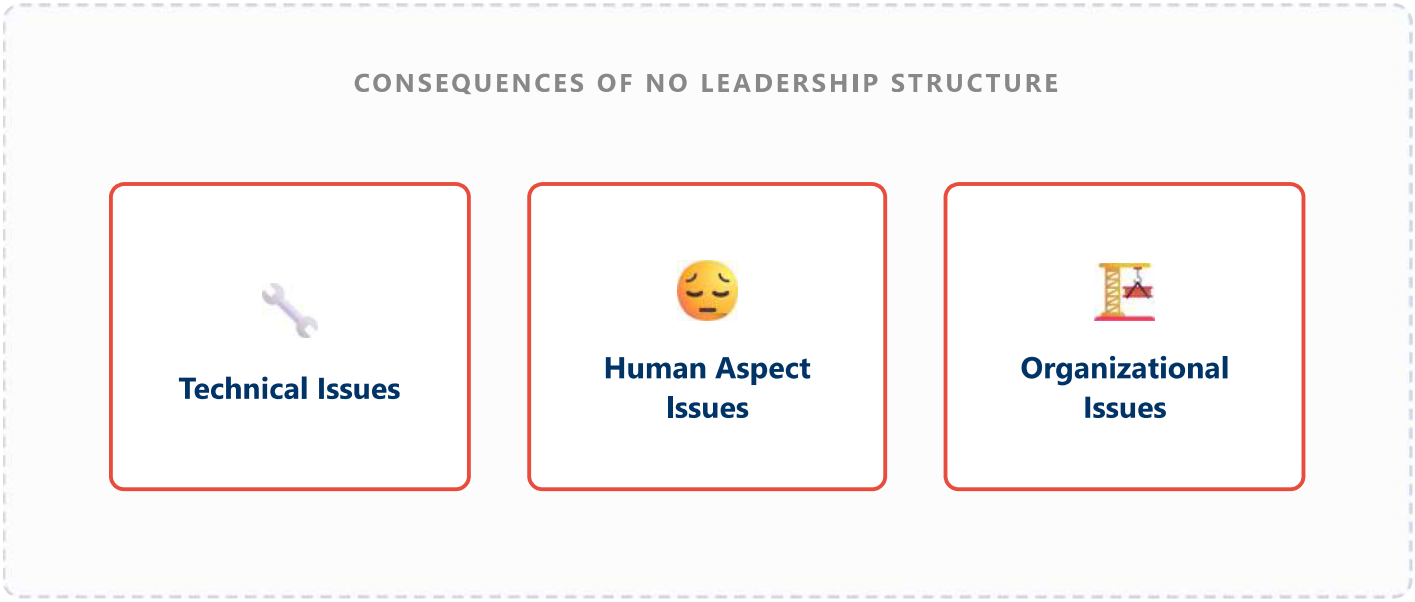
A proper structure must be put in place to spearhead, coordinate, and monitor the process. Ideally, this structure—called a Steering Committee—will include people from different groups of stakeholders affected by the implementation along with information system and technical experts. This team provides leadership and direction while addressing the concerns of all stakeholders.

Steering Committee Composition

- People from different groups of stakeholders affected by the implementation
- Information system experts
- Technical experts

This team provides **leadership and direction** while addressing the concerns of all stakeholders.

Failure to develop such a structure may create:



The leadership must include **strong line leadership** committed to the project's success, working alongside the **full support of top management**.

12.3.7 Lack of Change Management

Implementation is usually accompanied by varied levels of change, depending on the system's scope and technology exposure. Irrespective of how well a system is designed, the human potential in dealing with changes can be a limiting factor. Change management comprises structured approaches and management practices that help individuals, teams, and organizations transition from one system to another. An ill-designed change management strategy may spell doom for the project.

Two Components of Change Management

1. Change Management Approach

Refers to the established procedures, bureaucratic structures, and organizational culture that influence the way people work and think.

2. Management of Resistance to Change

Refers to managing employee resistance, which is one of the biggest barriers to successful change.

Why People Resist Change (Sacheva, 2009)

Studies list several factors for why people resist change in the context of information systems:

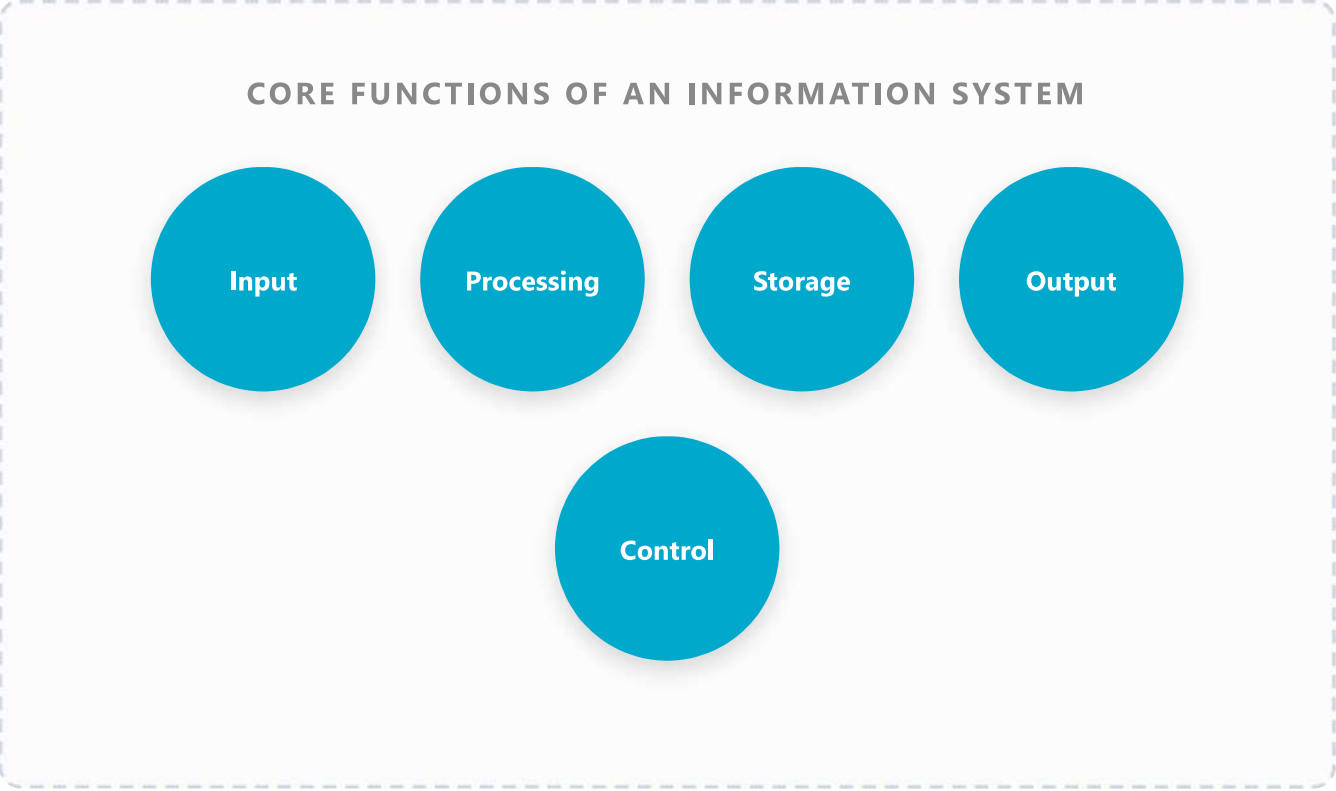
- Long implementation cycles
- Unknown drivers and lack of clarity of a vision
- Inadequate support from top management
- Process changes and official secrets
- Un-measurable benefits
- Disjointed systems and departments
- Fear of job loss or fear of loss of power
- Changes in job profile
- Cultural gaps and comfort with the status quo
- Work overload

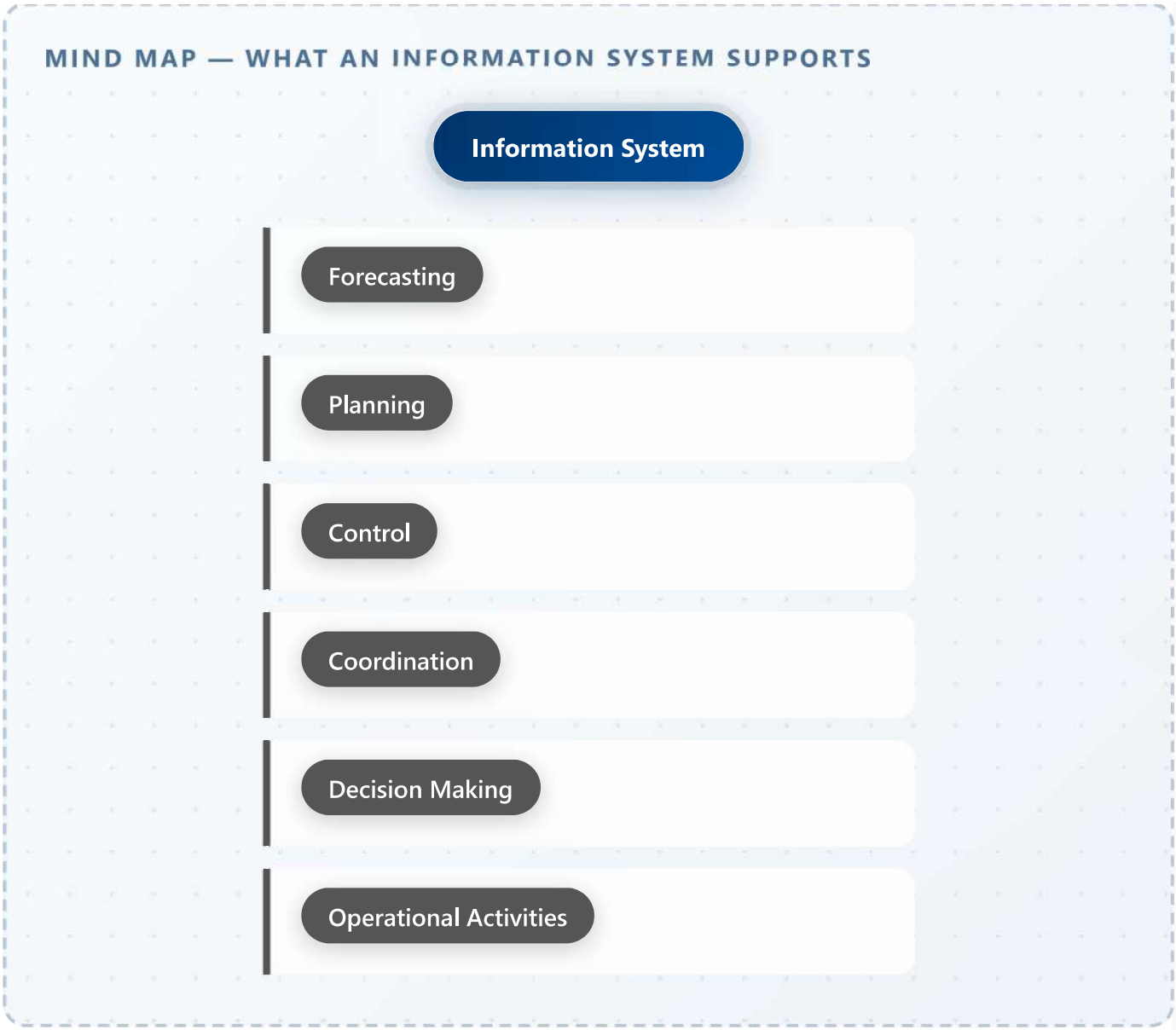
Information Systems in an Organization

Role, Enterprise Systems & Functional Information Needs

Introduction

An information system is a group of interrelated components that work to carry out input, processing, storage, output and control actions in order to convert data into information that can be used to support forecasting, planning, control, coordination, decision making and operational activities in an organization. Every business organization in this era needs an information system (IS) to keep track of all business activities, right from business planning, till the product delivery via manufacturing and quality cycles. Today, organizations of all size and field cannot exist without an effective information system. The level and type of information system needed would differ according to the product and type of information that is required to be assimilated.



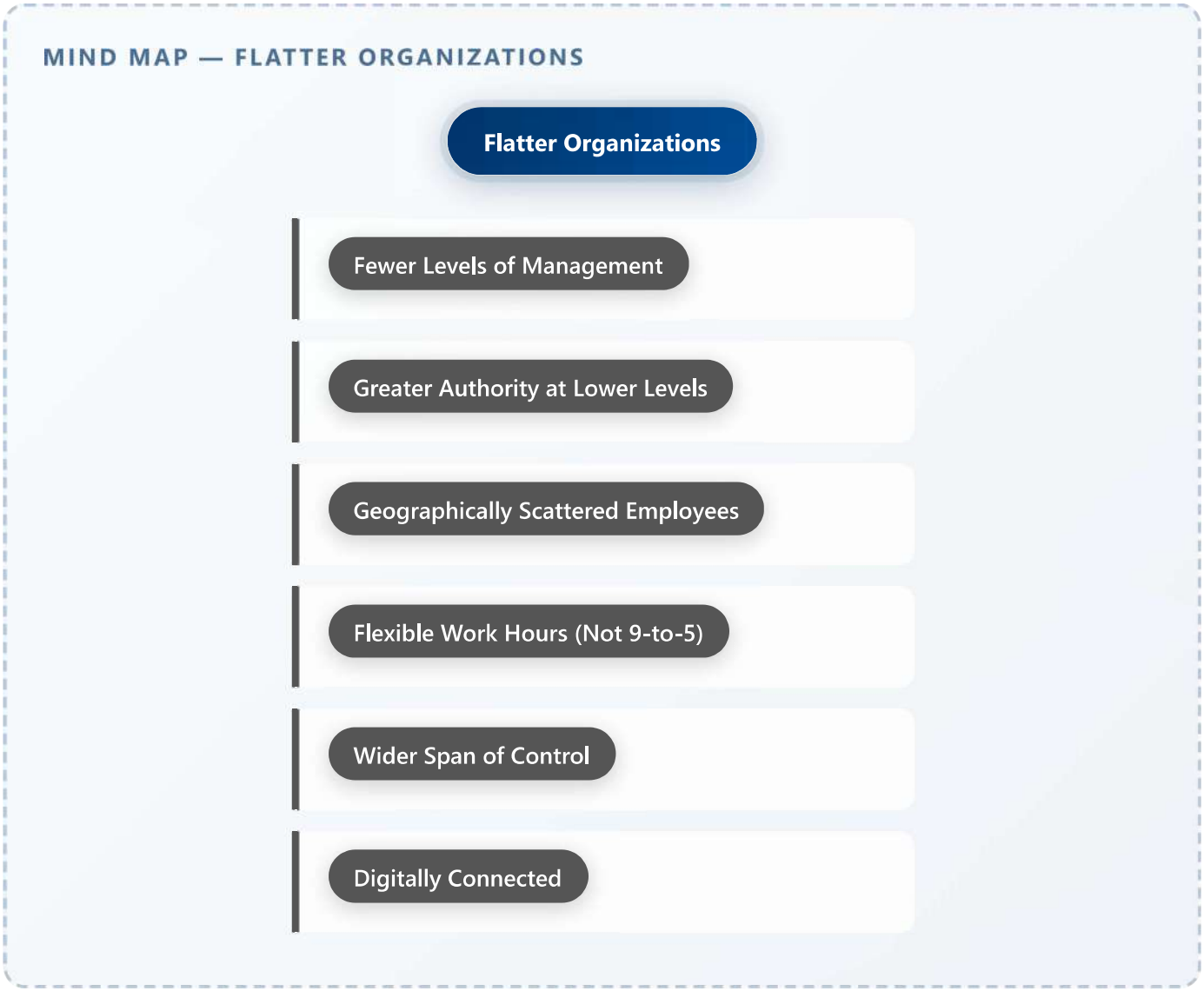


Key Takeaway

An information system converts data into information that supports **forecasting, planning, control, coordination, decision making** and **operational activities** in an organization. The level and type of IS needed differs according to the product and the type of information that is required to be assimilated.

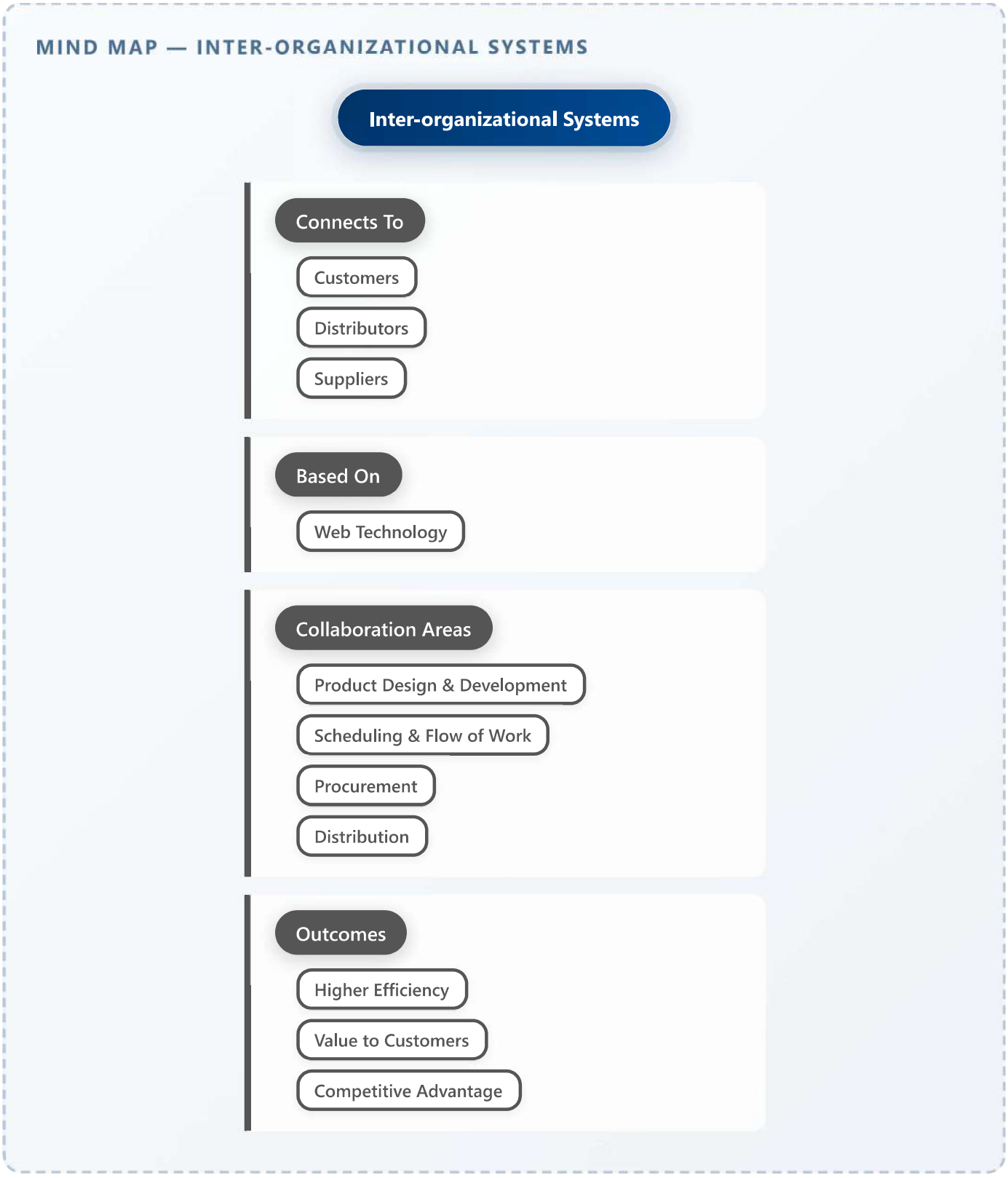
Role of an Information Systems in an Organization

Information Systems have become an integral part of any organization and managers cannot ignore them. This excessive usage of IS has given birth to a new digitally connected organization having less space for the earlier hierarchy and organizational levels. Such organizations are often termed as **flatter organizations**. Flatter organizations have fewer levels of management and greater authority at lower-level employees in decision-making. These employees, assisted and aided with IS are empowered to make more decisions than in the past. They no longer work standard nine-to-five hours or necessarily work in an office, instead may be scattered geographically, sometimes working half a world away from the manager. These changes reflect the widening of the management span of control allowing high-level managers to manage and control more workers spread over greater distances.



Inter-organizational Systems

Inter-organizational systems are emerging that use links to connect a company to its customers, distributors, or suppliers in order to automate the flow of information across organizational boundaries. These systems are mainly based on web technology and provide greater sharing of knowledge, resources, and business processes than in the past. Firms using these systems jointly work with suppliers and other business partners in areas such as product design and development, scheduling and flow of work in manufacturing, procurement, distribution, etc. Information Systems, at almost all the levels of the organization, are an integral part of such collaboration and coordination that can lead to higher levels of efficiency, value to customers, and ultimately significant competitive advantage.



INTER-ORGANIZATIONAL SYSTEM CONNECTIONS



Company



Customers



Distributors



Suppliers



Collaboration Areas:

Firms using inter-organizational systems jointly work with suppliers and other business partners in areas such as

product design and development

,

scheduling and flow of work in manufacturing

,

procurement

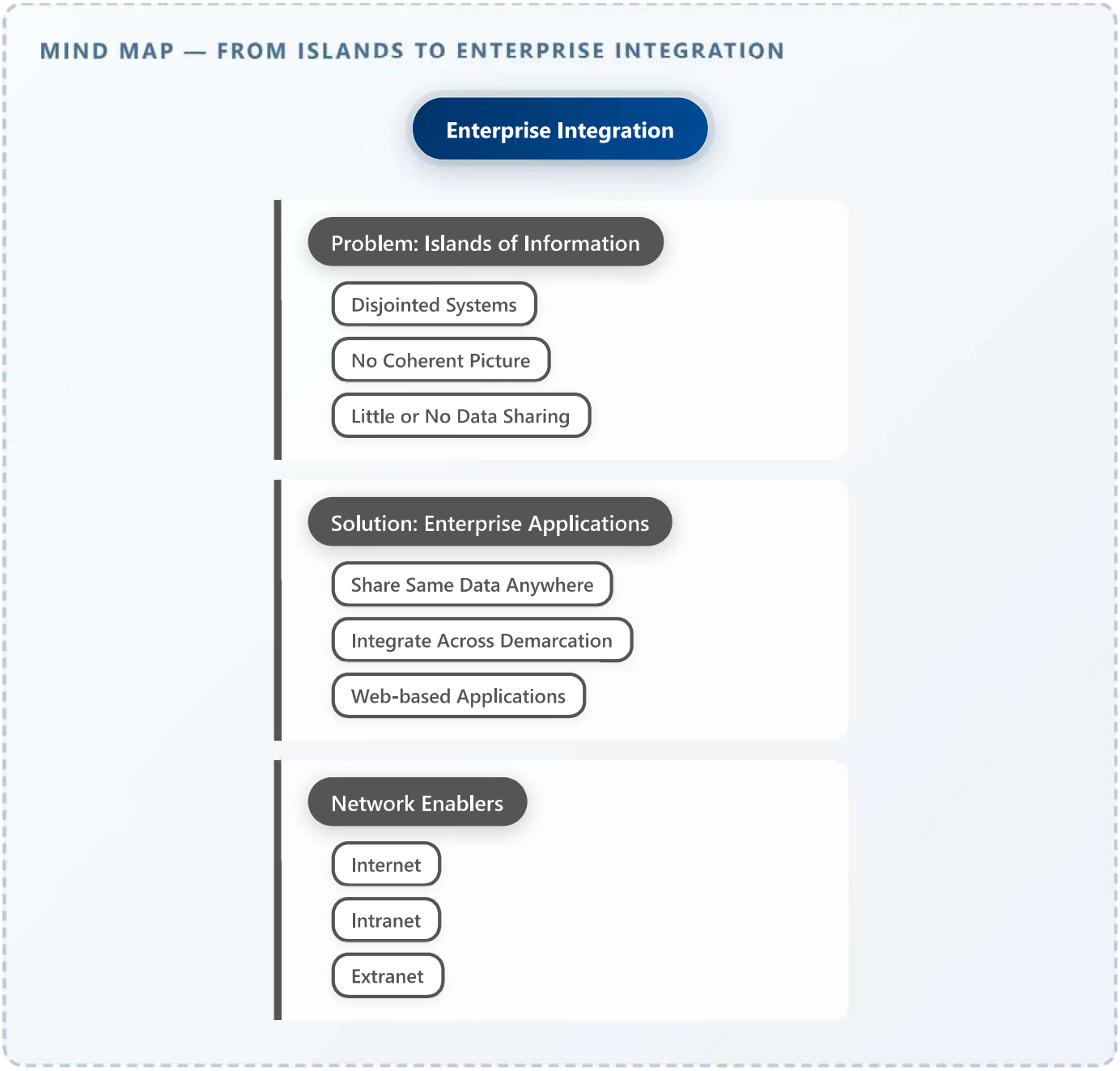
, and

distribution

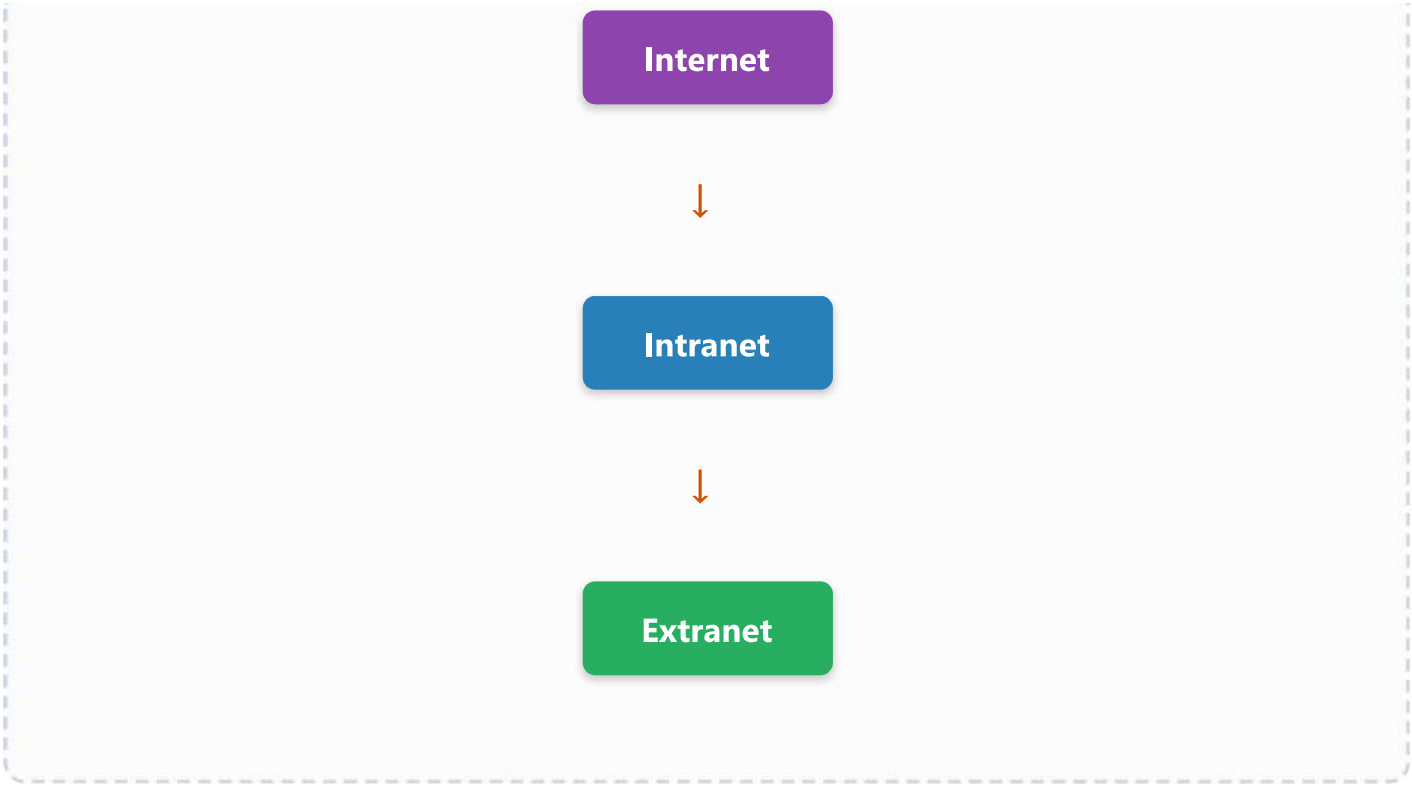
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Systems that Span an Enterprise

It's not unusual to find an organization with many different information systems that act as islands. No business can afford disjointed information systems that don't work together to produce a coherent picture of the entire organization. All the functions of a business need to be integrated across traditional lines of demarcation. Islands of information can prove to be devastating to a company if there is no or little sharing of data throughout the company. These kinds of problems are what gave rise to enterprise applications that share the same data anywhere it's needed in an organization. As networks of all kinds take hold, from the Internet to intranets to extranets, Web-based enterprise applications are increasingly widespread.



ENTERPRISE APPLICATION NETWORK TYPES



The Problem of "Islands of Information"

When different information systems act as isolated islands with little or no sharing of data throughout the company, the results can be **devastating**. Enterprise applications solve this by sharing the same data anywhere it's needed, integrating all business functions across traditional lines of demarcation.

Prevalent Enterprise Applications

Some of the most prevalent enterprise applications covered in the following sections are:

- Sales and Marketing Systems
- Finance and Accounting Systems
- Manufacturing and Production Systems
- Human Resource Management Systems
- Material / Inventory Management Systems

Sales and Marketing Systems

The sales and marketing function is responsible for selling the organization's product or service. At the strategic level, sales and marketing systems monitor trends affecting new products and sales opportunities, support planning for new products and services, and monitor the performance of competitors. At the management level, sales and marketing systems support market research, advertising and promotional campaigns, and pricing decisions. They analyze sales performance and the performance of the sales staff. Knowledge-level sales and marketing systems support marketing analysis workstations. At the operational level, sales and marketing systems assist in locating and contacting prospective customers, tracking sales, processing orders, and providing customer service support.



Strategic Information

1. Search for new markets and marketing strategies.
2. Analysis of competitor's strategy.
3. Technology and demographic forecasts and product changes.

Tactical Information

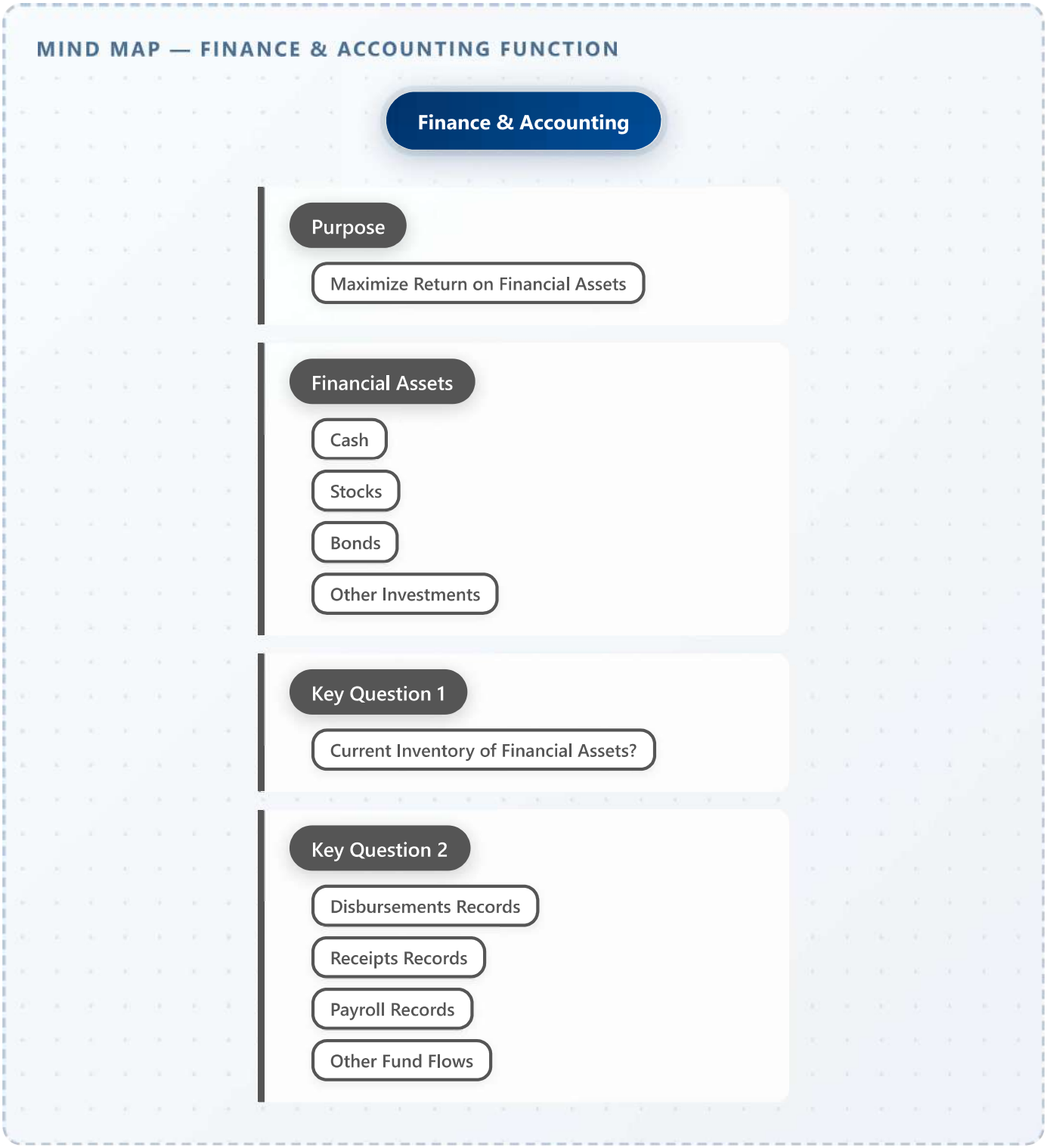
1. Advertising techniques and analysis of their impact.
2. Customer preference surveys.
3. Correlation of prices and sales.
4. Sales force deployment and targets.
5. Exploring alternate marketing channels.
6. Timing of special sales campaigns.

Operational Information

1. Sales analysis by regions, customer class, sales person.
2. Sales target versus achievement.
3. Market share and trends.
4. Seasonal variations.
5. Effect of model changes.
6. Performance of sales outlets.
7. Costs of campaigns and benefit.

Finance and Accounting Systems

The finance function is responsible for managing the firm's financial assets, such as cash, stocks, bonds, and other investments in order to maximize the return on these financial assets. Finance and accounting share related problems how to keep track of a firm's financial assets and fund flows. They provide answers to questions such as these: What is the current inventory of financial assets? What records exist for disbursements, receipts, payroll, and other fund flows?



Strategic Information

1. Methods of financing.
2. Pricing policies.
3. Tax planning.

Tactical Information

1. Variations between budget and expenses.
2. Large outstanding payments/Receipts.
3. Credit and payment status.
4. Cost increases and pricing.
5. Impact of taxation on pricing.

Operational Information

1. Periodic financial report.
2. Budget status to all functional managers.
3. Tax returns.
4. Share transfers.
5. Profit and loss account.
6. Payments and receipts.
7. Payroll, provident fund accounts.



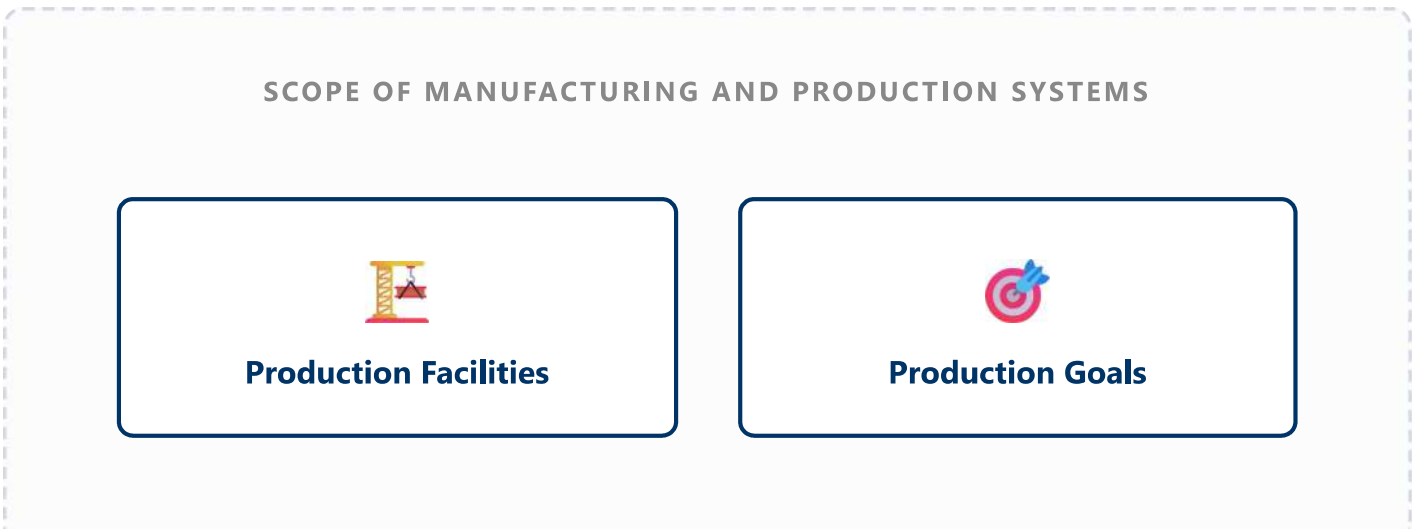
Core Questions Addressed:

Finance and accounting systems answer: What is the current inventory of financial assets? What records exist for disbursements, receipts, payroll, and other fund flows?

Manufacturing and Production Systems

The manufacturing and production function is responsible for actually producing the firm's goods and services. Manufacturing and production systems deal with the planning, development, and maintenance of production facilities; the establishment of production goals; the acquisition, storage, and availability of production materials; and the scheduling of equipment, facilities, materials, and labor required to fashion finished products.







Production Materials



Scheduling

Strategic Information

1. Yearly and monthly production quotas and alternate schedules.
2. Policies on machine replacement, augmentation and modernization.
3. Identifying best product mix.

Tactical Information

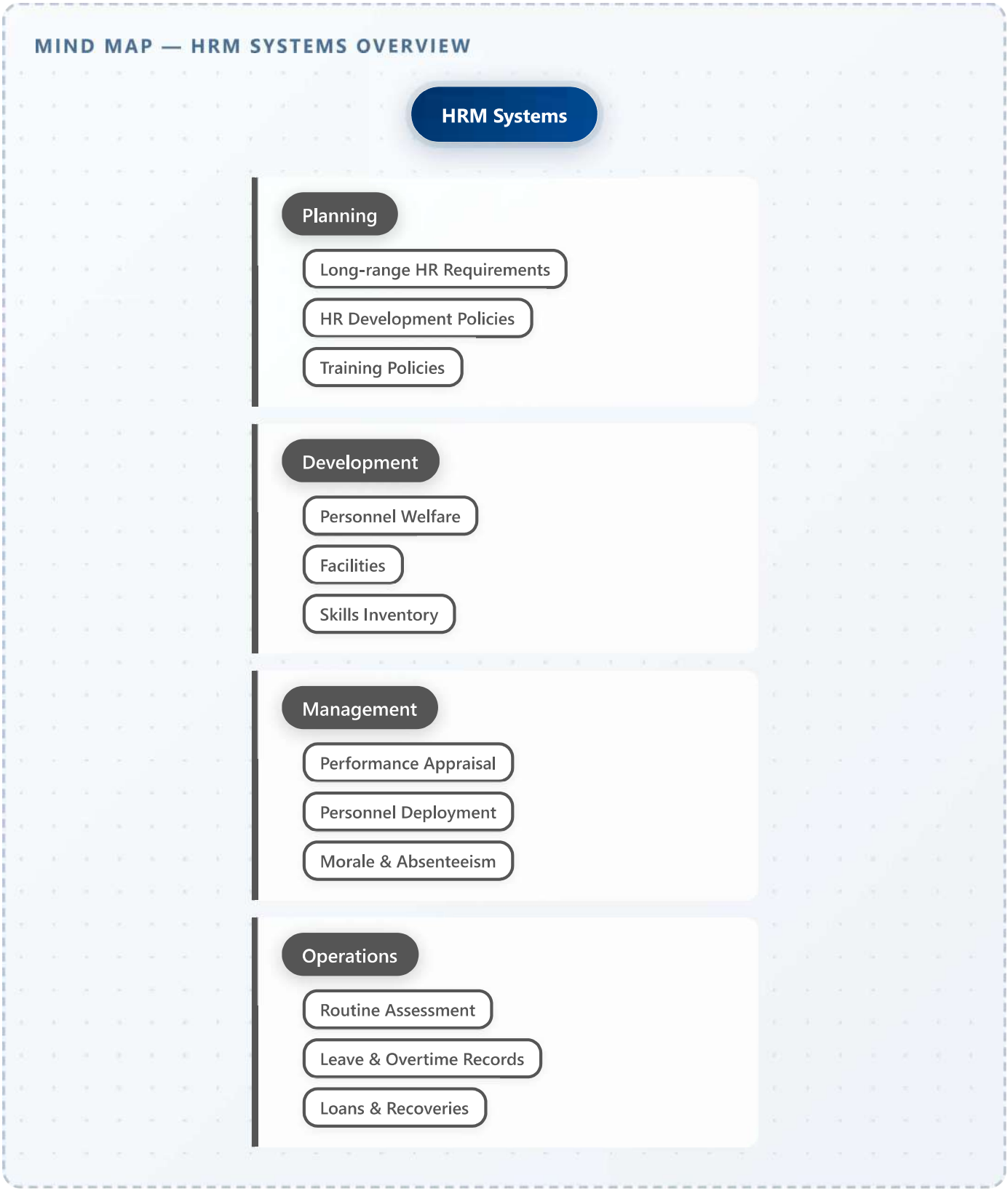
1. Identifying and controlling areas of high cost.
2. Identifying critical bottlenecks in production.
3. Identifying alternate production schedules based on tools, machines etc.
4. Performance measures of machines to decide replacement.

Operational Information

1. Monitoring up to date production information by examining assemblies, detecting likely shortages and giving early warning.
2. Scheduling better production dynamically.
3. Preventive maintenance schedules.
4. Monitoring tool, machine and personnel availability.

Human Resource Management

Human Resource Management systems support the planning, development, and management of the organization's human resources across all levels.



Strategic Information

1. Long range human resource requirements at different levels.
2. Policies on human resource development and training.
3. Policies on personnel welfare and facilities.

Tactical Information

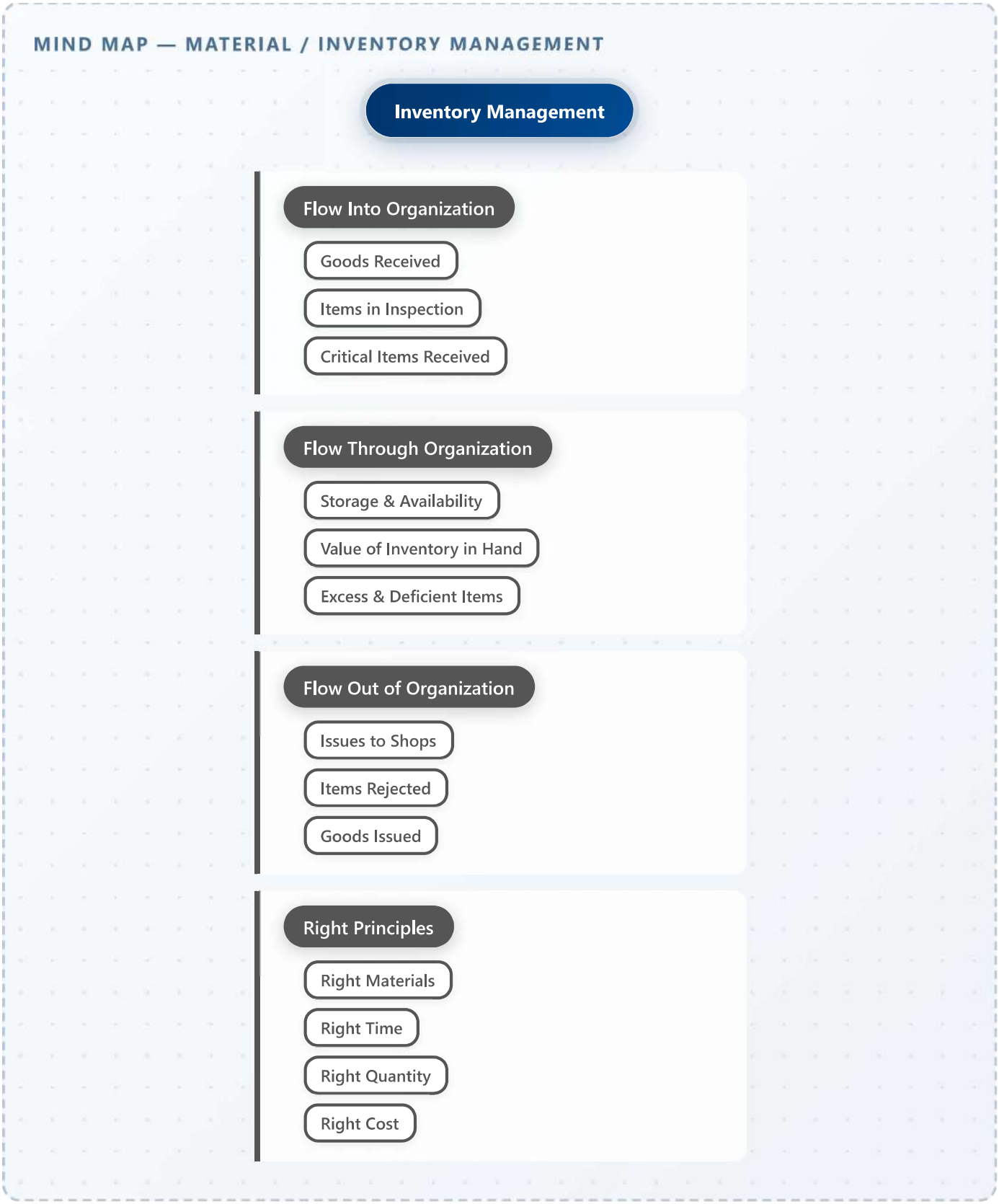
1. Performance appraisal.
2. Demographic make-up of personnel and its impact on retirement.
3. Production incentives.
4. Morale of personnel.
5. Absentee reduction.
6. Leave and overtime policies.
7. Personnel deployment policies.

Operational Information

1. Routine assessment.
2. Skills inventory.
3. Loan/advances and recoveries.
4. Leave record.

Material / Inventory Management

Material and Inventory Management systems are responsible for managing the flow of materials into, through, and out of the organization. These systems ensure that the right materials are available at the right time, in the right quantity, and at the right cost — supporting both production and distribution activities.



Strategic Information

1. Developing vendors for critical items.
2. Determining optimal levels of inventory.
3. Determining proportion of material needed.
4. Reducing varieties of inventory.

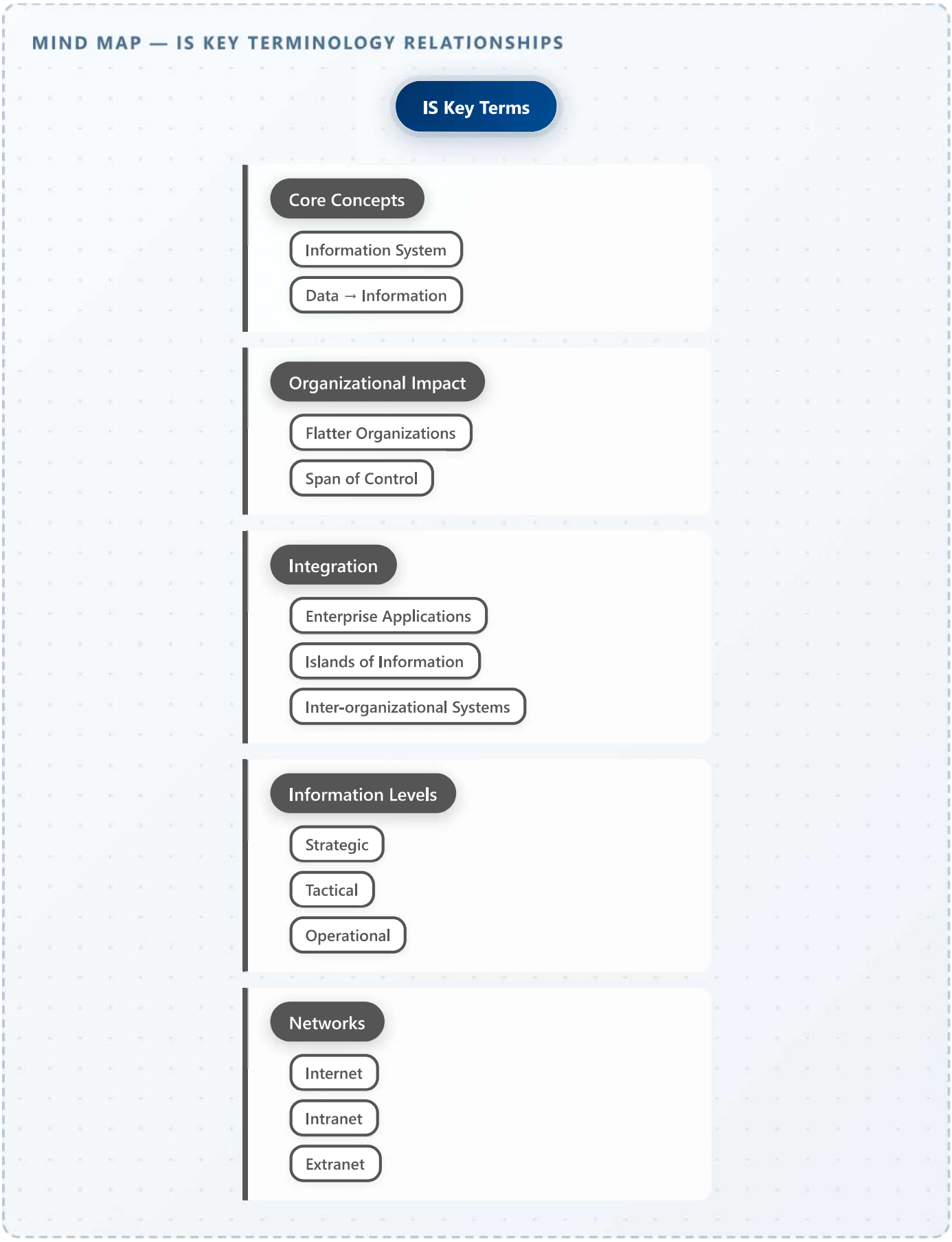
Tactical Information

1. Developing vendor performance measures.
2. Determining optimal reorder levels.
3. Determining issues of items to shops versus standard needs.
4. Controlling high value of inventory.
5. Determining impact on material cost and procurement with design changes and new product introduction.

Operational Information

1. List of excess and deficient items received.
2. List of items rejected.
3. Critical items received.
4. Stores in transit and in inspection.
5. Value of inventory in hand.
6. Goods received, rejected and issued.

Key Terms


MIND MAP — IS KEY TERMINOLOGY RELATIONSHIPS
IS Key Terms
Core Concepts
Information System
Data → Information
Organizational Impact
Flatter Organizations
Span of Control
Integration
Enterprise Applications
Islands of Information
Inter-organizational Systems
Information Levels
Strategic
Tactical
Operational
Networks
Internet
Intranet
Extranet

Information System (IS)

A group of interrelated components that work to carry out input, processing, storage, output and control actions in order to convert data into information that can be used to support forecasting, planning, control, coordination, decision making and operational activities in an organization.

Flatter Organizations

Organizations that have fewer levels of management and greater authority at lower-level employees in decision-making, enabled by the excessive usage of Information Systems. These organizations are digitally connected with less space for earlier hierarchy and organizational levels.

Inter-organizational Systems

Systems that use links to connect a company to its customers, distributors, or suppliers in order to automate the flow of information across organizational boundaries. These systems are mainly based on web technology and provide greater sharing of knowledge, resources, and business processes.

Enterprise Applications

Applications that share the same data anywhere it's needed in an organization. They arose from the need to eliminate "islands of information" — disjointed information systems that don't work together to produce a coherent picture of the entire organization.

Strategic Information

Information at the highest level of management that supports long-range planning, policy formulation, and overall organizational direction. Examples include methods of financing, pricing policies, tax planning, and identification of best product mix.

Tactical Information

Information at the management level that supports medium-term planning, monitoring, and control activities. Examples include variations between budget and expenses, sales force deployment and targets, and vendor performance measures.

Operational Information

Information at the operational level that supports day-to-day transactions and routine activities. Examples include periodic financial reports, sales analysis by regions, monitoring production information, and goods received/rejected/issued records.

Islands of Information

A situation where an organization has many different information systems that act as isolated islands with no or little sharing of data throughout the company. This can prove devastating to a company and gave rise to enterprise applications.

Span of Control

The number of workers a high-level manager can manage and control. Information Systems have widened this span, allowing managers to manage and control more workers spread over greater distances, including those scattered geographically.

Intranet

An internal network within an organization that facilitates enterprise-wide data sharing and communication, forming part of the network infrastructure supporting web-based enterprise applications.

Extranet

An extended network that connects an organization to its external business partners, enabling controlled access to specific information and supporting inter-organizational systems.

Advanced Architectural Frameworks in MIS

MBA / M.Sc. IT — Structural Subsystems, Descriptions & Inter-Organizational Hierarchical Levels

Lecture Overview & Learning Objectives

Aligning with the mandates of NEP 2026, this lecture transcends introductory definitions to investigate the structural dynamics of modern Management Information Systems (MIS). Today, we analyze the symbiotic relationship between advanced MIS subsystems and contemporary organizational hierarchies. By the end of this session, you will be able to map cross-functional data pipelines, evaluate systemic integration challenges, and architect information flows tailored to different management tiers.

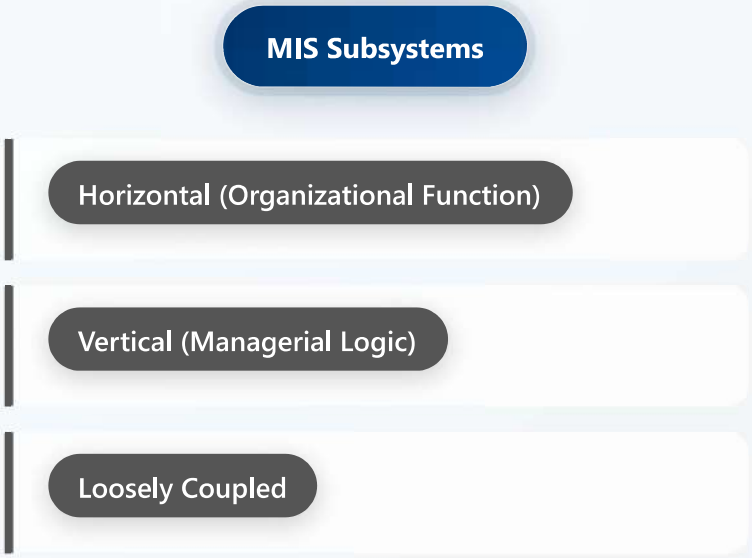
LEARNING OBJECTIVES



Core Premise

An enterprise-grade MIS is not a monolithic application; it is a **federation of loosely coupled, highly integrated subsystems**. These subsystems are categorized by **Organizational Function** (Horizontal) and **Managerial Logic** (Vertical).

MIND MAP — MIS SUBSYSTEMS ARCHITECTURE



Horizontal Subsystems (Functional Perspective)

Functional subsystems cater to the distinct domain requirements of the enterprise, acting as the primary nodes for specialized data generation.



FUNCTIONAL SUBSYSTEM NODES



Financial & Accounting



Marketing & Sales



Logistics & Supply Chain



Human Resource (HRIS)

Financial & Accounting Information Subsystems

Manages automated general ledgers, algorithmic cash-flow forecasting, tax compliance engines, and automated auditing frameworks (e.g., leveraging continuous compliance protocols mandated by regulations like SOX).

Marketing & Sales Information Subsystems

Driven by cloud-native CRM engines, real-time sentiment analysis, multi-channel attribution modeling, and dynamic price-optimization algorithms.

Logistics & Supply Chain Subsystems

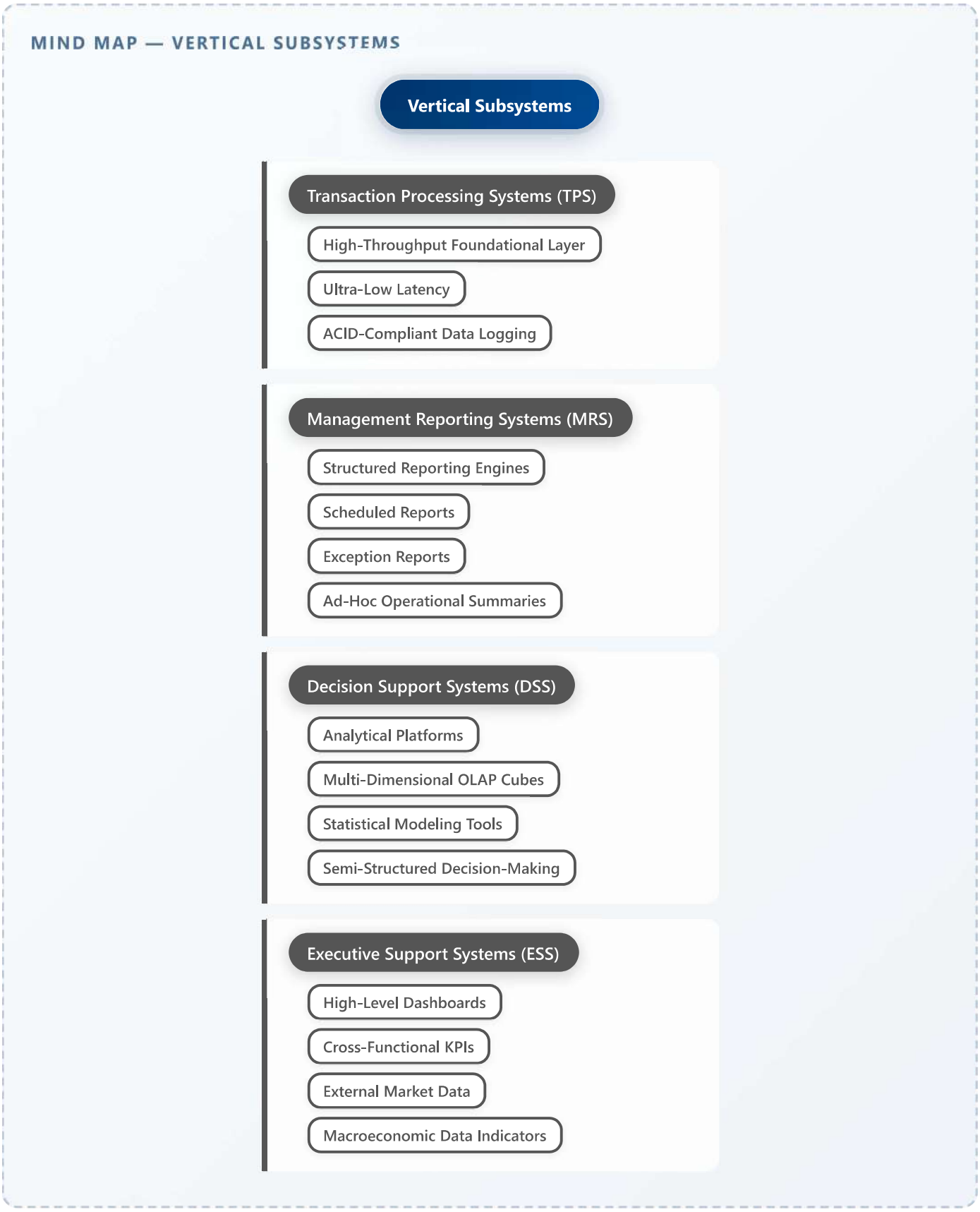
Orchestrates real-time inventory optimization, multi-echelon predictive routing, automated procurement, and fleet analytics powered by IoT-enabled sensors.

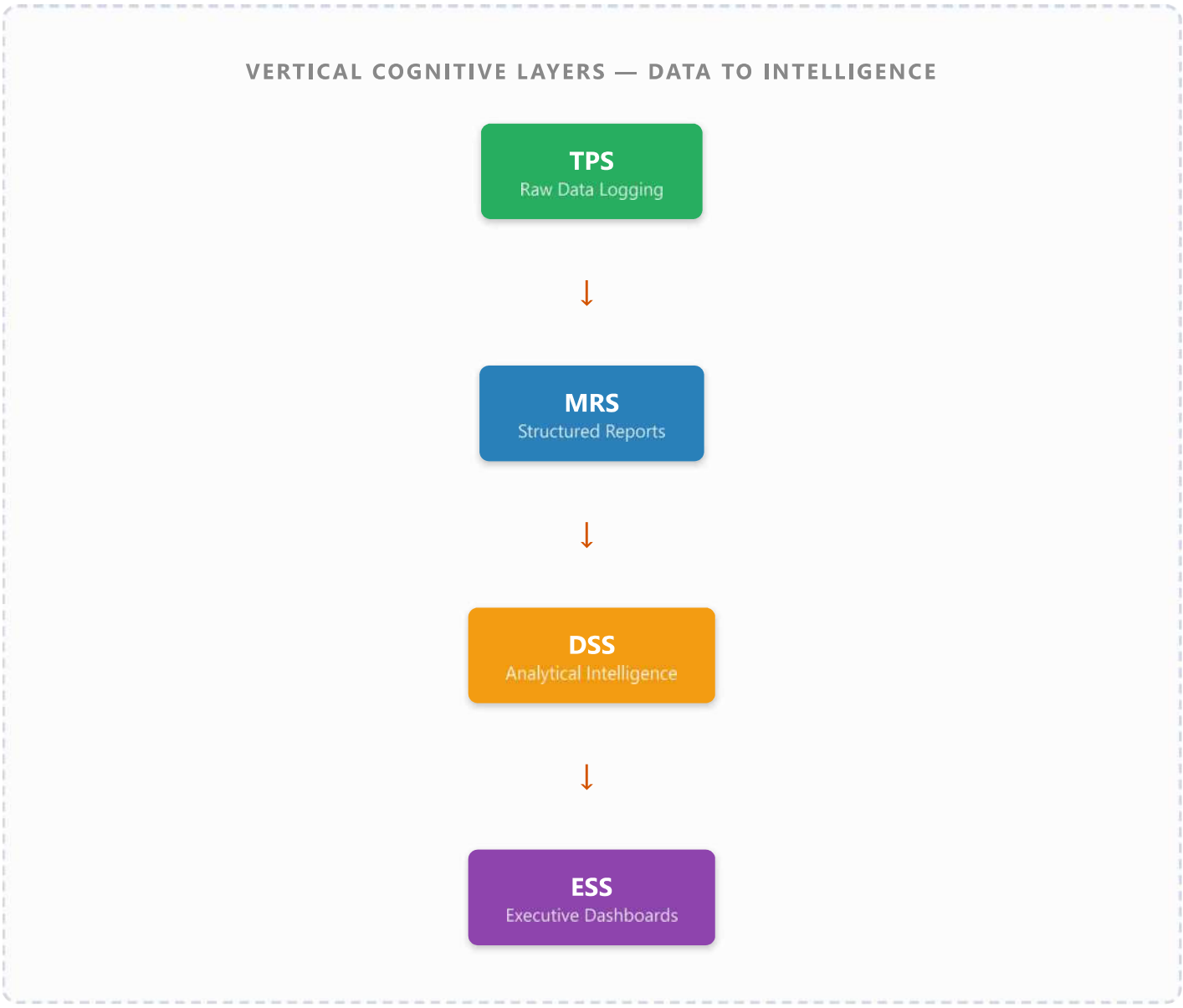
Human Resource Information Subsystems (HRIS)

Powers automated payroll, AI-driven talent acquisition, predictive attrition modeling, and continuous workforce performance tracking.

Vertical Subsystems (Logical & Processing Perspective)

Vertical subsystems act as the cognitive layers processing raw inputs into actionable intelligence.





Transaction Processing Systems (TPS)

The high-throughput foundational layer designed for ultra-low latency, ACID-compliant data logging.

Management Reporting Systems (MRS)

Structured reporting engines generating scheduled, exception, and ad-hoc operational summaries.

Decision Support Systems (DSS)

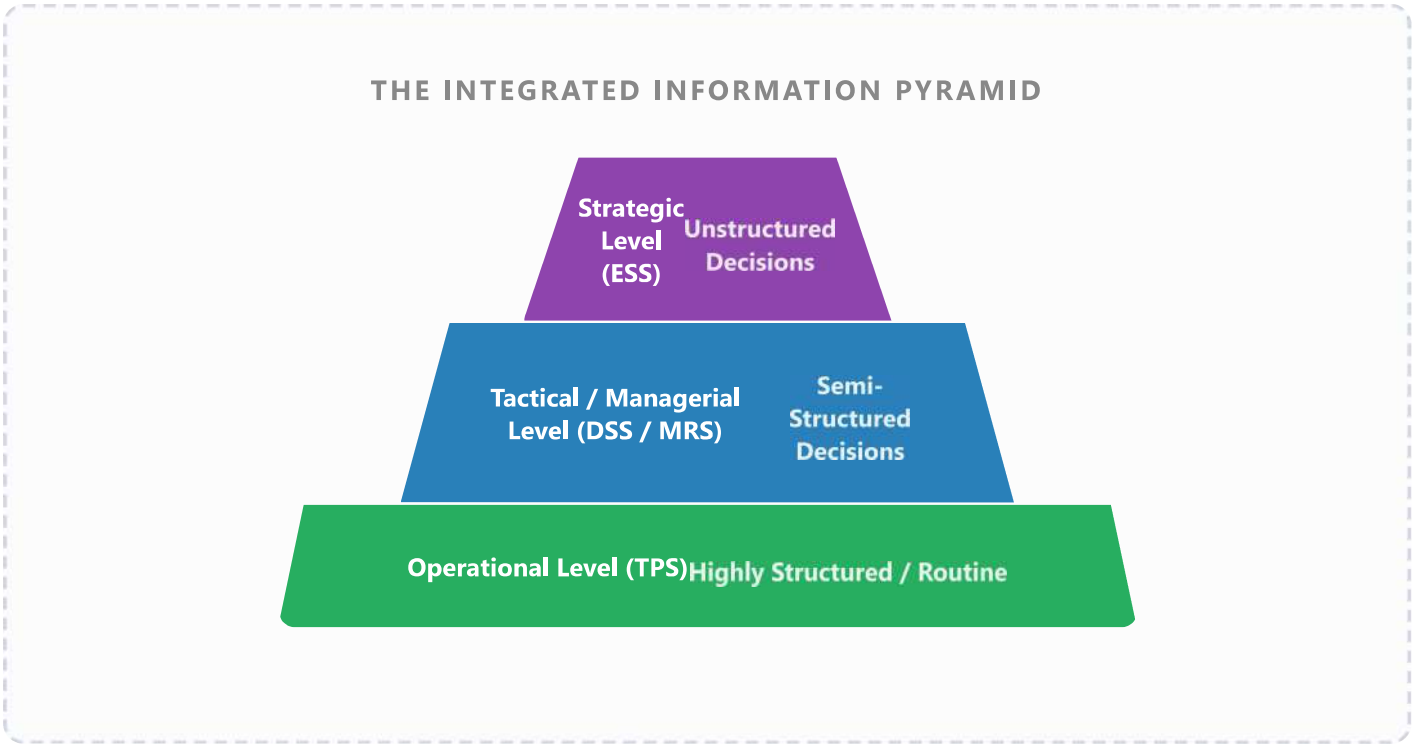
Analytical platforms equipped with multi-dimensional **OLAP cubes** and statistical modeling tools for semi-structured decision-making.

Executive Support Systems (ESS)

High-level dashboards visualizing cross-functional **KPIs** alongside external market and macroeconomic data indicators.

MIS Alignment Across Organizational Levels

The foundational architecture of an MIS is conceptualized as an integrated information pyramid. The system's structure must strictly map to the cognitive demands, decision frequency, and risk tolerance of each organizational level.



Structural Mapping Principle

The system's structure must strictly map to the **cognitive demands**, **decision frequency**, and **risk tolerance** of each organizational level. Moving from the base to the apex, information becomes more aggregated, less structured, and longer-term in horizon.

Operational Level — TPS

- Highly Structured / Routine Decisions
- High transactional throughput
- Ultra-fine, raw data parameters
- Present / Immediate Past time horizon
- Risk Focus: Process failure / Downtime

Tactical / Managerial Level — DSS / MRS

- Semi-Structured Decisions
- Resource optimization, variance analysis
- Aggregated, summarized data
- Short to Medium Term (Months)
- Risk Focus: Resource misallocation

Strategic Level — ESS

- Unstructured (Heuristic) Decisions
- Long-term visioning, competitive differentiation
- Highly synthesized trends
- Long Term (Years)
- Risk Focus: Market obsolescence / Valuation

The Operational Level (Structured Control Layer)

Profile

- **Primary Users:** Frontline supervisors, section leads, operations coordinators, and automated transactional workflows.
- **Nature of Information:** Highly granular, real-time, internally sourced, deterministic, and historical.
- **System Anchor:** Transaction Processing Systems (TPS).
- **Core Objective:** Operational efficiency, internal process auditing, and maintaining high transactional throughput.

Advanced Description

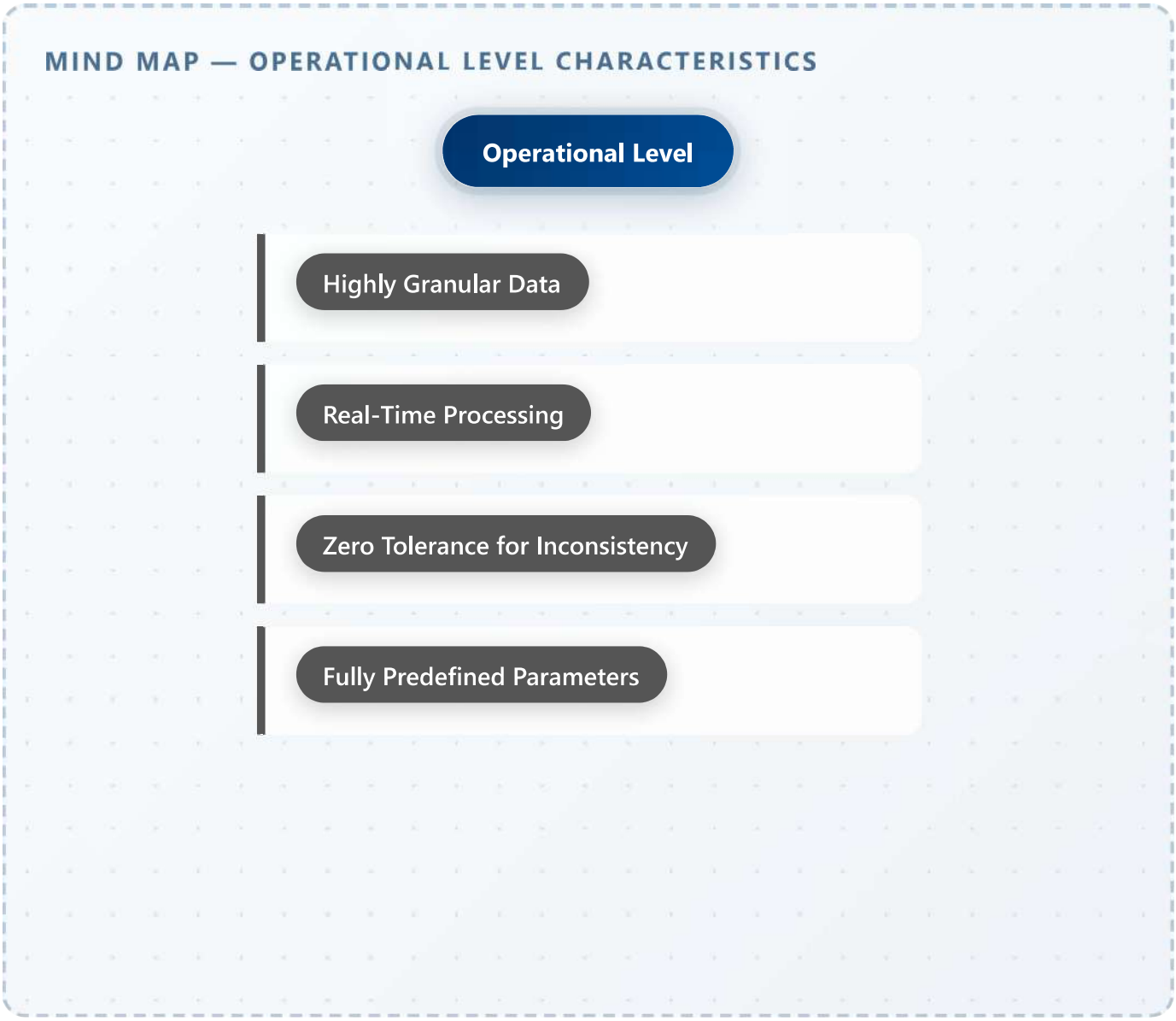
At this stratum, the MIS processes massive datasets with zero toleration for data inconsistency. The system focuses on routine, highly structured decisions where the parameters are fully predefined (e.g., "*If inventory of Component X drops below 15% of safety stock, trigger a purchase order to Supplier Y.*")



2026 Real-World Context:

Consider an automated fulfillment center for an e-commerce giant like **Amazon** or **Flipkart**. The operational MIS continuously streams telemetry from warehouse IoT sensors and RFID tags. The moment an item crosses the sorting belt, the TPS processes the scan, updates the inventory database, assigns an optimal packaging lane, and generates a real-time shipping manifest without human intervention.

MIND MAP — OPERATIONAL LEVEL CHARACTERISTICS



The Tactical / Managerial Level (Semi-Structured Control Layer)

Profile

- **Primary Users:** Departmental Heads, Regional Managers, and Business Unit Directors.
- **Nature of Information:** Aggregated, semi-structured, diagnostic, comparison-driven, and near real-time.
- **System Anchor:** Management Reporting Systems (MRS) and Decision Support Systems (DSS).
- **Core Objective:** Resource optimization, variance analysis, performance evaluation, and tactical forecasting.

Advanced Description

The managerial layer acts as the bridge transforming high-volume operational data into contextualized business intelligence. Decisions here are semi-structured—partially governable by standard algorithms, but requiring human intervention and domain expertise to account for volatile variables.

Key Analytical Modalities

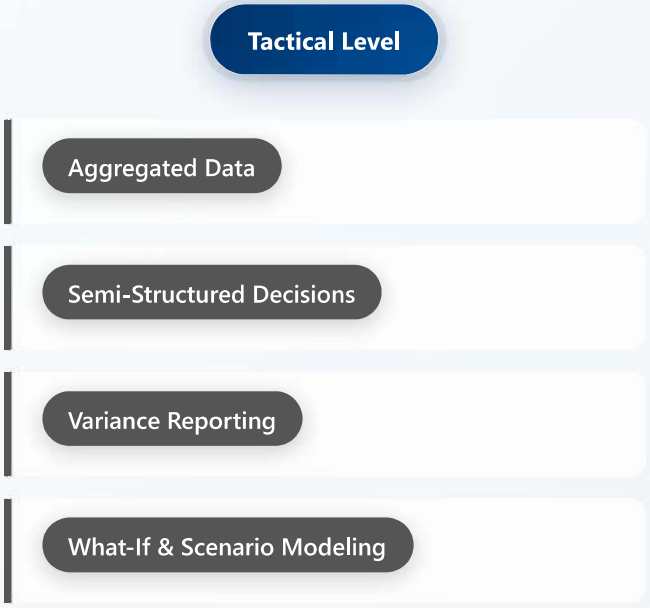
Variance Reporting: Automatically highlighting deviations between budgeted performance and actual execution metrics.

What-If & Scenario Modeling: Simulating the impact of tactical changes (e.g., adjusting regional logistics routes or promotional discount percentages).



2026 Real-World Context:
In a modern banking environment (such as **HDFC Bank** or **HSBC**), regional managers utilize a managerial DSS to analyze credit portfolio risks. If the system flags an unexpected surge in **Non-Performing Assets (NPAs)** within a specific retail sector, the manager runs predictive scenario models to test how raising the credit-score threshold by 50 points would impact new loan conversions and default rates over the subsequent two quarters.

MIND MAP — TACTICAL / MANAGERIAL LEVEL



The Strategic Level (Unstructured Value Layer)

Profile

- **Primary Users:** Chief Executive Officer (CEO), Chief Financial Officer (CFO), Board of Directors, and Corporate Strategists.
- **Nature of Information:** Highly summarized, holistic, forward-looking (predictive/prescriptive), heavily reliant on external data (competitor intelligence, regulatory changes, macroeconomic shifts).
- **System Anchor:** Executive Support Systems (ESS) / Executive Information Systems (EIS).
- **Core Objective:** Long-term visioning, competitive differentiation, capital allocation, and risk management.

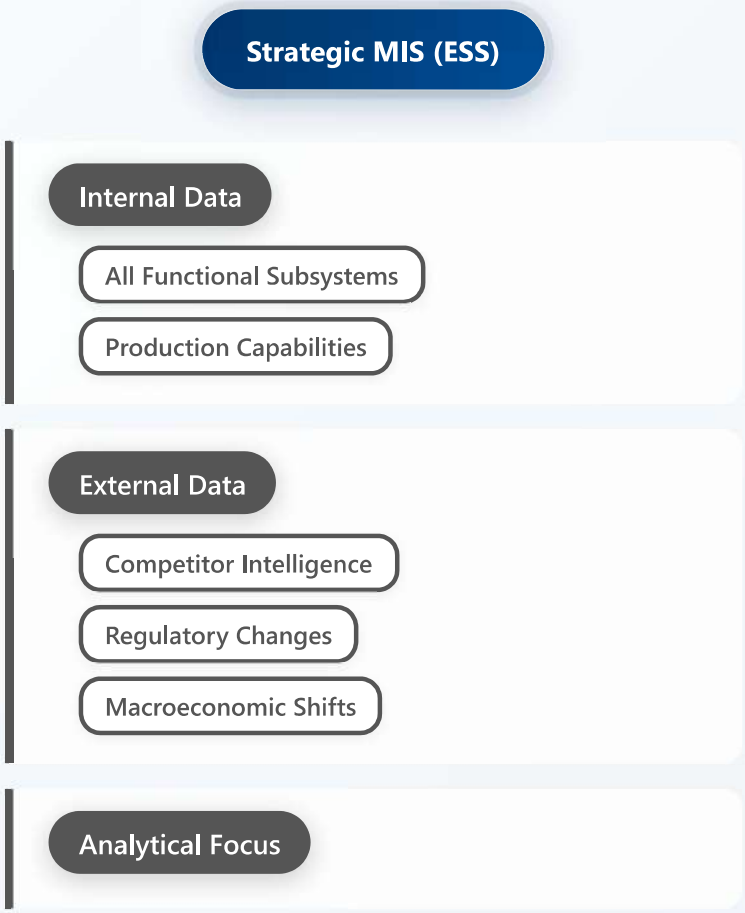
Advanced Description

Strategic management deals with unstructured, non-routine challenges where the path forward is ambiguous and information is incomplete. The strategic MIS must unify data from all internal functional subsystems and overlay it with complex global market feeds, policy tracking, and alternative datasets.



2026 Real-World Context:
Imagine the executive boardroom of a global automotive manufacturing company navigating the transition to autonomous electric vehicles. The executive team relies on an AI-powered ESS dashboard. To decide whether to invest $2B in a new solid-state battery manufacturing plant, the ESS aggregates internal production capabilities alongside external datasets:
global lithium futures volatility
,
changing environmental subsidies under regional policies
, and
patent filing indices of key competitors
.

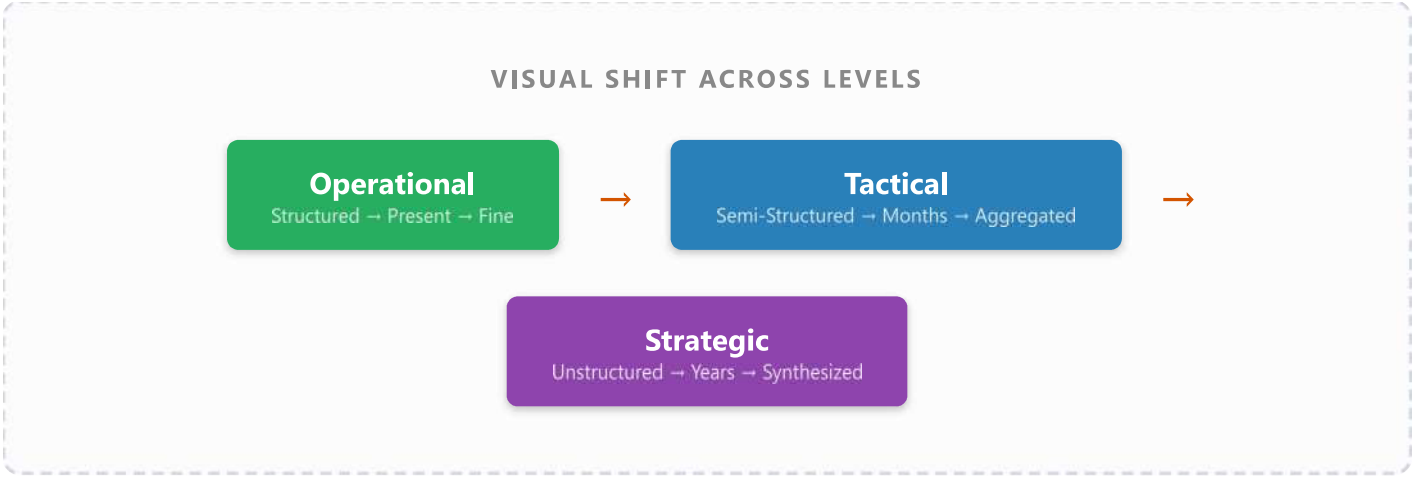
MIND MAP — STRATEGIC LEVEL DATA SOURCES



Comparative Architecture Framework

To synthesize how information characteristics shift across these layers, review this systematic structural matrix:

Architectural Metric	Operational Level	Tactical / Managerial Level	Strategic Level
Decision Structure	Structured (Algorithmic)	Semi-structured	Unstructured (Heuristic)
Time Horizon	Present / Immediate Past	Short to Medium Term (Months)	Long Term (Years)
Data Scope	Narrow, domain-specific	Focused cross-functional slices	Broad, enterprise-wide & global
Data Granularity	Ultra-fine, raw parameters	Aggregated, summarized	Highly synthesized trends
Primary System Type	TPS / Operational Databases	MRS / OLAP Cubes / DSS	ESS / Prescriptive Dashboards
Risk Focus	Process failure / Downtime	Resource misallocation	Market obsolescence / Valuation



Key Takeaway

As one moves up the organizational hierarchy from Operational to Strategic, information becomes **less structured**, covers a **broader scope**, shifts from **ultra-fine to highly synthesized** granularity, extends over a **longer time horizon**, and the **risk focus transitions** from process-level failures to market-level obsolescence and valuation risks.

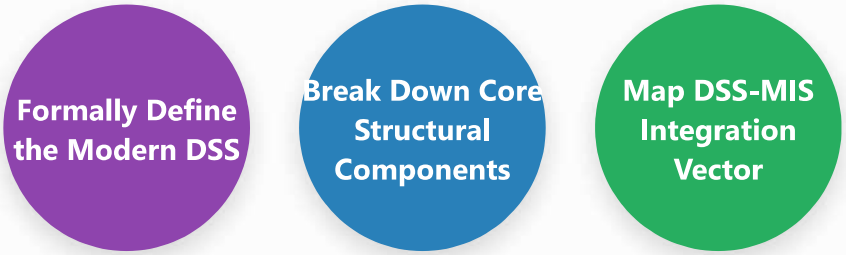
Advanced Decision Support Architectures and the MIS Integration Matrix

MBA / M.Sc. IT — DSS: Core Definitions, Structural Architecture & Functional Relationship with MIS

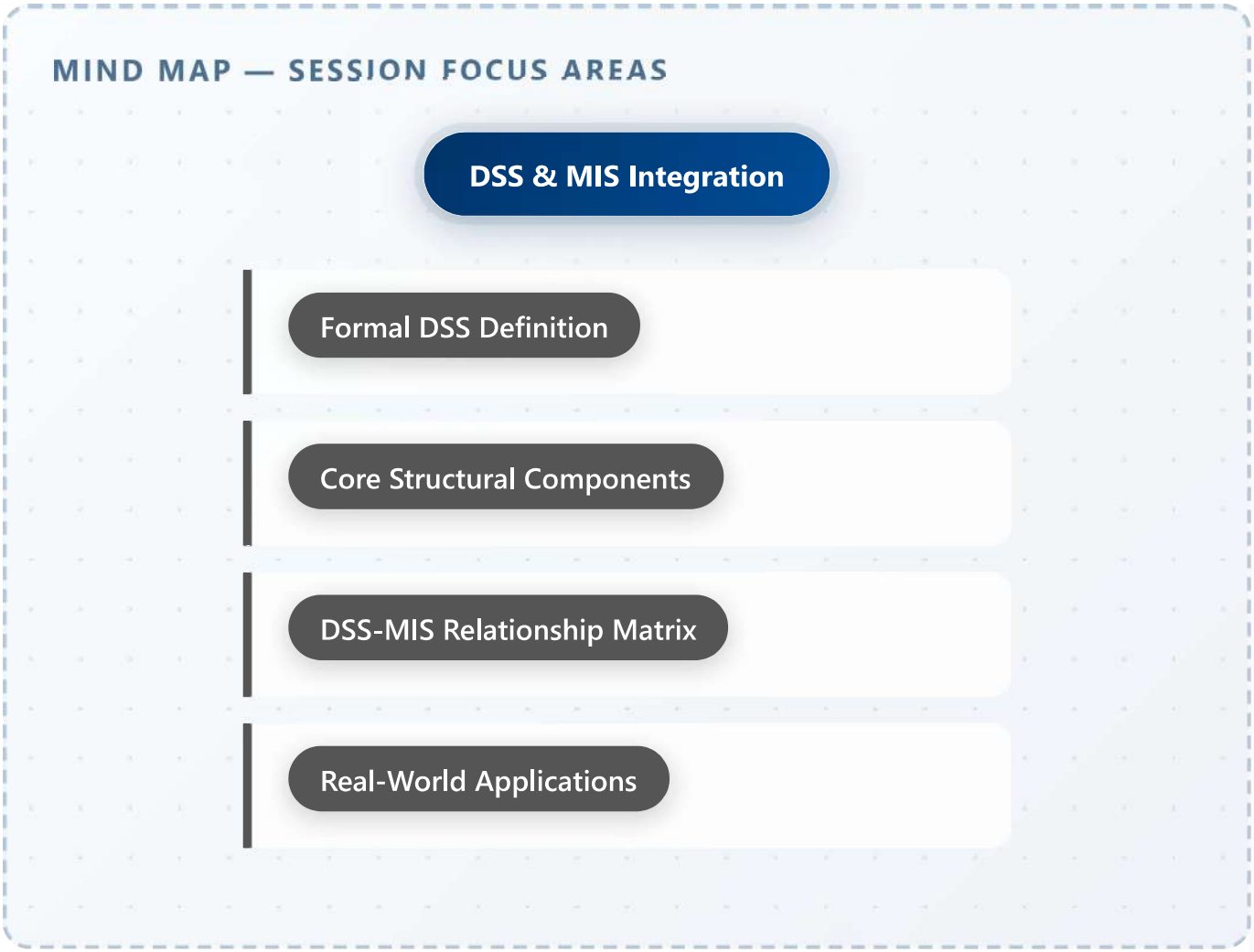
Lecture Overview & Learning Objectives

In alignment with the advanced learning verticals of NEP 2026, this session moves past entry-level conceptualizations to analyze the mechanics of Decision Support Systems (DSS). As organizations navigate increasingly VUCA global markets, relying purely on periodic reporting is insufficient. Today, we will formally define the boundary metrics of a modern DSS, break down its core structural components, and map its exact integration vector within the broader enterprise MIS ecosystem.

LEARNING OBJECTIVES



MIND MAP — SESSION FOCUS AREAS



Deconstructing the Modern Decision Support System (DSS)

Formal Definition

A **Decision Support System (DSS)** is an interactive, computer-based information framework that utilizes data pipelines, specialized analytical models, and user-centric interfaces to facilitate semi-structured and unstructured decision-making processes.

Key Distinction

Unlike systems designed to automate routine procedures, a DSS serves as an **intellectual augment** for human decision-makers. It handles complex data processing, allowing managers to apply heuristic insights to multi-variable business challenges.

DSS CORE CHARACTERISTICS



Interactive



Data Pipelines



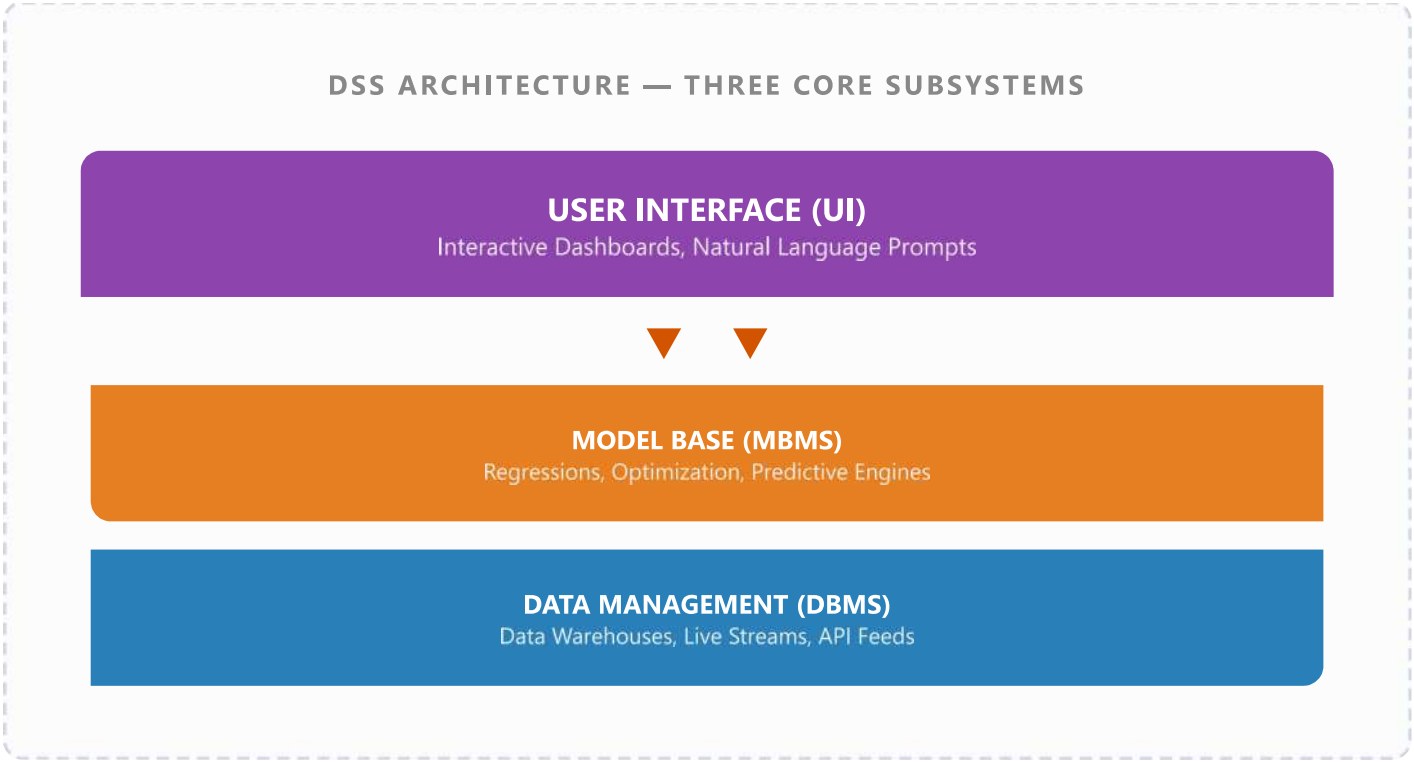
Analytical Models

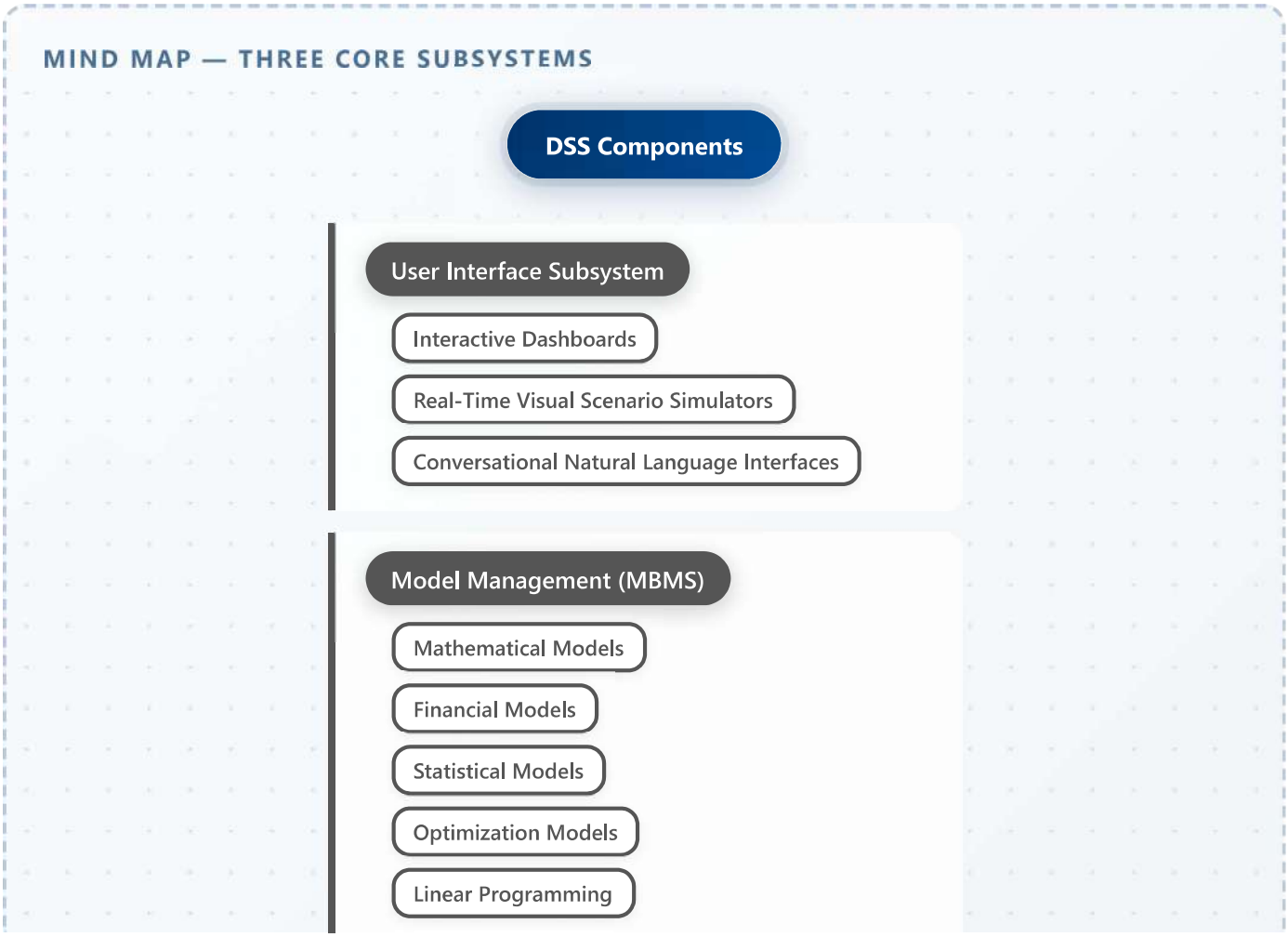


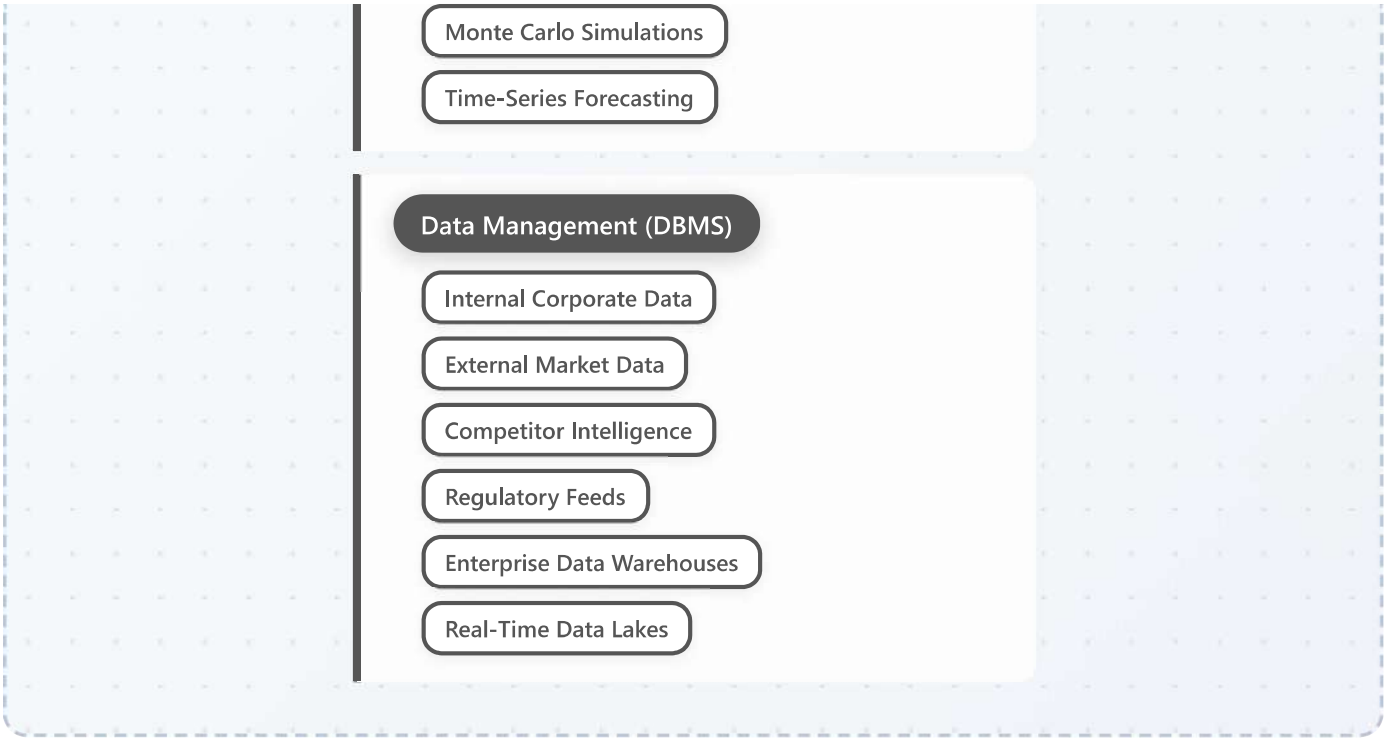
User-Centric Interfaces

Core Structural Components of a DSS

To understand a DSS, it helps to look at its underlying technical components. Every enterprise DSS relies on three core subsystems working together seamlessly:







1. The Data Management Subsystem (DBMS)

This serves as the system's foundational repository. It houses not only internal corporate data drawn from transactional systems but also external market data, competitor intelligence, and regulatory feeds. In modern environments, this layer is represented by optimized enterprise data warehouses or real-time data lakes.

2. The Model Management Subsystem (MBMS)

This acts as the analytical brain of the DSS. It contains a library of mathematical, financial, statistical, and optimization models (e.g., linear programming, Monte Carlo simulations, time-series forecasting). The MBMS allows users to apply these formulas to the underlying data without needing to hardcode the mathematics themselves.

3. The User Interface Subsystem (Dialog Generation and Control)

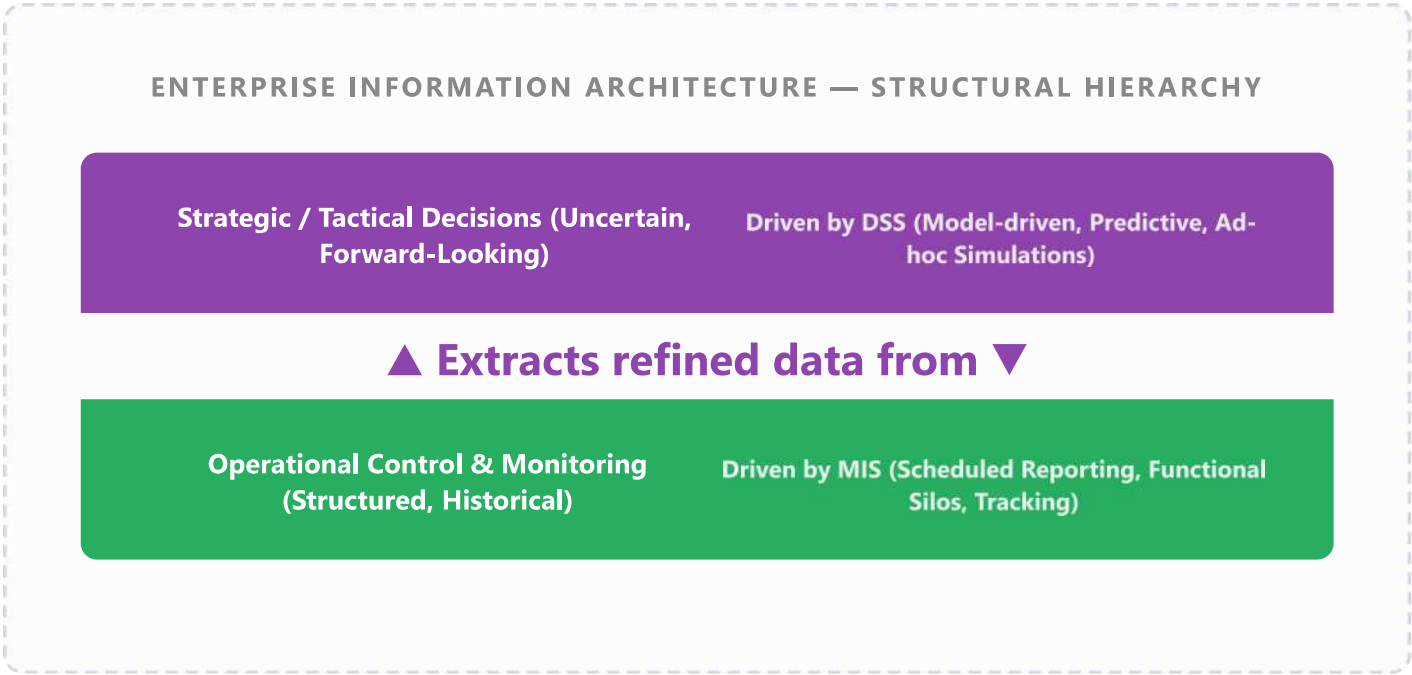
This is the portal through which the human decision-maker interacts with the data and models. Today, this transcends simple static command boxes, taking the form of low-latency interactive dashboards, real-time visual scenario simulators, and conversational natural language interfaces.

The Relationship Matrix: How DSS and MIS Coexist

A common point of confusion in systems architecture is the distinction between a standard MIS and a dedicated DSS. Under modern enterprise frameworks, they do not compete; rather, they operate in a symbiotic, tiered relationship.

The Structural Hierarchy

Think of the enterprise information architecture as a continuum of data refinement. The **MIS** serves as the structural foundation, establishing data governance, capturing transactions, and organizing internal operations. The **DSS** acts as an analytical layer built *on top* of the MIS, specifically designed to process that foundational data into tactical and strategic actions.

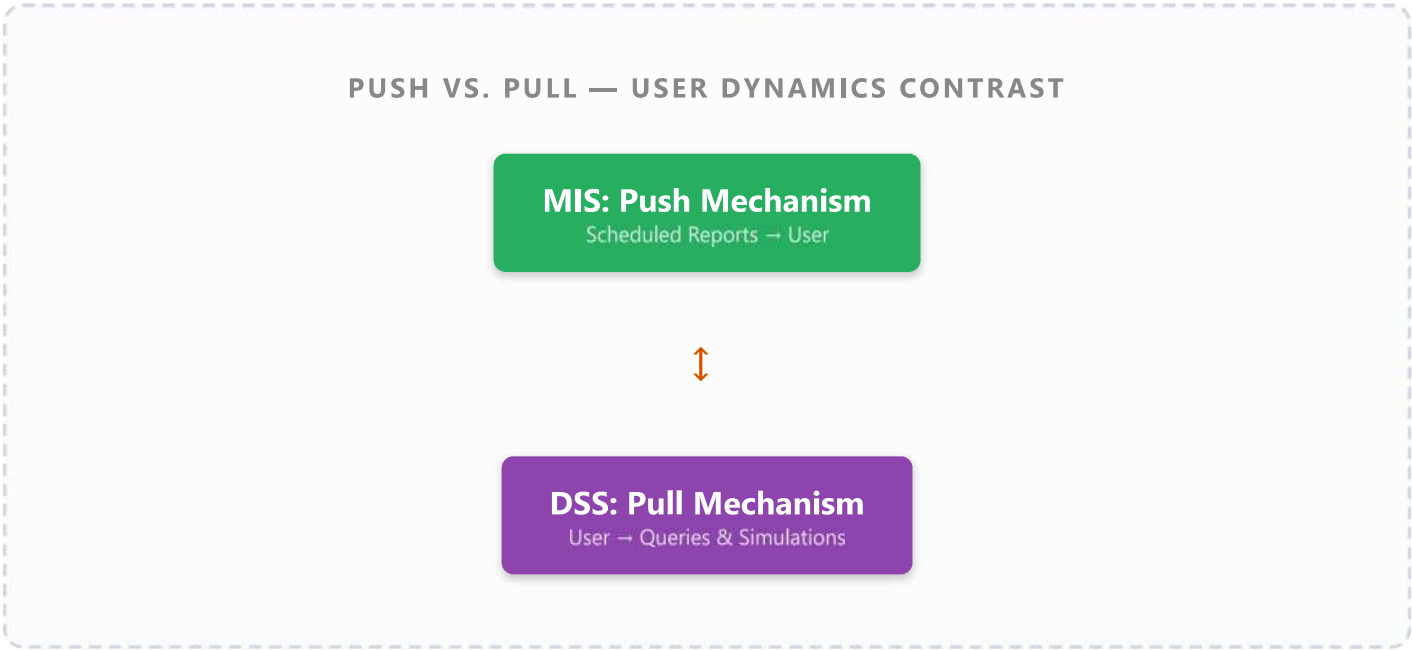


Continuum of Data Refinement

The enterprise information architecture is a **continuum of data refinement**. The MIS establishes the foundation — data governance, transaction capture, and internal operations organization. The DSS is the analytical layer built on top of that foundation, processing the foundational data into tactical and strategic actions.

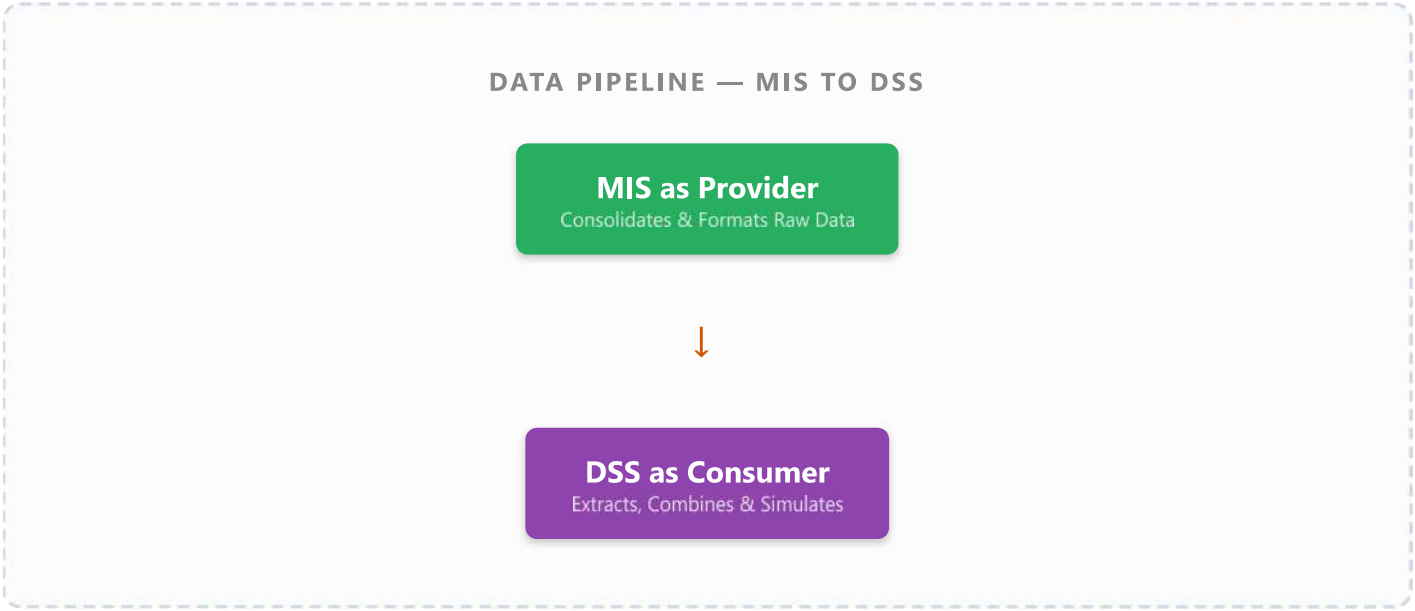
Key Architectural Differences

Architectural Metric	Management Information System (MIS)	Decision Support System (DSS)
Primary Focus	Information flow, process monitoring, and operational efficiency.	Decision effectiveness, problem-solving, and strategic agility.
Decision Context	**Structured Decisions:** Routine, repetitive problems governed by standard operating procedures.	**Semi-structured to Unstructured:** Unique, fluid issues with shifting dependencies.
Data Orientation	Historically focused, internal, and highly standardized.	Forward-looking (predictive), blending internal history with external variables.
Output Type	Periodic summaries, exception reports, and fixed dashboards.	Iterative query responses, "What-If" simulations, and optimization paths.
User Dynamics	Structured information delivery (push mechanism).	Highly interactive, user-driven exploration (pull mechanism).



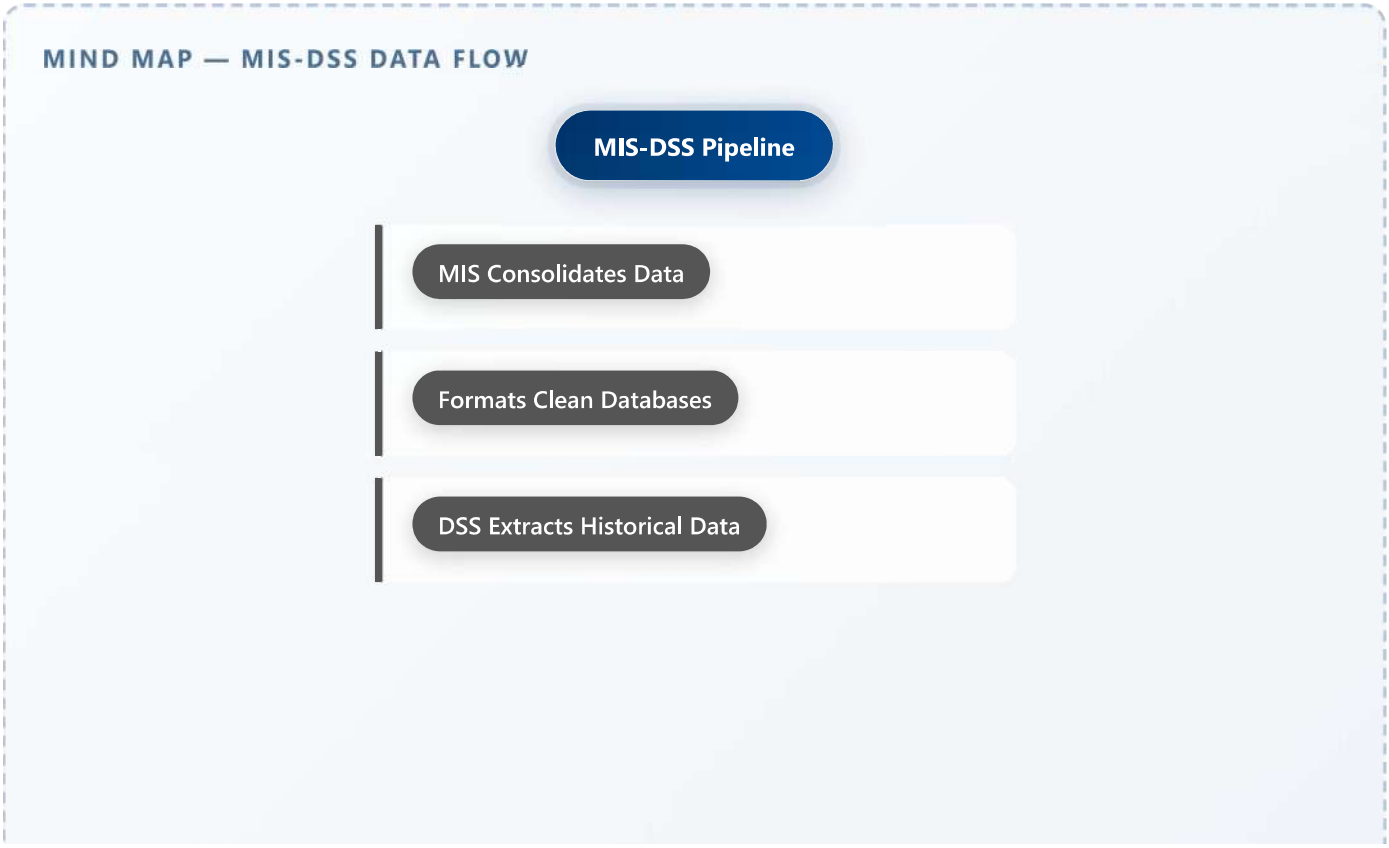
The Functional Intersect

The relationship is deeply operationalized through data pipelines:



The MIS as the Provider: The MIS consolidates raw transactional data from various operational points (e.g., Sales, HR, Logistics) and formats it cleanly into databases.

The DSS as the Consumer: The DSS taps into these clean databases, extracts the relevant historical data, combines it with external variables (like market inflation or competitor pricing), and runs simulations to guide complex future decisions.



Modern Real-World Application: Global Supply Chain Optimization

To better understand how these systems function in a modern business environment, consider the example of global supply chain optimization (e.g., **Unilever** or **DHL**).

🏭 **The MIS Foundation:**

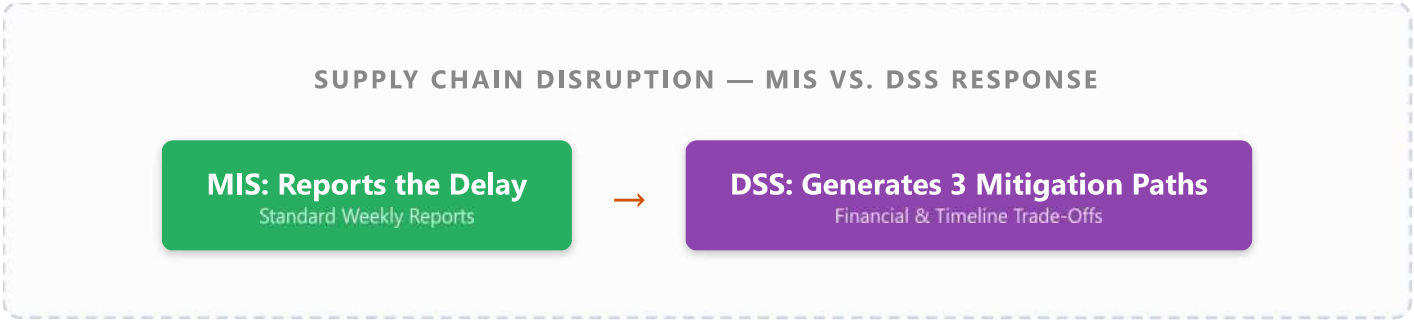
The enterprise resource planning (ERP) system logs daily tracking codes, factory outputs, current fleet locations, and raw material costs. It outputs standard weekly inventory status reports.

⚡ **The DSS Intervention:**

When a sudden geopolitical trade dispute or severe weather anomaly shuts down a major port, the operational MIS can only report the delay. The supply chain director then activates the
DSS
.

🎯 **The Decision Output:**

The manager inputs the port disruption parameters into the system's Model Base. The DSS runs a multi-commodity flow network model, analyzing alternative air-freight costs, tariff updates, and production capacities across alternate factories. It instantly generates three optimized risk-mitigation paths, showing the financial and timeline trade-offs for each choice.



Dynamic Retail Pricing

A second industry example is dynamic retail pricing systems (e.g., **Uber** or **Reliance Retail**).

🛒

The MIS Foundation:

Point-of-Sale (POS) systems continuously record transactional volumes, historical sales trends, and real-time inventory counts across hundreds of digital and physical storefronts.

💰

The DSS Intervention:

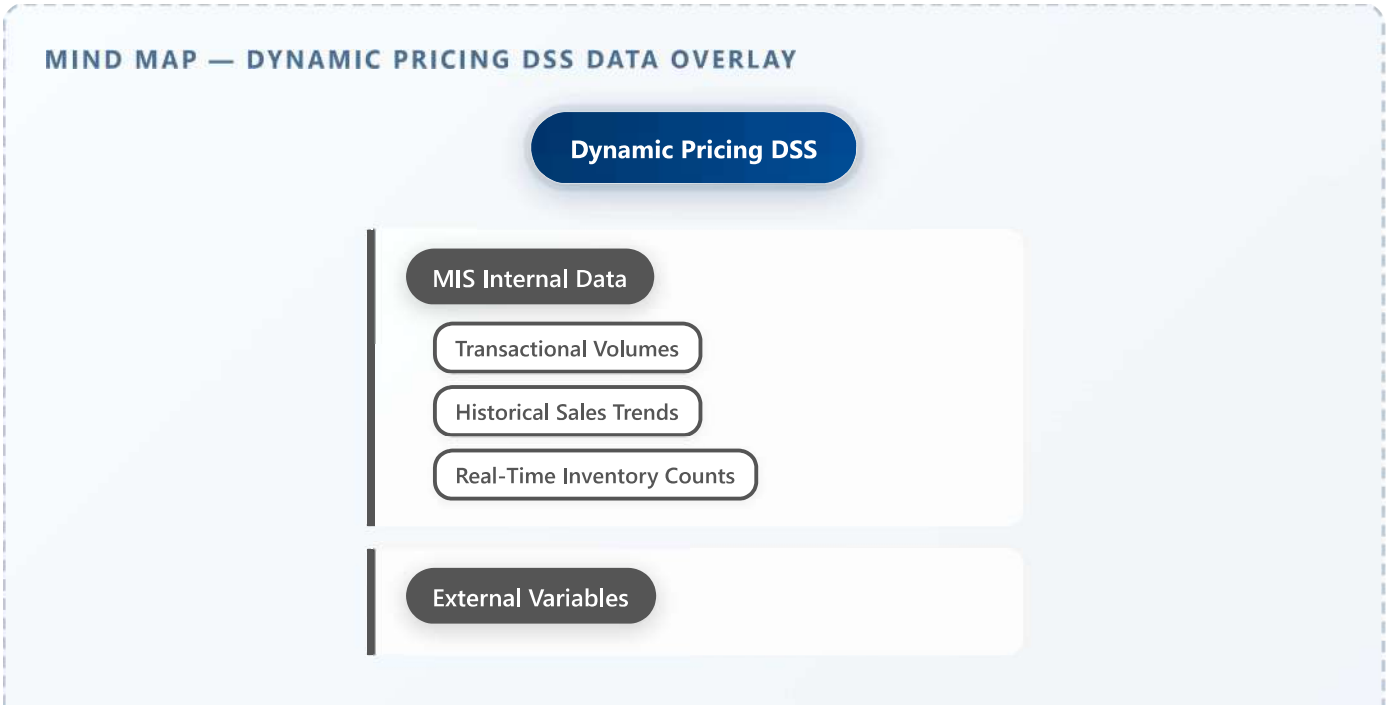
To maximize margins during a peak shopping season or sudden demand surge, a dynamic pricing **DSS**

actively polls the MIS database.

📊

The Decision Output:

The system overlays external variables—such as localized weather shifts, competitor digital pricing feeds scraped in real time, and local event schedules—onto the historical baseline. Using algorithmic pricing models, it suggests real-time price optimizations to the category manager, allowing them to balance consumer demand against supply constraints.



Enterprise Resource Planning (ERP)

Unit 8 — Basics, Benefits, Challenges & Evolution of Enterprise Systems

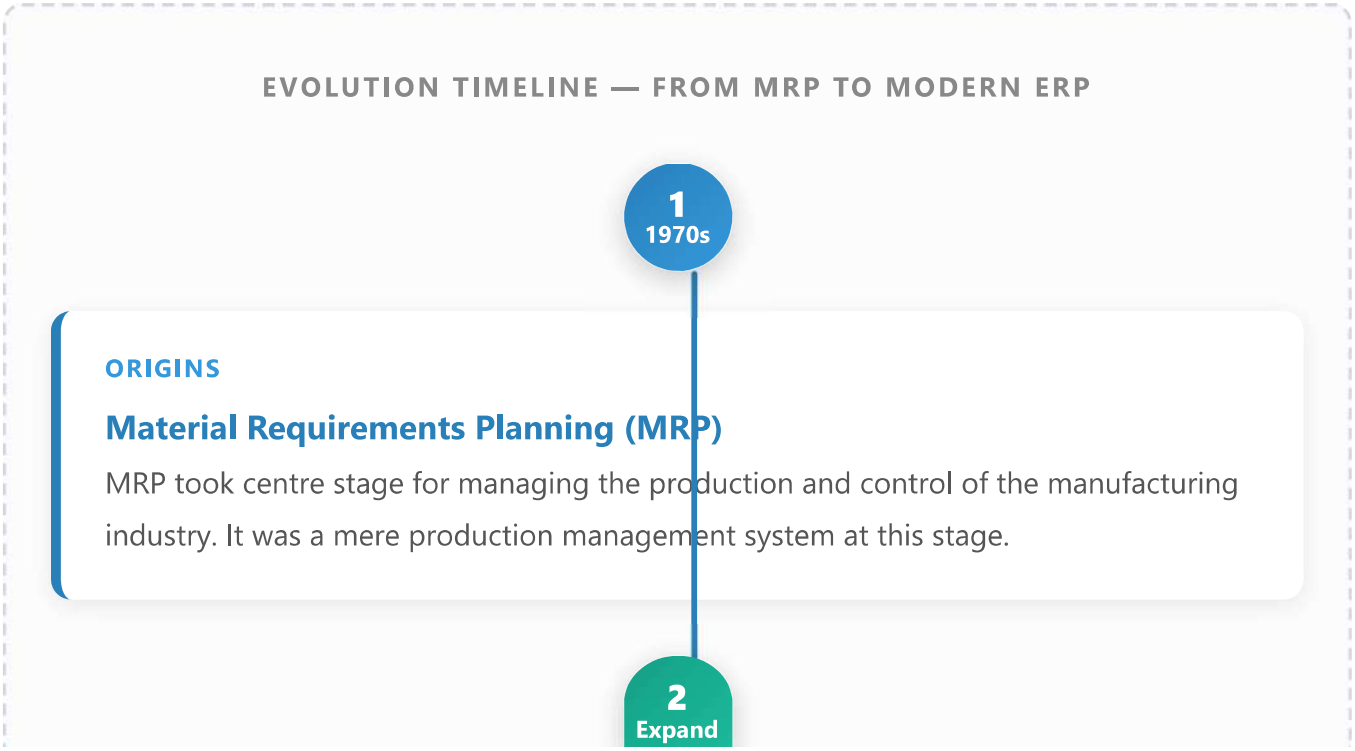
Introduction & Evolution of ERP

UNIT 8 — SELF-INSTRUCTIONAL MATERIAL

It was in the 1970s that Material Requirements Planning (MRP) took the centre stage for managing the production and control of the manufacturing industry. MRP evolved from a mere production management system to a system that can be used for personnel and distribution planning. It came to be known as Manufacturing Resource Planning-II (MRP-II).

After being introduced, MRP-II functions like human resources, distribution management and accounting were clubbed in one category. This system covered all areas of the enterprise business and thus, it became popular as Enterprise Resource Planning (ERP). The first forms of ERP came into existence in the Gartner Group in 1990. The best period for ERP growth was in the 1990s, as companies started moving from traditional to ERP systems.

Nowadays, ERP has evolved into a new generation practice. It allows Web-based access to the data to the employees and people outside the enterprise, irrespective of the time and location. In this unit, we will discuss the basics of ERP, along with its benefits, challenges, and evolution. We will also focus on enterprise systems in large organizations.



EVOLUTION

Manufacturing Resource Planning-II (MRP-II)

MRP evolved to include personnel and distribution planning. Functions like human resources, distribution management and accounting were clubbed in one category.

3
1990

GARTNER GROUP

Enterprise Resource Planning (ERP) Coined

The system now covered all areas of the enterprise business and became popular as ERP. The first forms of ERP came into existence in the Gartner Group in 1990.

4
1990s

GROWTH ERA

Mass Adoption of ERP Systems

The best period for ERP growth was in the 1990s, as companies started moving from traditional to ERP systems in large numbers.

5
Modern

NEW GENERATION

Web-Based ERP

ERP has evolved into a new generation practice. It allows Web-based access to the data to the employees and people outside the enterprise, irrespective of the time and location.

Unit Objectives

After going through this unit, you will be able to:

- Explain the basics of ERP
- Discuss benefits, challenges, and evolution of ERP
- Describe the enterprise systems in large organizations

Basics of ERP: Benefits and Challenges of Enterprise Systems

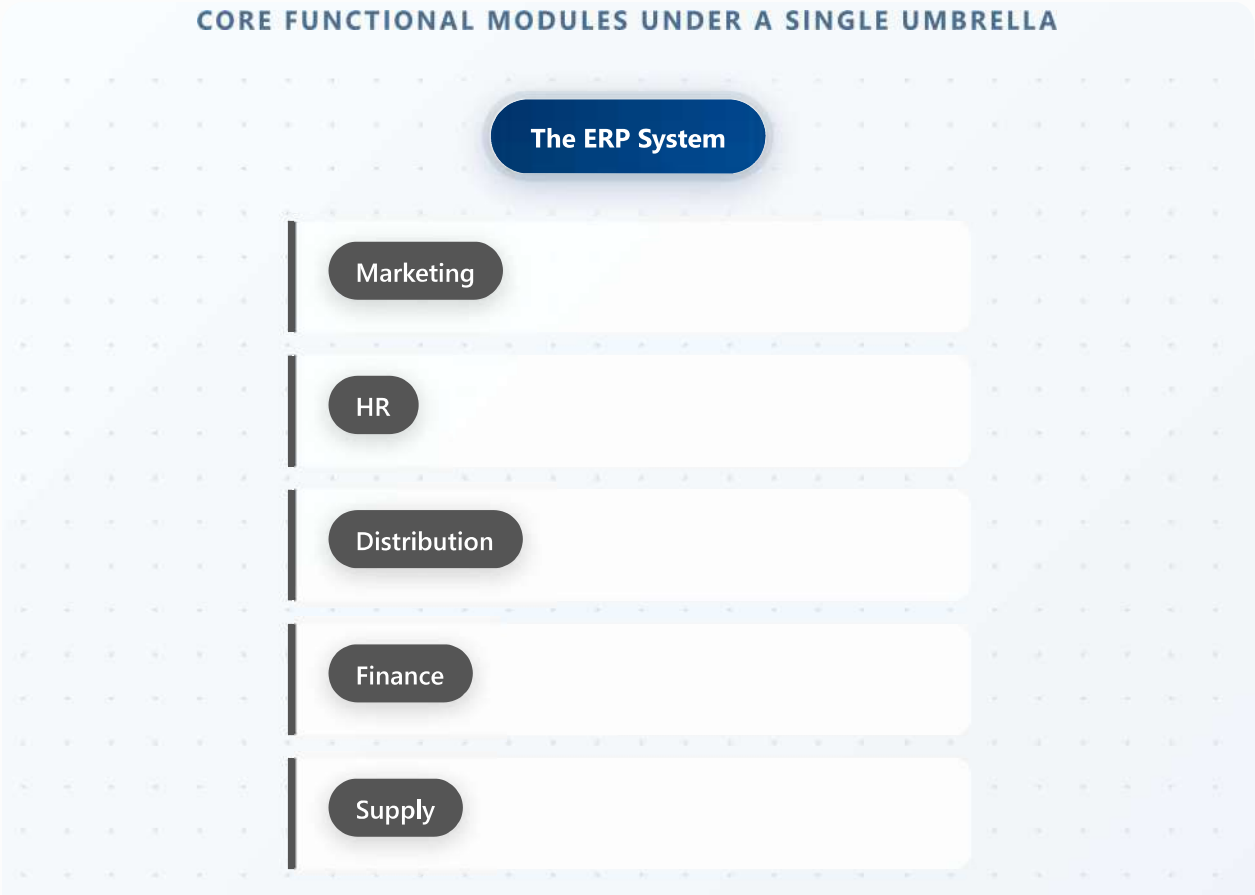
Enterprise Resource Planning (ERP) is a self-sufficient, closely knit system used for managing different resources within and outside the enterprise (vendors, etc.) in order to improve the overall efficiency of the enterprise. ERP is a computer based system that can be used for finance, HR or material resource management. It has the capability to bring various functions of an enterprise under a single umbrella.

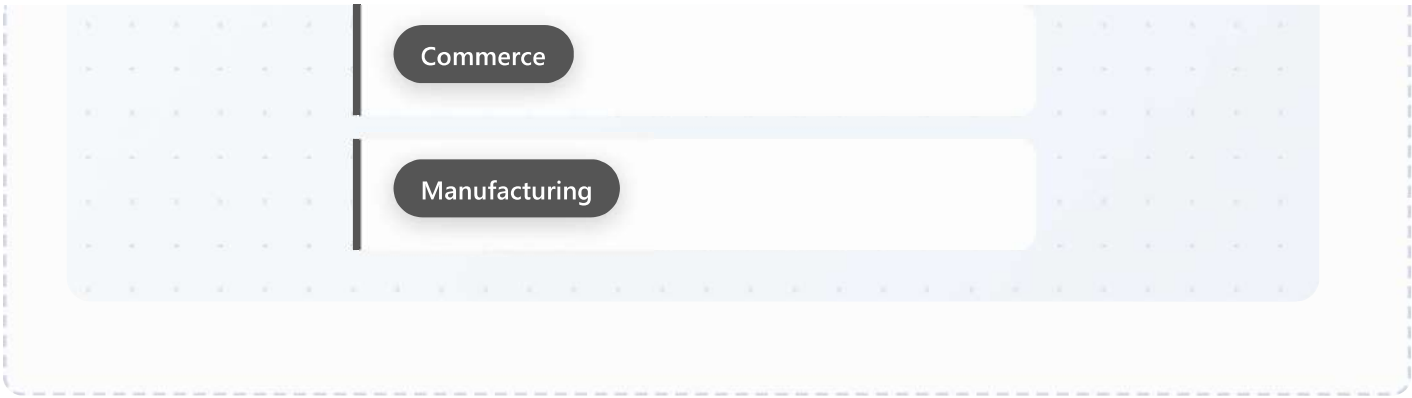
It is possible to have the entire ERP system running on a single server or in clusters of multiple servers, connected in a local area network. Using the second approach often simplifies things and provides more stability and robustness to the system.

Original Scope vs. Modern Scope

ERP was developed keeping the manufacturing industry in mind and was limited to a few functions, such as sales, production and finance. But these days more functions are covered by the latest ERP systems since these systems are being adopted by a large number of industries.

FIG. 8.1 — THE ERP SYSTEM (FUNCTIONAL MODULES)





We can think of ERP software as that which automates major business processes, like manufacturing and distribution. Not all ERP system implementations are successful. At times the legacy software would prove to be better as it can be written to suit the specific needs of an organization and it can also be tailored better to suit its needs. So, care should be exercised while going for the implementation of an ERP system within the enterprise.

Implementation Risk

Not all ERP system implementations are successful. At times the legacy software would prove to be better as it can be written to suit the specific needs of an organization and it can also be tailored better to suit its needs. So, care should be exercised while going for the implementation of an ERP system within the enterprise.

Deployment Architecture

It is possible to have the entire ERP system running on a single server or in clusters of multiple servers, connected in a local area network. Using the second approach often simplifies things and provides more stability and robustness to the system.

Why Companies Switch from Legacy Systems to ERP

The first question that arises at this point of time is 'Why are the companies willing to change their legacy business systems for ERP?' ERP has been in demand in recent times and the reasons for the continuous growth of the ERP market are detailed in the next section.

⚠

The Legacy Software Question

Not all ERP system implementations are successful. At times the legacy software would prove to be better as it can be written to suit the specific needs of an organization and it can also be tailored better to suit its needs. Care should be exercised while going for the implementation of an ERP system within the enterprise.

Reasons for Continuous Growth of the ERP Market

ERP has been in demand in recent times and the reasons for the continuous growth of the ERP market are:

1 Speed

PERFORMANCE

Better Performance by Businesses

The time for a complete cycle reduces, the business becomes more agile and orders can be taken care of faster.

2 Growth

SCALABILITY

Facilitation in the Growth of Business

The same ERP system can support multiple languages, currencies, new products and new customers.

3 Decide

RESPONSE TIME

Faster Decision-Making

It helps improve the overall response time in an organization and hence facilitates faster decision-making.

4
Efficient

SUPERIORITY OVER LEGACY

More Efficient Than Legacy Systems

It is more efficient than the legacy systems as it solves various issues such as scalability, data loss prevention and faster data processing.

The Expanding ERP Market

ERP vendors have commendable bank accounts owing to the popularity of the ERP systems. Gradually, the focus is moving towards ERP implementations in smaller and medium enterprises as well. There are a number of players in the arena competing to get a larger share of the profits. As the competition increases, the customers get better prices and services.

Reason	How It Drives ERP Adoption
Better Performance	The time for a complete cycle reduces, the business becomes more agile and orders can be taken care of faster.
Business Growth	The same ERP system can support multiple languages, currencies, new products and new customers.
Faster Decision-Making	It helps improve the overall response time in an organization and hence facilitates faster decision-making.
Efficiency Over Legacy	It solves various issues such as scalability, data loss prevention and faster data processing.

Advantages of ERP Systems

ERP systems have manifold advantages. They enhance the efficiency of an organization, help in the decision-making processes and speed up the response time. In addition to these, there is an increase in the number of satisfied and repeated customers. The most visible benefits of using ERP systems are discussed below:

1 Integrate

ADVANTAGE 1

Integration of Business Processes

Earlier, various business units in an organization had their own information systems. Exchange of information amongst them was not only tough but a challenging task as well. In large organizations, exchange of data amongst the various departments is more frequent. The ERP system plays a key role here, as after implementation of this system, data updated for a single transaction at one location, simultaneously gets updated at multiple locations. This helps in the integration of various business processes across the organization.

2 Flexible

ADVANTAGE 2

Flexibility

A single ERP system is able to cope with different languages and currency values across the globe, for a company. Hence ERP systems help in creating a truly global environment for organizations. This flexibility of the system helps in management of the organization in a better manner and it also aids the maintenance and development of systems across the organization.

3 Insight

ADVANTAGE 3

Better Insight

ERP systems help organizations to plan better. They can provide a runtime data analysis on the basis of various parameters defined in the system. They help the management by providing a better insight and the ability to take decisions in a shorter span of time.

4
Tech

ADVANTAGE 4

Technological Advancements

ERP systems grow better day by day with advancements in the software used to run them. The technological advancements are always advantageous to ERP vendors and help make the systems more adaptable and flexible for further development and a better user experience.

5
Speed

ADVANTAGE 5

Lower Turnaround Times

A successful ERP implementation helps in reduction of time in receiving and the completion of the order (this time is also known as lead time). Using ERP systems the demand can be anticipated well in advance and the production managers can be ready with the materials needed to meet the demand. The shortage of materials can also be estimated well in advance and they can be ordered before they are actually exhausted, in order to meet the targets on time. In today's world, the number of raw materials needed by an organization is huge and maintaining the inventory for such a large number of materials manually is practically impossible. This is where ERP implementation comes to the rescue of the management.

6
Ship

ADVANTAGE 6

On-Time Shipment

In today's changing times, customers have become more demanding and manufacturers have to be a step ahead to remain in competition, by enabling themselves to be able to fulfill the needs of the customers in a better and more efficient time frame. Using technology, ERP systems help to simulate different production scenarios and we can select the best one out of all available production methods. This helps to be able to ship the finished product at the earliest.

7
Utilize

ADVANTAGE 7

Better Utilization of Resources

ERP systems can provide good estimates of the amount and quantities of materials required for manufacturing a product. This helps in reducing the wastage of raw materials and hence helps in better utilization of resources. Various modules of an ERP system (manufacturing, material management, maintenance, sales and distribution, etc.) work together to ensure that the wastage of raw materials is minimal and the machine down time is low.

8
Satisfy

ADVANTAGE 8

Better Customer Satisfaction

The increased efficiency of the organization in production and shipping times, using the ERP system, helps the organization in increasing customer satisfaction. The customers are better satisfied when the products delivered to them contain the features they had requested for. The company's response should be timely and satisfactory and the final product delivered should be usable and free from defects. If there is a Web based ERP system, customers can actually place, track and check status of their order in the comfort of their homes. Since the ERP systems consolidate all resources at a single place, resolving customer-care queries becomes easier for the support departments.

More Accurate Information and Better Decision-Making

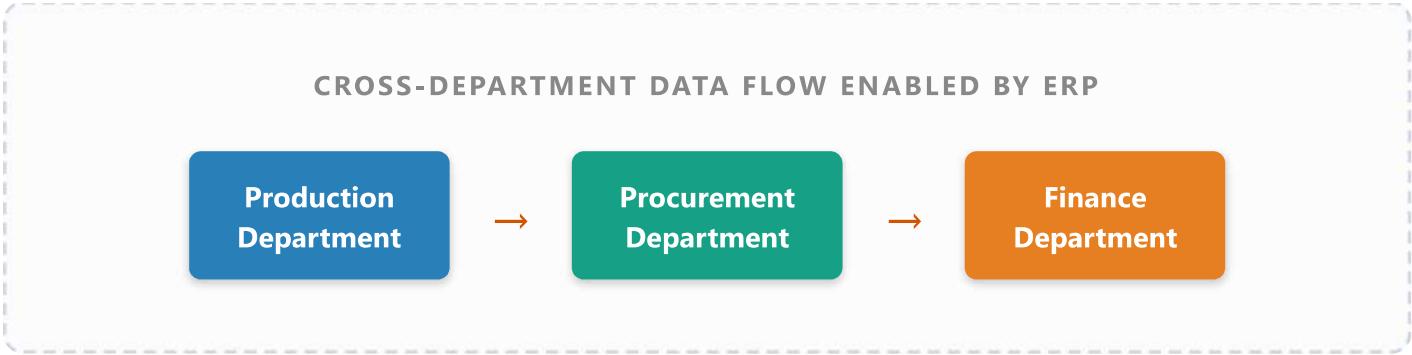
Information management has become very important in an organization's lifecycle to be able to compete in the current scenario. ERP systems enable an organization to manage huge amount of data, extract relevant information from it and collate the necessary information to be presented to the user. The availability of information in a timely and prompt manner helps in making better decisions.

ERP systems provide great integration of various systems and help in better collation of information from different sources to enable the employee or the management to take better decisions. Before ERP came into existence, all departments use to function independently in an organization. Now, with ERP in place, the production department can enter their requirement, which when seen by the procurement department, enables them to procure materials earlier. The entry of the procurement department enables the finance department to manage finances accordingly, for that month.

Cross-Department Example: Before vs. After ERP

Before ERP came into existence, all departments use to function independently in an organization. Now, with ERP in place:

- The **production department** can enter their requirement
- Which when seen by the **procurement department**, enables them to procure materials earlier
- The entry of the procurement department enables the **finance department** to manage finances accordingly, for that month



ERP vs. Conventional Systems — Three Key Differentiators

Using ERP systems over conventional systems has its own advantages:

1. Multi-Department Span

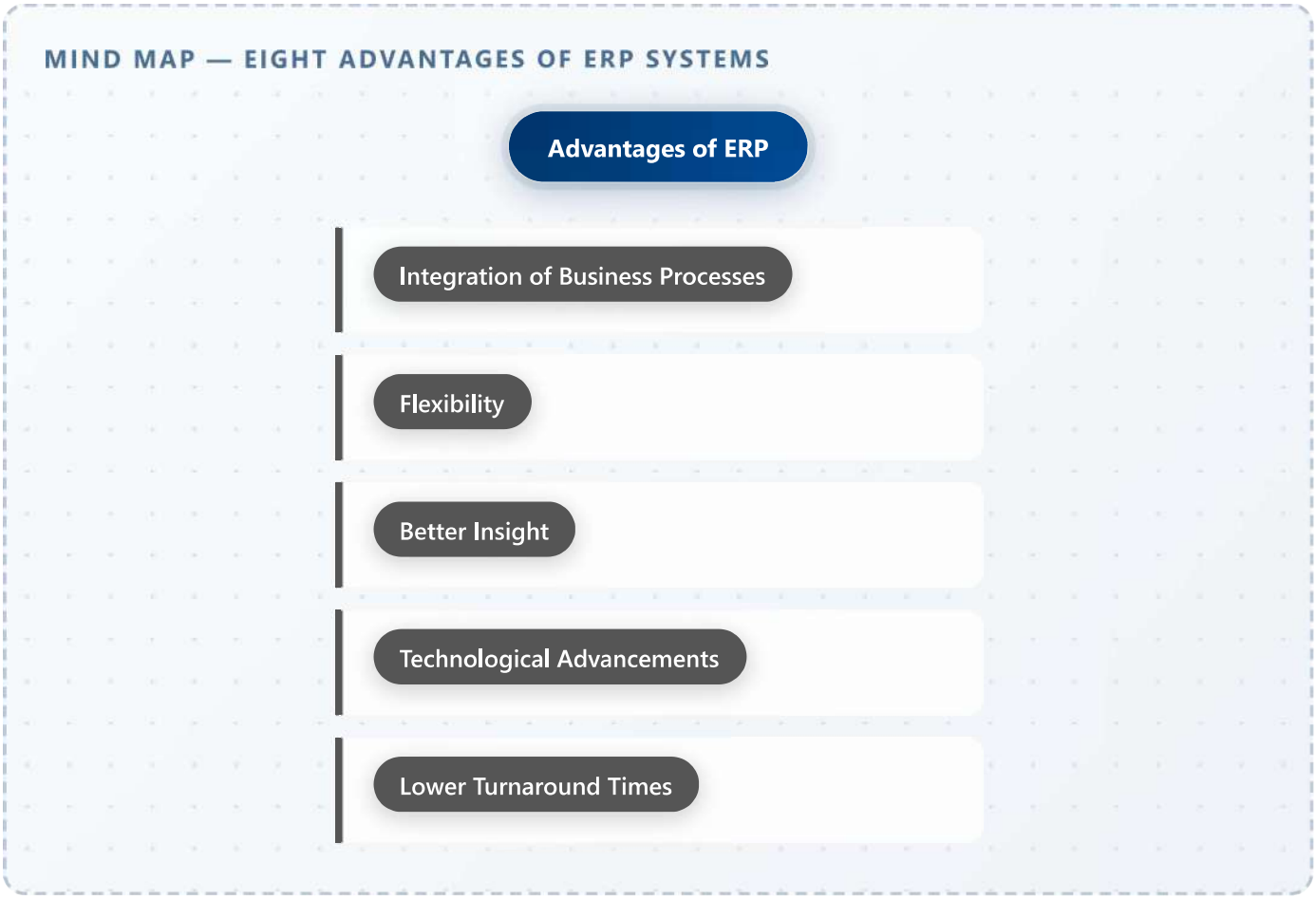
ERP systems are not restricted to a single business function, rather, they span to multiple departments across the organization.

2. Enterprise Size Flexibility

They are flexible enough to be used by small or large enterprises.

3. Global Viability

The support for multiple languages and currencies makes ERP more viable for the organizations.



Introduction to ERP Architecture

Enterprise Resource Planning (ERP) systems have become the operational backbone of modern businesses. From managing finances and inventory to streamlining sales, procurement, manufacturing, and human resources, ERP software helps organizations connect critical business functions through a single platform.

But what makes an ERP system capable of handling thousands of transactions, users, and business processes every day?

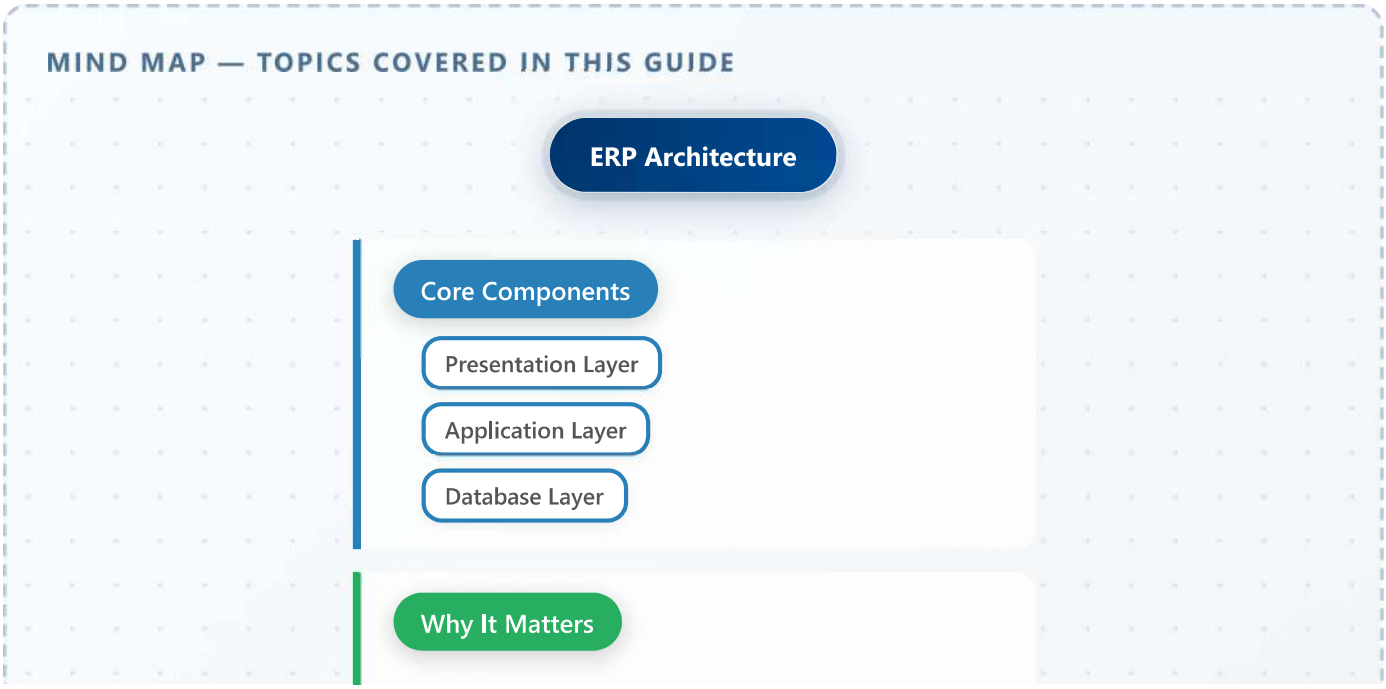
The answer lies in **ERP architecture**.

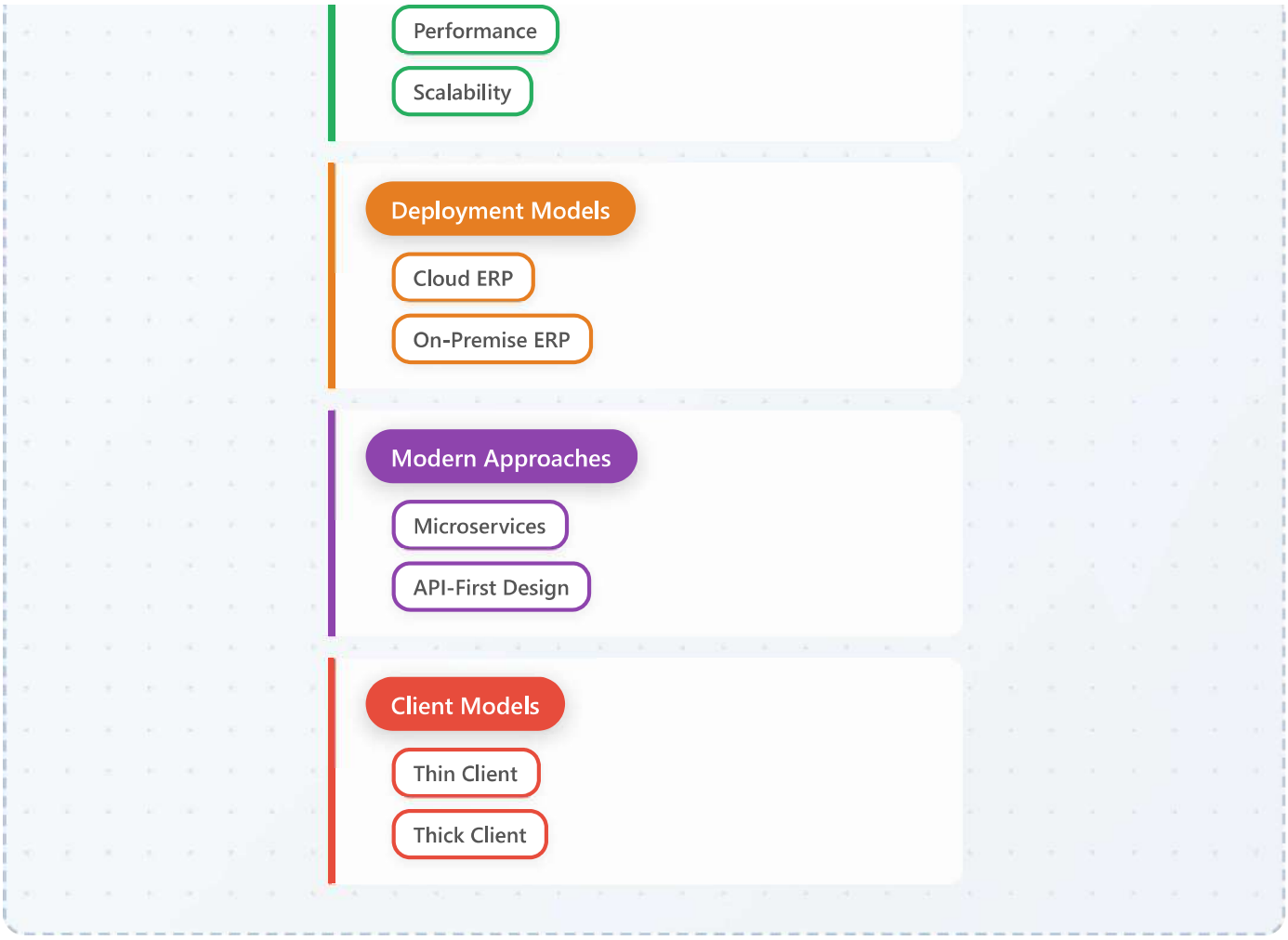
ERP architecture defines how an ERP system is designed, structured, and deployed. It determines how data flows through the system, how different modules communicate, how users access information, and how the platform scales as a business grows.

What a Well-Designed ERP Architecture Delivers

- **Centralize business data** into a single source of truth
- **Improve operational efficiency** by eliminating redundant processes
- **Enable real-time reporting** for faster decision-making
- **Strengthen data security** with layered access controls
- **Support business growth** without performance degradation
- **Simplify system maintenance** with modular, upgradeable components

Whether you're planning a new ERP implementation, evaluating ERP software vendors, or simply trying to understand how enterprise systems work, understanding ERP architecture is essential.







The Building Blueprint Analogy

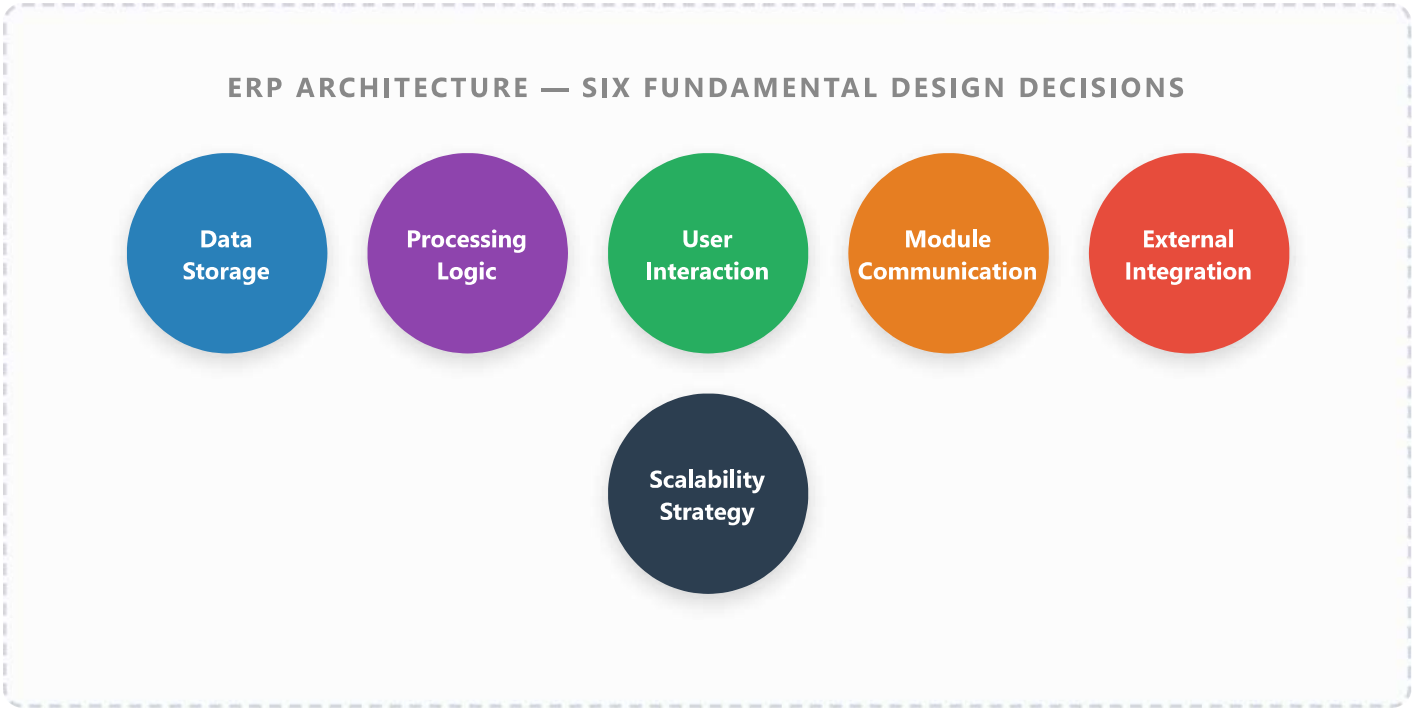
Think of ERP architecture as the blueprint of a building. Just as a building requires a strong foundation, electrical systems, plumbing, and structural support, an ERP system requires databases, application logic, user interfaces, integrations, and security mechanisms working together in a coordinated manner. A beautiful facade with a weak foundation will eventually collapse — the same is true for ERP systems with poor architectural foundations.

What Is ERP Architecture?

ERP architecture refers to the structural framework that defines how an ERP system is organized and how its components work together.

What ERP Architecture Determines

- **How data is stored** — relational databases, NoSQL, data lakes, or distributed storage
- **How information is processed** — batch processing, real-time event streams, or hybrid approaches
- **How users interact with the system** — web browsers, mobile apps, desktop clients, or portals
- **How modules communicate** — direct calls, message queues, APIs, or service meshes
- **How external applications connect** — point-to-point integrations, middleware, or API gateways
- **How the system scales over time** — vertical scaling, horizontal scaling, or auto-scaling



The Cost of Poor Architecture

Without a proper architectural foundation, ERP systems become difficult to maintain, slow to operate, and expensive to expand. Organizations that ignore architecture during vendor selection often face painful migrations within 3–5 years as outgrown systems cannot adapt to new business requirements.



🛠

Real-World Example

When a sales rep enters a new order in the ERP, the presentation layer captures the input, the application layer validates inventory availability and calculates pricing, and the database layer records the transaction, updates inventory counts, and creates an accounts receivable entry — all within seconds, across all three layers working in concert.

Why ERP Architecture Matters

Many organizations focus on ERP features when selecting software. While functionality is important, architecture plays an equally critical role. A powerful ERP with poor architecture can create long-term challenges that undermine the entire investment.

Risks of Poor ERP Architecture

Performance bottlenecks — slow response times during peak usage as the system cannot distribute workload efficiently

Difficult upgrades — tightly coupled components mean upgrading one module risks breaking others

Security vulnerabilities — monolithic systems with single points of failure are harder to secure at every layer

Integration limitations — inability to connect with modern tools, IoT devices, or third-party platforms

Higher maintenance costs — proprietary, closed architectures require specialized (and expensive) talent to maintain

Business Benefits of Modern ERP Architecture



Improved Performance

Efficient system design allows ERP platforms to process large volumes of transactions quickly and accurately, even during peak periods.



Better Scalability

As businesses grow, ERP systems must support additional users, locations, products, and transactions without sacrificing performance.



Enhanced Security

Modern architectures include advanced security controls, encryption, authentication, and role-based access management at every layer.



Easier Maintenance

Well-structured, modular systems simplify upgrades, bug fixes, and feature enhancements without disrupting operations.



Greater Flexibility

Organizations can adapt to changing business requirements without completely redesigning their ERP environment.

ARCHITECTURE IMPACT — FROM RISK TO REWARD

Poor Architecture
Bottlenecks & Rigidity

→

Modern Architecture
Agility & Growth

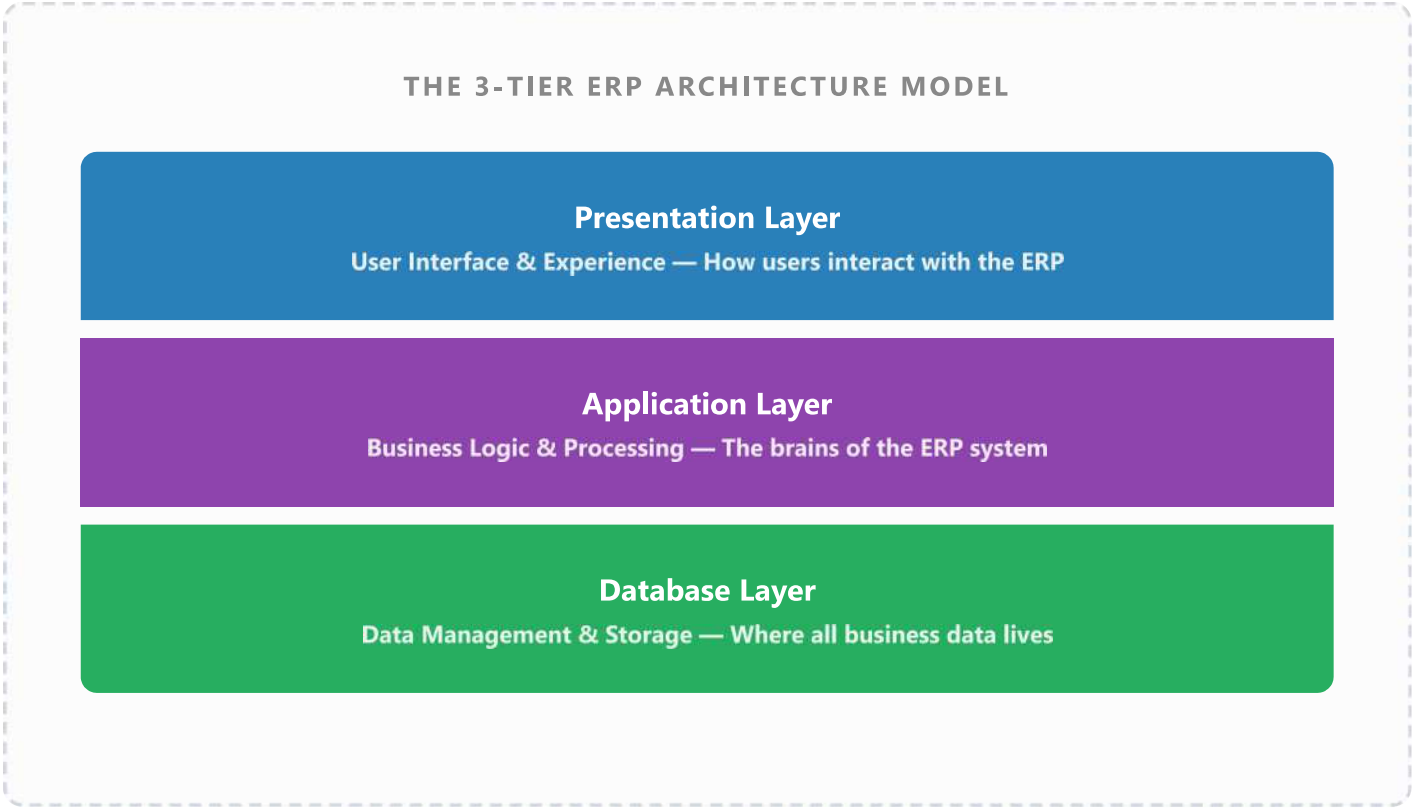


The Hidden Cost Trap

Two ERPs may look identical on a demo screen, but the one built on a 3-tier microservices architecture will cost significantly less to own over 5 years. Why? Faster upgrades (hours vs. weeks), simpler integrations (APIs vs. custom code), and the ability to scale individual modules independently. Architecture is the difference between a system that works on day one and a system that still works on day 1,825.

Core Components of ERP Architecture

Although ERP platforms vary by vendor and industry, most modern systems are built around three core architectural components. Together, these layers create a structured framework that enables efficient data processing and seamless business operations.



1 Present

LAYER 1 OF 3

Presentation Layer — The User Interface

The part users interact with directly. Employees access this layer to view dashboards, create transactions, approve requests, generate reports, and monitor business performance. Its primary purpose is converting complex business data into understandable, actionable information through web browsers, mobile applications, desktop applications, and self-service portals.

2 Process

LAYER 2 OF 3

Application Layer — The Business Logic Engine

This is where all the processing happens. The application layer contains business rules, workflow automation, validation logic, calculations, and module-to-module communication. When a user submits an order, this layer checks inventory, applies pricing rules, calculates taxes, triggers notifications, and coordinates updates across finance, sales, and inventory modules.

3
Store

LAYER 3 OF 3

Database Layer — The Data Vault

The foundation that stores, retrieves, and secures all business data. This layer typically includes relational databases (for structured transactional data), data warehouses (for analytical reporting), and caching systems (for performance optimization). It handles data integrity, backup, recovery, and ensures that every module reads from and writes to a single source of truth.

2-Tier vs. 3-Tier Architecture

A 2-tier ERP architecture combines presentation and application logic on the client side, while a 3-tier architecture separates all three layers, providing better scalability, security, and performance. The 3-tier model is the standard for all modern ERP platforms.

Aspect	2-Tier Architecture	3-Tier Architecture
Structure	Presentation + Application on client; Database on server	Presentation, Application, and Database each on separate tiers
Scalability	Limited — adding users strains client devices	High — each tier scales independently
Security	Lower — business logic resides on client machines	Higher — logic stays on secured application servers
Maintenance	Difficult — updates require redeploying to every client	Easier — application updates are centralized

Aspect	2-Tier Architecture	3-Tier Architecture
Performance	Degrades with heavy business logic on clients	Optimized — processing distributed across dedicated servers
Modern Usage	Legacy systems only	Standard for all modern ERP platforms



The Restaurant Kitchen Analogy

Think of the 3-tier model like a restaurant: the **Presentation Layer** is the dining room and menu (where customers interact), the **Application Layer** is the kitchen where chefs follow recipes, coordinate dishes, and ensure quality (business logic), and the **Database Layer** is the pantry and storage room where all ingredients are organized, tracked, and secured (data storage). Each area has a distinct role but works together to serve every order.

The Presentation Layer — User Interface Deep Dive

The presentation layer is the part of the ERP system users interact with directly. Employees access this layer to view dashboards, create transactions, approve requests, generate reports, and monitor business performance.

The primary purpose of the presentation layer is to provide a user-friendly interface that converts complex business data into understandable information.

Presentation Layer Interfaces

Interface Type	Description	Primary Users
Web Browser	Accessed via URL in Chrome, Firefox, Safari — the most common modern approach	All employees, remote workers
Mobile App	Native iOS/Android apps for on-the-go tasks like approvals and time tracking	Executives, field sales, warehouse staff
Desktop App	Installed software — mostly in legacy on-premise ERP systems	Back-office accounting, power users
Customer Portal	External-facing interface for customers to place orders and track shipments	Customers, partners
Self-Service Portal	Employee-facing portal for HR tasks, leave requests, expense submissions	All employees

Why User Experience Matters in ERP

Modern ERP platforms prioritize intuitive user experiences because **user adoption is a critical factor in ERP success**. Studies consistently show that the #1 reason ERP implementations fail is not technical problems — it's that employees refuse to use the system because the interface is confusing, slow, or frustrating. A well-designed presentation layer directly impacts ROI by driving higher adoption rates.

Thin Client vs. Thick Client Architecture

ERP user interfaces are generally delivered through one of two architectural approaches. Understanding the difference is crucial when evaluating ERP platforms.

Thick Client Architecture

- Software installed directly on each user's computer
- Significant processing occurs on the local device
- Regular software updates required per machine
- IT management becomes complex at scale
- Common in older, on-premise ERP systems
- Better offline capability (limited use case)

Thin Client Architecture

- Accessed via a web browser — no installation needed
- Most processing occurs on centralized servers
- Users can access from almost any device
- Updates deployed centrally — instant for all users
- Standard for all modern cloud ERP platforms
- Greater flexibility, accessibility, and scalability

Criterion	Thick Client	Thin Client
Installation	Required on every device	None — browser-based
Processing Location	Local device (client-side)	Centralized servers (server-side)
Updates	Manual per-device deployment	Centralized — instant rollout
Device Flexibility	Limited to installed devices	Any device with a browser
IT Overhead	High — manage installations, versions, patches	Low — browser management only
Modern Relevance	Declining — legacy systems only	Dominant — cloud ERP standard

Vendor Evaluation Tip

If an ERP vendor requires installing desktop software for core functions, ask why. In 2025, there are very few legitimate reasons for a thick-client ERP. The thin-client, browser-based model provides superior security, lower TCO, and dramatically simpler IT management. Thick-client requirements often signal an older architecture that will be harder to maintain and integrate.



The Thin Client Advantage in Practice

A company with 500 employees across 12 offices needs to deploy an ERP update. With a thick client, the IT team must push updates to 500 machines individually — a multi-day process with inevitable failures. With a thin client, the update is deployed once on the server, and all 500 users see the changes the next time they refresh their browser. The difference is hours vs. days, zero vs. dozens of support tickets.

Modern ERP Architecture Approaches

The days of rigid, 2-tier, monolithic, on-premise systems are fading. The future belongs to scalable, 3-tier, cloud-based microservices that communicate intelligently via APIs and robust middleware.

1 Cloud

DEPLOYMENT MODEL

Cloud-Based Architecture

Hosted by a service provider and accessed over the internet. Cloud ERP offers lower upfront costs, easier maintenance, automatic updates, and elastic scaling. The provider manages infrastructure, security patches, and high availability — letting organizations focus on business processes rather than server management.

2 Micro

DESIGN PATTERN

Microservices Architecture

Breaks ERP functionality into independent services that can be updated, scaled, and maintained separately. Instead of one monolithic application, the ERP becomes a collection of small, focused services — one for inventory, one for billing, one for HR — each with its own database and logic. This improves flexibility, resilience, and business agility dramatically.

3 API

COMMUNICATION METHOD

API-First Design

Modern ERPs are built API-first — every function is accessible through well-documented REST or GraphQL APIs. This enables seamless integration with CRM systems, e-commerce

platforms, IoT devices, analytics tools, and custom applications. APIs replace fragile point-to-point connections with a clean, standardized integration layer.

4
Secure

SECURITY FRAMEWORK

Zero-Trust Security Architecture

Modern ERP security operates on a zero-trust model — no user, device, or connection is trusted by default. Every request is authenticated, authorized, and encrypted. Role-based access control, multi-factor authentication, data encryption at rest and in transit, and continuous monitoring protect sensitive business data at every architectural layer.

5
Scale

GROWTH STRATEGY

Elastic & Horizontal Scaling

Modern architectures support elastic scaling — the system automatically adds or removes resources based on demand. During month-end closing, the application layer scales out to handle peak processing. During quiet periods, it scales back in to reduce costs. Individual microservices scale independently based on their specific load patterns.

6
Smart

INTELLIGENCE LAYER

AI-Embedded Architecture

The newest ERP architectures embed artificial intelligence and machine learning directly into the application layer. Predictive analytics, anomaly detection, intelligent automation, and natural language processing are woven into workflows — not bolted on as afterthoughts. This transforms ERPs from record-keeping systems into proactive decision-support platforms.

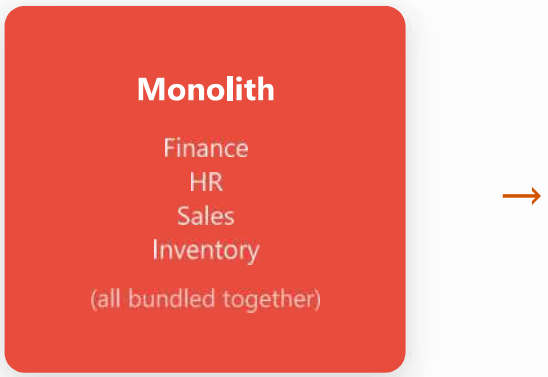
Cloud ERP vs. On-Premise ERP

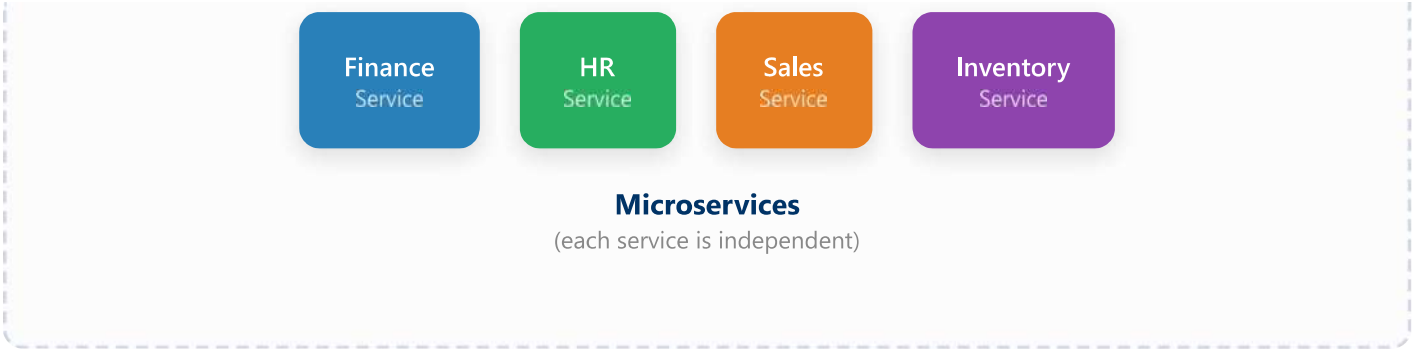
Aspect	Cloud ERP	On-Premise ERP
Hosting	Service provider's infrastructure	Company-owned servers and data centers
Upfront Cost	Lower — subscription-based pricing	Higher — hardware, licenses, implementation
Maintenance	Provider handles updates, patches, and infrastructure	Internal IT team manages everything
Data Control	Shared responsibility model	Full control over data and systems
Scalability	Elastic — scale up or down on demand	Fixed — requires hardware procurement to scale
Access	Any device, anywhere with internet	Typically limited to on-network or VPN
Updates	Automatic, continuous, and seamless	Manual, planned, and potentially disruptive

Understanding Microservices in ERP

In a traditional monolithic ERP, all functionality — finance, HR, sales, manufacturing — is bundled into a single, massive application. Changing one module can impact the entire system. Microservices architecture breaks this monolith into independent services. The finance service can be updated without touching the HR service. The inventory service can scale up during peak season while the payroll service stays at normal capacity. This architectural shift is the foundation of modern ERP agility.

MONOLITHIC VS. MICROSERVICES ARCHITECTURE





🌎

Building an Indestructible Digital Foundation

When you align a modern, well-architected ERP with smart business process management, you do more than just upgrade your software — you lay an indestructible digital foundation that will support your company's growth, adaptability, and success for decades to come. Choosing the right ERP solution is not just an IT project; it is a business transformation.

Key Takeaways for Modern ERP Architecture

- **3-tier separation** of presentation, application, and database layers is the modern standard
- **Cloud deployment** reduces TCO and eliminates infrastructure management burden
- **Microservices** enable independent scaling, updating, and deployment of each business function
- **API-first design** makes integrations clean, standardized, and future-proof
- **Thin-client delivery** through web browsers maximizes accessibility and minimizes IT overhead
- **Zero-trust security** protects data at every layer with continuous authentication and encryption
- **AI-embedded intelligence** transforms ERPs from passive record-keepers into active decision-support platforms

Introduction to ERP Modules

Enterprise Resource Planning System (ERP), just by considering name we can simply define ERP as System or software that used to manage all the resources of the whole enterprise. Right from employee payments to a single screw coming into the enterprise, everything can be managed & tracked by using ERP Systems.

ERP is cross-functional software that supports all the business processes within the organization.

In an organization, ERP helps to manage business processes of various departments & functions through the centralized application. We can make all the major decisions by screening the information provided by ERP.

There are many vendors in the market which are providing traditional ERP solutions or Cloud-based ERP solutions. Though implementation platforms or technologies are different, there are common & basic modules of ERP which can be found in any ERP System. Depending on organization's need required components are integrated & customized ERP system is formed.

All Basic Modules Found in an ERP System

- **Manufacturing**
- **Human Resource**
- **Inventory**
- **Sales & Marketing**
- **Purchase**
- **Finance & Accounting**
- **Customer Relationship Management (CRM)**
- **Engineering / Production**
- **Supply Chain Management (SCM)**

Each component mentioned above is specialized to handle the defined business processes of the organization. The following sections go through the introduction of the various modules in detail.

MIND MAP — ERP MODULE ECOSYSTEM



Module Overview — At a Glance

The following table summarizes each ERP module, its primary responsibility, and key functionalities as described in the course material.

Module	Primary Responsibility	Key Functionalities
Human Resource	Efficient management of human resources	Employee info, performance reviews, designations, skill matrix, time & attendance, payroll, travel expenses, training tracking
Inventory	Track stock of items using unique serial numbers	Inventory control, master units, stock utilization reporting, trace current location of items
Sales	Manage all sales transactions end-to-end	Sales queries, quotation drafting, sales orders, invoices with taxation, dispatch/shipment, pending order tracking
Purchase	Procurement of items/raw materials	Vendor listing, item linking, RFQ to vendors, quotation analysis, purchase orders, GRN preparation, stock updates
Finance & Accounting	Manage all inflow & outflow of money	Expenditures, balance sheet, ledgers, budgeting, bank statements, payment receipts, tax management, financial reporting
CRM	Boost sales through better customer service	Communication history, calls, meetings, purchase details, contract duration, lead generation
Engineering / Production	Deliver products in manufacturing industry	Production planning, machine scheduling, raw material usage, BOM preparation, daily progress tracking, production forecasting
SCM	Manage flow of products from manufacturer to consumer	Demand & supply management, sales returns, replacing process, shipping & transportation tracking

Key Point — Module Integration

These modules do not work in isolation. The document emphasizes that **Purchase module is integrated with Inventory module & Engineering/Production module** for updating of stocks. Similarly, **CRM module can be integrated with the Sales module** to enhance sales opportunities, and the **Sales and Distribution module very actively interacts with the Material Management and Financial Accounting module** for delivery and billing.

Human Resource Module

Human Resource module helps to HR team for efficient management of human resources. HR module helps to manage employee information, track employee records like performance reviews, designations, job descriptions, skill matrix, time & attendance tracking.

Important Submodules in HR

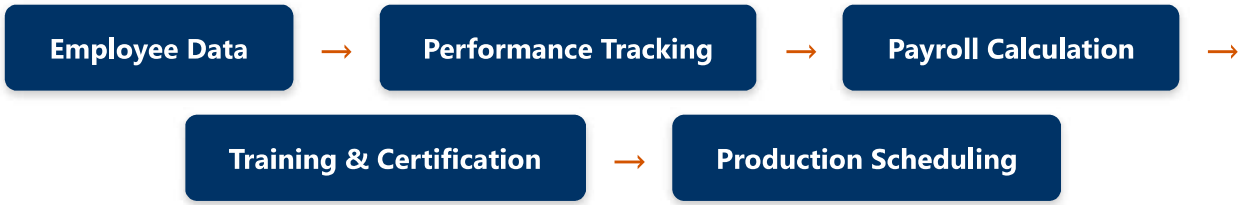
- **Payroll System** — Helps to manage salaries, payment reports etc.
- **Travel Expenses & Reimbursement Tracking**
- **Employee Training Tracking**

How HR Module Works in ERP

People produce the products all manufacturers sell. The human resources module in ERP supports those people. ERP helps by controlling the payroll rates and benefit packages each employee earns. Management knows when an employee is due for an appraisal or raise, and the new pay rate can be immediately used to calculate payroll costs.

The HR module works with the quality module tracking what training and certifications an employee needs to perform their current work or to become eligible for a promotion. Workers who fall behind in their certifications cannot be scheduled for work in the production scheduling module.

HR modules are where management uses future employee characteristics as a model helping locate and hire employees with the skills that will be required tomorrow.



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HR Integration with Production

If a worker's certification expires, the HR module flags this. The production scheduling module then prevents that worker from being scheduled — ensuring only certified personnel operate machinery. This cross-module integration is a core ERP capability.

Inventory Module

Inventory module can be used to track the stock of items. Items can be identified by unique serial numbers. Using that unique numbers inventory system can keep track of item and trace its current location in the organization.

Inventory Module Functionalities

- **Inventory Control** — Monitoring and managing stock levels
- **Master Units** — Defining units of measurement for items
- **Stock Utilization Reporting** — Reports on how stock is being used
- **Item Tracking via Serial Numbers** — Trace current location of each item



Practical Example

You have purchased 100 hard disks, so using inventory system you can track how many hard disks are installed, where they are installed, how many hard disks are remaining etc.

Integration with Other Modules

There may be an integration of the inventory module with the **purchase module** of ERP. When goods are received through the Purchase module (via GRN — Goods Receipt Note), the inventory is automatically updated. This integration is demonstrated in the Interactive Demo section of this page.

Sales & Purchase Modules

Sales Module

Typical sales process includes processes like Sales queries & inquiry analysis & handling, quotation drafting, accepting sales orders, drafting sales invoices with proper taxation, dispatch/Shipment of material or service, tracking pending sales order. All these sales transactions are managed by the sales module of ERP.

CRM module can take the help of the Sales module for future opportunity creation & lead generation.

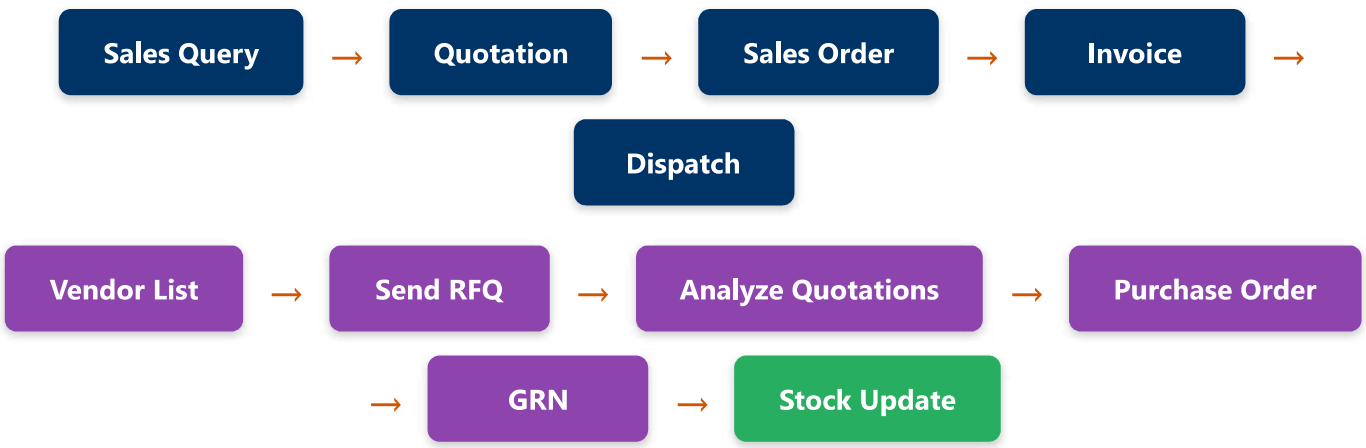
Purchase Module

As the name indicates, purchase modules take care of all the processes that are part of the procurement of items or raw materials that are required for the organization.

Purchase Module Functionalities

- **Supplier/Vendor Listing** — Maintain database of all vendors
- **Supplier & Item Linking** — Map which vendor supplies which items
- **Sending Quotation Request to Vendors** — RFQ process
- **Receiving & Recording Quotations**
- **Analysis of Quotations** — Compare vendor prices and terms
- **Preparing Purchase Orders**
- **Tracking the Purchase Items**
- **Preparing GRNs (Goods Receipt Notes)** & updating stocks & various reports

Purchase module is integrated with **Inventory module & Engineering/Production module** for updating of stocks.



Finance & Accounting Module

Whole inflow & outflow of money/capital is managed by the finance module. This module keeps track of all account-related transactions like expenditures, Balance sheet, account ledgers, budgeting, bank statements, payment receipts, tax management etc.

Financial reporting is an easy task for this module of ERP. Any Financial data that is required for running the business is available on one click in Finance module.

Finance Module Covers

- Expenditures tracking
- Balance sheet preparation
- Account ledgers
- Budgeting
- Bank statements reconciliation
- Payment receipts
- Tax management
- Financial reporting (one-click access)

Customer Relationship Management (CRM) Module

CRM department is helping to boost the sales performance through better customer service & establishing a healthy relationship with customers. All the stored details of the customer are available in the CRM module.

CRM module helps to manage & track detailed information of the customer like communication history, calls, meetings, details of purchases made by the customer, contract duration etc.

CRM module can be integrated with the **Sales module** to enhance sales opportunities.

Manufacturing Module

Manufacturing is the process by which raw materials are transformed into finished products. The role of the ERP manufacturing module is to complete the inventory management by implementing the operations specific to a streamlined manufacturing process.

Manufacturing Recipes Management in ERP

The first phase of the process is represented by the entry of a project into manufacturing process and it involves the generation of recipes, based on which the final product will be created. In the ERP system it is named composed product and it can have one or more attached recipes.

Recipe Types in ERP

- **Promotion** — Used if there is no manual labor involved. E.g., if you buy 10 pencils you get a free sharpener. When such a promotional product is selected, the system automatically discharges the components.
- **Manufacturing** — Used in specific manufacturing processes
- **Composition** — Used in specific processes
- **Decomposition** — Used in specific processes

Engineering / Production Module

Production module is a great help for the manufacturing industry for delivering the product. This module consists of functionalities like:

- Production planning
- Machine scheduling
- Raw material usage
- Bill of Material (BOM) preparation
- Track daily production progress
- Production forecasting
- Actual production reporting

Supply Chain Management (SCM)

SCM module manages the flow of product items from manufacturer to consumer & consumer to manufacturer.

Common roles involved are: manufacturer, Super Stockiest, Stockiest, distributors, retailers etc.

SCM Involves

- Demand & supply management
- Sales returns & replacing process
- Shipping & transportation tracking

Relevance for SMBs

Today many SMBs face challenges in their process automation. ERP is a great help for such organizations. ERP can efficiently streamline the business operations of the organization. The introduction of modules can help you to choose & customize the ERP modules depending on your organization's requirements.

Plant Maintenance Module

The Plant and Machine Maintenance module in resource ERP provides an integrated solution for supporting the operational needs of an enterprise-wide system. The module includes an entire family of products covering all aspects of plant/machine maintenance and becomes integral to the achievement of process improvement.

Technical System Structure

Resource ERP Plant & Machine Maintenance module supports various options for structuring technical systems with its object, type and function-related views, and enables flexible navigation. Data concerning the planning processing and history of maintenance tasks is documented in the system and complies with business verification requirements.

All maintenance tasks such as **inspection, servicing and repair activities** are saved in a historical database. In addition to standard indicators, diverse analysis options are also available in the system for evaluating this data.

Reports & Analysis

Plant and Machine Maintenance module provides you with technical and business reports and various presentation options, according to the criteria used: for example organizational unit, location, execution period for tasks, or system manufacturer.

This information helps you to:

- Reduce the duration and costs of plant down times as a result of damage
- Recognize possible weak points within your technical system in good time
- Define an optimum maintenance strategy in the sense of 'Total Productive Maintenance (TPM)' or risk-optimized maintenance

Major Sub-systems

- Breakdown Repair Log
- Equipment Master Register
- Machine Breakdown Log
- Maintenance Type

Man Power for Machine Details

Preventive Maintenance Record

Spare Part Installation Log

Materials Management Module

Materials Management consists of all master data, system configuration, and transactions to complete the Procure to Pay process.

MM Module Components

- Vendor Master and Material Master Data
- Consumption Based Planning
- Purchasing
- Inventory Management
- Evaluation of Materials
- Invoice Verification

Materials Management contains many aspects of SAP functionality, including purchasing, goods receiving, material storage, inventory, and invoicing. Materials handling, packaging, warehouse planning, accounting, scrap, surplus and obsolete materials disposal, finished goods safety and care are the activities managed by the materials management department.

Data Warehousing

Enterprise resource planning (ERP) systems integrate and automate internal and external management of information. They allow for basic financial planning primarily by aggregating departmental budgets. They tend to provide **static reporting rather than analysis**, which often involves transforming the data to answer the question.

Why Data Warehouses Are Required

Data warehouses are required because ERP systems don't store the data in a manner that supports analysis and reporting. They also do not always gather all available data because of system incompatibilities and increasing amounts of information from systems external to the enterprise.

Purpose of Data Warehouses

- **Extract** data from disparate sources
- **Cleanse** the data
- **Align** it so that it can be aggregated, compared, and analyzed
- **Enable business decisions** through analysis
- **Store** in a single common platform optimized to support enterprise-wide data analysis

Data Mining

Data mining is the computational process that involves a wide variety of techniques in statistics being applied to big data sets usually to discover patterns. It is normally applied to predict events or end results and also detect trends by making use of methods that involve artificial intelligence, database systems, machine learning, and statistics.

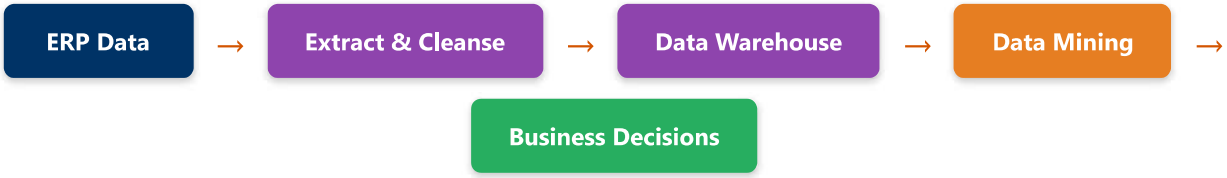
This is often done with the general goal of extracting information from a set of data to transform it into a structure that is understandable for further use. The concise and valuable knowledge of interest that has been discovered can be incorporated into a decision support system and the results are used to make informed business decisions by firm owners.

Data Mining ERP Software

Data Mining ERP software is what results from this integration — specific applications meant to ease the input of data and the output of sensible information for business owners.

This software has become a great industry, producing components that flourish a variety of business functions. Such software is now considered a key organizational tool because of its ability to put together diverse organizational systems and facilitate transactions and production free from error.

The ERP software runs on several networks and computer hardware configurations, usually making use of a database as an information source.



Quality Management

Quality management includes customer-driven quality, top management leadership and commitment, continuous improvement, fast response, actions based on facts, employee participation and above all a quality management culture. Each part of the company is involved in total quality, operating as a customer or supplier to one another.

> **Survival Imperative**
>
> Ultimately whatever be the industry in which you're present or the size, to survive and succeed in the present scenario, it is imperative that you maintain a very high quality at all stages of manufacturing and distribution.

Areas Addressed by Quality Management Module

Quality Checks Across the Value Chain

- **Procurement & Subcontract Process** — During Goods receipt or on completion of Services, quality parameters and attributes can be captured. This serves as the basis for clearance, usage and supplier payment.
- **Inventory & Storage** — During storage of materials, a periodic quality check can be done to assess the quality.
- **Production** — During the start of a batch or a new product, or periodically — required to check and record Quality at different stages based on which action pertaining to re-work, recalibration, tool setting etc. would be undertaken.
- **Sales & Dispatch** — Quality inspection can also be carried out before dispatch of goods.
- **Project Tasks** — Quality check could be done for tasks being executed as part of a project.

Specific Quality Features

Facility for In-process Inspection and Quality Clearance

Ability to define control points based on control factors (Warehouse, Master Recipe, Supplier, Customer, etc.)

Ability to define inspection plan for each control point

Ability to have attribute / non-attribute based (check list) inspection plan

Facility to define standard operating procedures

Ability to capture Analysis methods for attributes

Ability to specify attribute values for each inspection plan

Ability to override sample result status

Ability to suggest lot change / item change based on Quality Clearance Feedback

Sales and Distribution

To keep pace with rapid changes in the business world, companies need an integrated and flexible enterprise system that supports all aspects of their business with state-of-the-art functionality. This innovative solution should upgrade effortlessly and interface easily with third-party applications as well as have the ability to incorporate existing systems while extending its reach to the Internet and e-commerce.

In today's competitive business environment, companies are increasingly being forced to streamline business processes. In a world where it is no longer enough to simply have the best product; companies are focusing on core competencies and closer partnerships over the whole supply chain.

Here, increased efficiency in sales and distribution is a key factor to ensure that companies retain a competitive edge and improve both profit margins and customer service. The Sales and Distribution module of resource ERP systems offers a comprehensive set of best-of-breed components for both order and logistics management.

Resource ERP system is tightly integrated with the Sales and Distribution module. This integration enables the mapping and supply of single-site or multi-site organizations. Developing precise logistics planning for just-in-time deliveries, this system can also generate replenishment orders by using defined warehouse requirements.

Sales Related Business Transactions

- Sales queries (inquiries and quotations)
- Sales orders
- Outline agreements (contracts and scheduling agreements)
- Delivery/Shipment
- Invoicing/billing
- After sales support

Basic Functions During Sales Order Processing

1. **Inquiry handling** — Process and respond to customer inquiries
2. **Quotation preparation and processing** — Draft and manage quotations
3. **Contracts and contact management** — Order management activities

4. **Monitoring the sales transaction** — Track progress of each sale
5. **Checking for availability** — Verify stock availability
6. **Transferring requirements to materials planning (MRP)**
7. **Scheduling the delivery**
8. **Calculating pricing and taxes**
9. **Checking credit limits**
10. **Invoicing/billing**
11. **Creating printed or electronically transmitted documents**

Depending on how your particular system is configured, these functions may be completely automated. The data that results from these basic functions is stored in the system where it can be displayed. Resource ERP's Sales and Distribution module very actively interacts with the **Material Management and Financial Accounting module** for delivery and billing.

Case Study: ERP in Banking Sector

Enterprise resource planning (ERP) systems integrate the organizations business functions allowing efficient information sharing across all business divisions. Through the information sharing is achieved not only better coordination but also faster and more efficient adjustment to the potential risks and business opportunities alike.

Current State of ERP Systems in Banking Sector

On a daily basis the banks fulfil several obligations towards their customers. ERP systems with their capabilities can provide useful aid towards that direction. A financial institution as part of its typical responsibilities needs to:

- Receive a customer's money and cheques as collection and credit his account with them
- Provide a statement of account and a statement of the balance on request
- Honour their cheque up to the credit balance or overdraft limit, provided they are in order and there is no stop

Needless to say, that all the aforementioned processes must be executed with **strict secrecy about customer affairs**.

Why Banking is Ideal for ERP

The banking industry is highly data-oriented and offers enormous potential for ERP applications. An ERP system offers wide-ranging integration between different banking system modules. ERP integrates users, information, processes, and applications for higher productivity.

ERP Capabilities in Banking

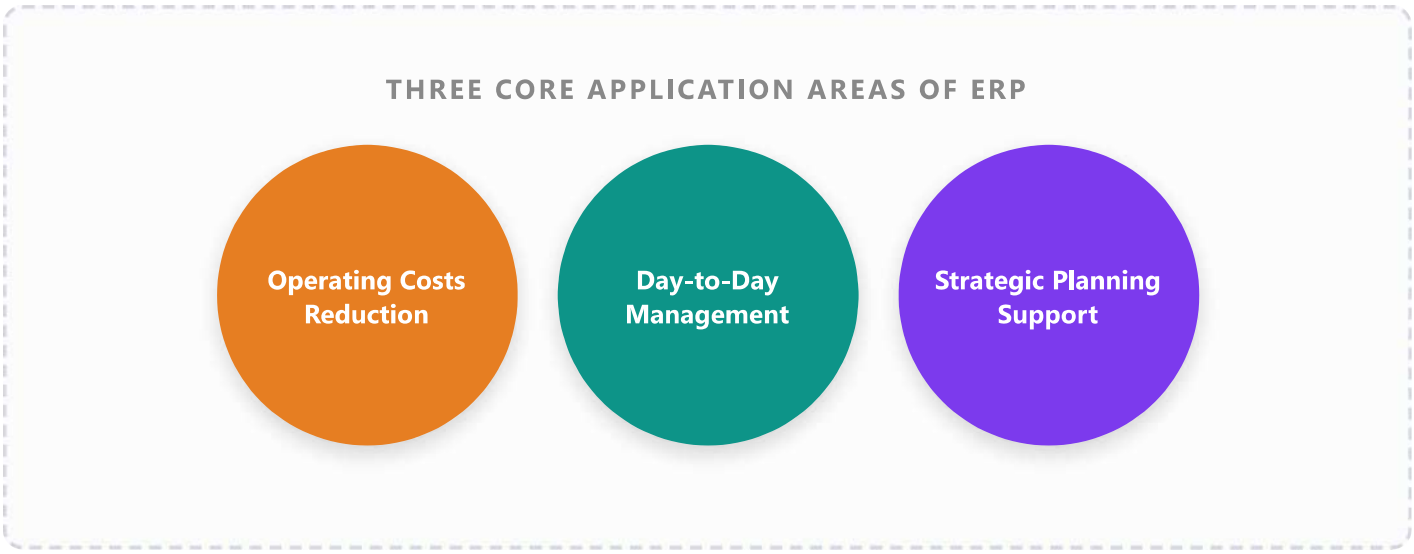
- Facilitates decision making with simulations for enhanced responsiveness and change
- Uses portal technology, business intelligence, knowledge management, and mobile technologies that save time and reduce costs
- Enables banking employees interact with bank's top management for reduced time and effort
- Reverses the usual communication 'top-down' to 'bottom-up'
- Provides complete end-to-end solution covering payment processing, cash accounting, cash management, technical analysis, index tracking and portfolio optimization

Key Banking Processes Supported by ERP

Process	Description
Payment Processing	Revamping payment processing needs a high quality understanding on how banks make payments. Complexity increases while dealing with multiple branches in different countries and operating under country specific legislations.
Cash Accounting	Cash accounting complexity increases with multiple locations and currencies.
Cash Management	Need to administer available account on regular basis so that investments of funding decisions can be made in an appropriate manner.
Loans & Advances	Principal components of bank assets and main source of income. Includes demand loans (temporary, heavy interest), term loans (2-30 years, secured, for renovation/expansion/modernization), cash credits, and overdrafts.
Portfolio Management	Fundamental analysis tools focus on company's performance. Technical analysis focuses exclusively on stock price movement. Modern ERP uses Markowitz's Mean-Variance model — maximizing portfolio expected return for a given amount of portfolio risk.

Topic Overview: Applications of ERP

The aim of ERP software is to optimize processes, improve communication between departments, and enhance their efficiency. But the main reason entrepreneurs choose these solutions is the opportunity to **save time and money** while making a profit and growing. ERP applications can cost higher than expected, but it is worth every single cent.





Why ERP Matters

- ERP unites all business processes into a **holistic system**, breaking down departmental silos
- It provides the backbone for data-driven decision making across all levels of the organization
- Despite high initial investment, the long-term ROI through cost savings and efficiency gains justifies the expense
- However, success depends on **organizational willingness to change** — not just technology adoption

3
Core Application Areas

100%
Process Integration Goal

High
ROI When Done Right



Operating Costs Reduction

ERP helps unite all the business processes in a holistic system. It leads to the improvement of coordination and raises the company's efficiency. After implementation, you will be able to see a crucial difference — employees will have more time for other tasks, or you may even cut staff because some manual processes will be fully or partly automated. You will also be able to reduce the cost of marketing, production, and office maintenance.

Conceptual Explanation

Operating cost reduction through ERP is not about cutting corners — it's about **eliminating waste**. When processes are automated, data doesn't need to be re-entered across departments, reducing errors and the labor needed to fix them. When all departments share the same data, redundant reports and reconciliation tasks disappear. Marketing budgets become data-driven, production schedules optimize resource use, and physical office overhead shrinks as digital workflows replace paper-based ones.

EXAMPLE :
Imagine a retail company handling order processing manually versus with an ERP:
Before ERP (Manual & Wasteful)
A customer orders online. A customer service rep manually types the order details into an Excel sheet and emails it to the warehouse. A warehouse worker re-types the address into a shipping system, makes a typo in the zip code, and ships the package to the wrong address.
The Resulting Waste: Hours spent re-entering data across teams, extra shipping fees to re-send the item, customer service time spent handling the complaint, and paper invoices printed and filed in cabinets.

After ERP (Automated & Lean)
The customer orders online, and the ERP instantly updates inventory, notifies the warehouse to pack the item, generates the shipping label automatically, and sends the customer an invoice.
The Saved Cost: Zero re-entry errors, no wasted labor fixing mistakes, reduced paper overhead, and faster fulfillment using a fraction of the operating effort.

Interactive Dashboard: Cost Reduction Mechanisms

Click a component from the pool below, then click the matching gray zone on the dashboard to reveal how each mechanism reduces costs.

ERP Cost Reduction Dashboard 0 / 5 Placed

PROCESS AREA

COORDINATION HUB

MARKETING ZONE

PRODUCTION ZONE

OVERHEAD ZONE

AVAILABLE COMPONENTS

Click a component, then click the matching gray zone on the dashboard above.



Facilitation of Day-to-Day Management

The ERP application provides **better corporate planning** and **easier data access**. It helps employees make decisions based on available information rather than guesswork. ERP can also control the actual operations' costs and calculate the spendings in real time, giving managers the visibility they need for daily operational decisions.

Conceptual Explanation

Day-to-day management in any organization suffers from **information fragmentation —** sales data lives in one system, inventory in another, finance in a third. ERP collapses these silos by providing a single source of truth. Managers no longer wait for weekly reports; they access real-time dashboards showing current stock levels, pending orders, cash flow status, and employee workloads. This transforms reactive management into **proactive, data-driven management**.

EXAMPLE :
Imagine a coffee shop chain preparing for a busy weekend:

Without ERP (Fragmented Silos)

- Sales sees a sudden surge in cold brew orders on Friday morning.
- Inventory doesn't know about the spike, so they don't order extra coffee beans or dairy.
- Finance is working off last week's report and assumes normal budget needs.
- The Result: On Saturday, stores run out of beans mid-morning. Managers lose thousands in potential sales while frantically making emergency retail-price runs to local grocery stores.

With ERP (Single Source of Truth)

- As Friday's cold brew sales surge, the ERP automatically alerts the manager on a real-time dashboard that bean stock will deplete by Saturday morning.
- The system auto-generates a supplier reorder immediately, matching current cash flow limits pre-approved by finance.
- The Result: The store receives a bean delivery before opening Saturday, avoids stockouts, and maximizes weekend revenue proactively.

Interactive Dashboard: Daily Management Facilitation

Click a component from the pool below, then click the matching gray zone on the dashboard to reveal how ERP facilitates daily management.

ERP Daily Management Dashboard

0 / 5 Placed

PLANNING ZONE

DATA ZONE

DECISION ZONE

COST ZONE

FINANCE ZONE

AVAILABLE COMPONENTS

Click a component, then click the matching gray zone on the dashboard above.

Real-World Example

A warehouse manager using ERP can see real-time stock levels, pending inbound shipments, and current order backlog on a single screen. Without ERP, they would need to check three separate systems, make phone calls, or wait for end-of-day reports — wasting hours and risking stockouts or overordering.

Strategic Planning Support

Strategic planning processes determine the **goals, tasks, needs, assets, and results** that entrepreneurs use for analysis. ERP helps plan the company's opportunities and future actions by providing a data-rich foundation for strategic decisions.

Conceptual Explanation

While day-to-day management is about *running* the business, strategic planning is about *directing* the business. ERP supports this by aggregating historical and real-time data that reveals trends, patterns, and opportunities. Leaders can analyze which product lines are most profitable, which markets show growth potential, where resources are underutilized, and what the likely outcomes of different strategies would be — all backed by data rather than estimates.

EXAMPLE :
Imagine a sportswear company deciding which products to launch next year:

Without ERP (Strategy Based on Estimates)

- The executive team assumes winter jackets are their best product because total revenue was high last year.
- They decide to double winter jacket production for next season based on gut feeling.
- Finance is working off last week's report and assumes normal budget needs.
- The Result: They missed that raw material costs and returns for jackets were so high that profits were actually minimal—wasting millions in factory capacity that could have gone to higher-margin items.

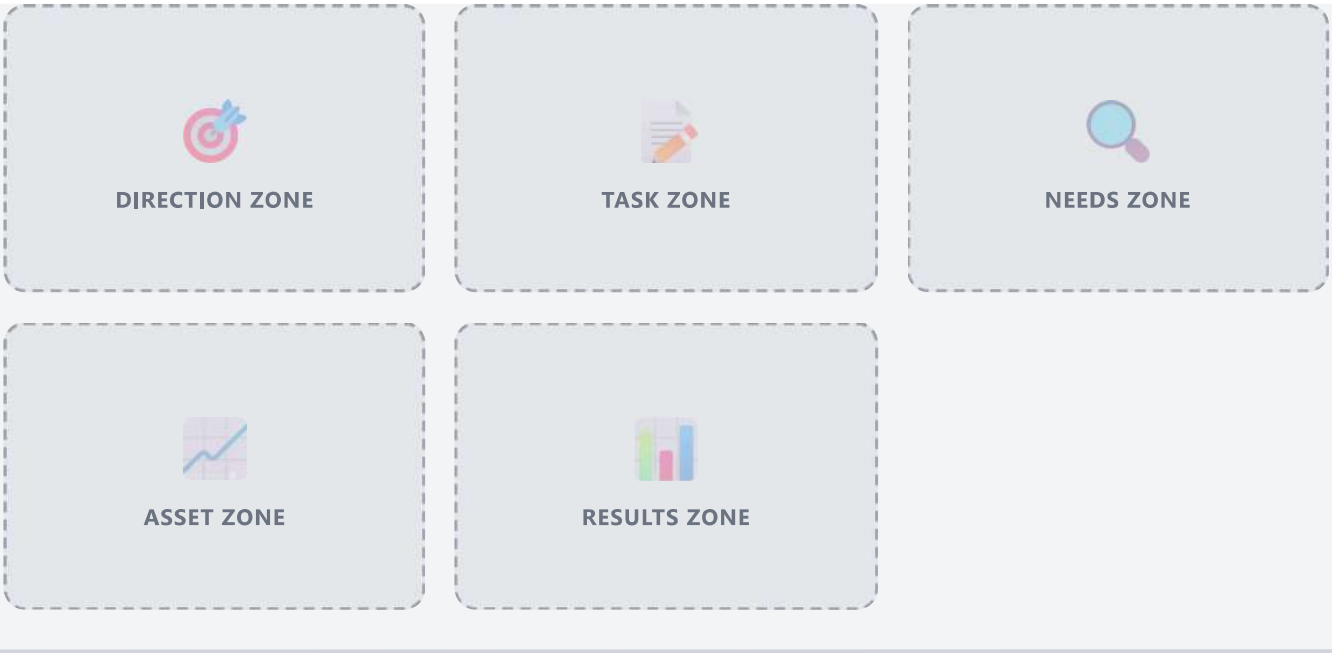
With ERP (Strategy Driven by Data)

- Executives pull up an ERP profitability report that combines multi-year sales, return rates, production costs, and shipping expenses across all regions.
- The data reveals that while winter jackets bring in high top-line revenue, custom yoga leggings actually generate a 45% higher profit margin and have growing year-over-year demand.
- The Result: Leadership allocates budget to expand the yoga apparel line and optimize jacket production, maximizing long-term profitability based on clear financial realities.

Interactive Dashboard: Strategic Planning Components

Click a component from the pool below, then click the matching gray zone on the dashboard to reveal how ERP supports strategic planning.

ERP Strategic Planning Dashboard 0 / 5 Placed



AVAILABLE COMPONENTS

Click a component, then click the matching gray zone on the dashboard above.

Important: Why ERP Implementations Fail

An ERP system doesn't always eliminate inefficiencies within the business. The company needs to **rethink the way it's organized**, or else it will end up with incompatible technology. ERP systems usually fail to achieve the objectives that influenced their installation because of a company's **reluctance to abandon old working processes** that are incompatible with the software. Some companies are also reluctant to let go of old software that worked well in the past. The key is to **prevent ERP projects from being split into many smaller projects**, which can result in cost overruns.

- **Organizational resistance:** Employees and managers cling to familiar processes rather than adapting to ERP workflows
- **Legacy software attachment:** Companies keep old systems running alongside ERP, creating data conflicts and duplication
- **Piecemeal implementation:** Splitting ERP into too many small projects leads to fragmentation, inconsistent data, and budget overruns
- **Lack of reengineering:** Implementing ERP without first rethinking business processes means automating inefficiencies rather than eliminating them
- **Inadequate training:** Without proper change management, users find workarounds that bypass the ERP system entirely

Implementation of ERP

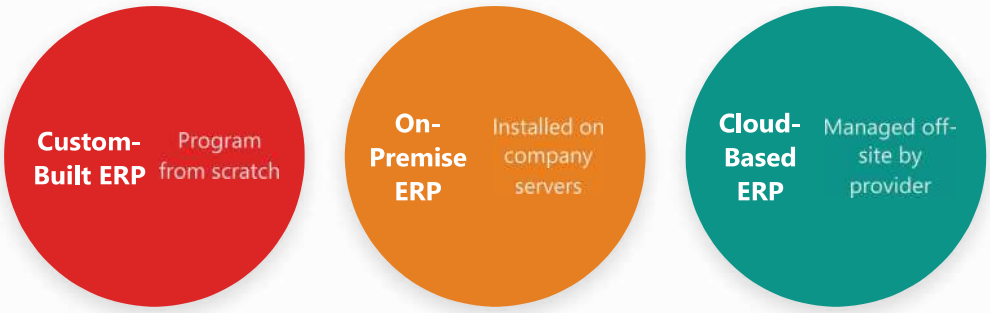
Enterprise Resource Planning — MCA 608-EA | Unit 11

Topic Overview: Implementation of ERP

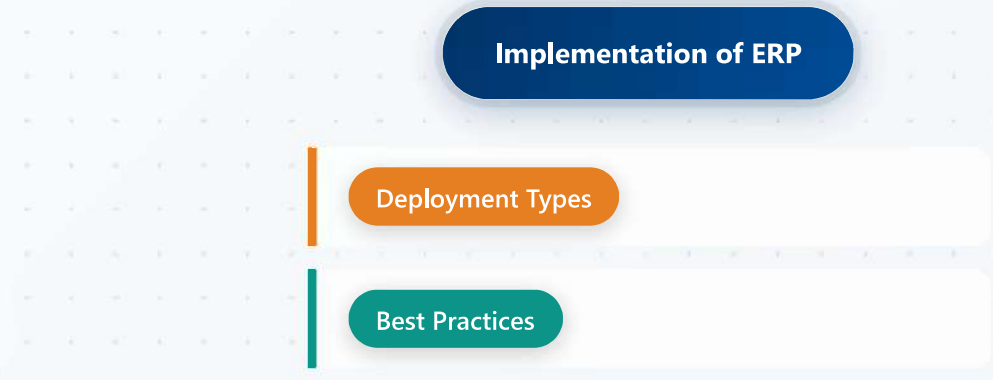
ERP implementation is a critical project that must be taken seriously regardless of business type. From concerns about cost and demonstrating ROI, to practical considerations about who will manage the process, ERP implementation efforts require commitment by a project team and support by business leaders to ensure success.

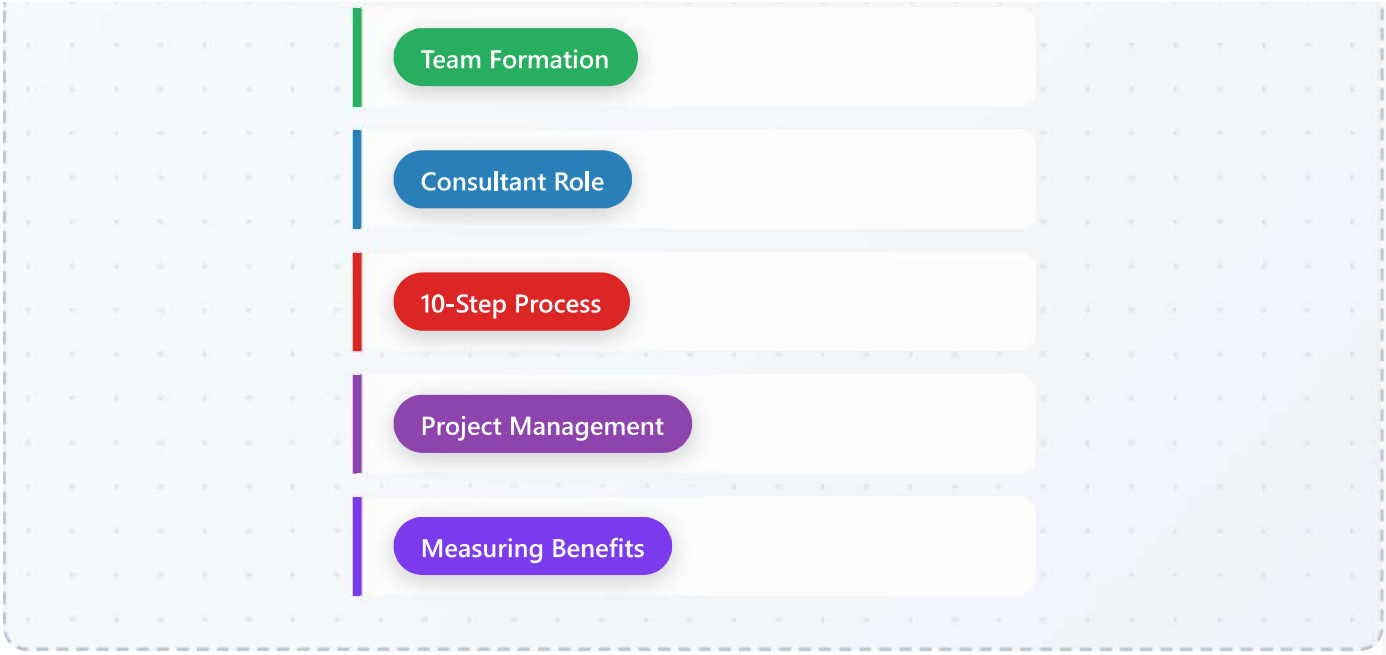
With proper planning and execution, implementing an ERP system should be a smooth process that quickly improves efficiency across your business. However, the sheer size and complexity of ERP implementations makes managing these projects difficult — it touches the entire organization and can affect nearly every employee.

THREE DEPLOYMENT OPTIONS FOR ERP



MIND MAP — IMPLEMENTATION OF ERP





Why ERP Implementation Matters

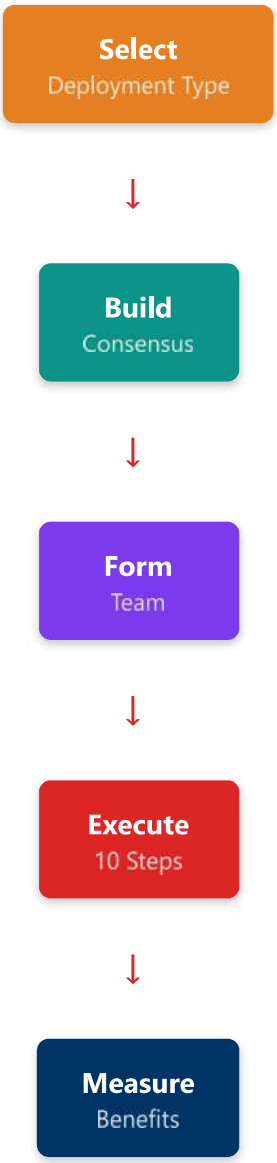
- ERP deployment is a **major undertaking** — but with proper planning it doesn't have to be painful. Animation
- Two-thirds of mid-size businesses run **old versions** of their on-premise ERP software because upgrades are too painful. Animation
- Cloud-based implementations are typically much **faster and easier** to deploy — average 3 months vs 18 months for on-premise. Animation
- The gap between ERP promise and delivered business value can be **enormous** without proper management. Animation
- Success depends equally on **people and technology** — not just the software itself. Animation



Mid-Size Firms on Old ERP

Critical Warning

Enormous cost overruns, deadlines missed by years, and even abandoned implementations make clear that managing ERP projects is a complex task. One mismanaged ERP implementation left a southeastern electronics manufacturer unable to accept deliveries and nearly closed a plant.



Before - After

BEFORE IMPLEMENTATION OF ERP

Before an ERP system, there were different databases of different departments which they managed on their own. The employees of one department does not know anything about the other department.



AFTER IMPLEMENTATION OF ERP

After the ERP system, databases of different departments are managed by one system called the ERP system. It keeps track of all the databases within the system. In this scenario, employees of one department have information regarding the other departments.



ERP Implementation Best Practices

Implementing an ERP solution is a critical project that requires commitment by a project team and support by business leaders. With proper planning and execution, the process should be smooth and quickly improve efficiency. Here are the **four critical best practices** to consider when selecting and deploying a solution.

Conceptual Explanation

Best practices in ERP implementation are not optional guidelines — they are **survival strategies**. The statistics are sobering: enormous cost overruns, deadlines missed by years, and even abandoned implementations are common. The root cause is almost never the software itself, but rather failures in *how* the implementation was approached. These four best practices address the most common failure points: choosing the wrong deployment model, lacking leadership support, setting unrealistic timelines, and under-resourcing the project team.

Interactive Dashboard: Four Best Practices

Click a component from the pool below, then click the matching gray zone on the dashboard to reveal each best practice.

ERP Best Practices Dashboard 0 / 4 Placed

DEPLOYMENT ZONE

LEADERSHIP ZONE

TIMELINE ZONE

RESOURCE ZONE

AVAILABLE COMPONENTS

Click a component, then click the matching gray zone on the dashboard above.

Real-World Insight

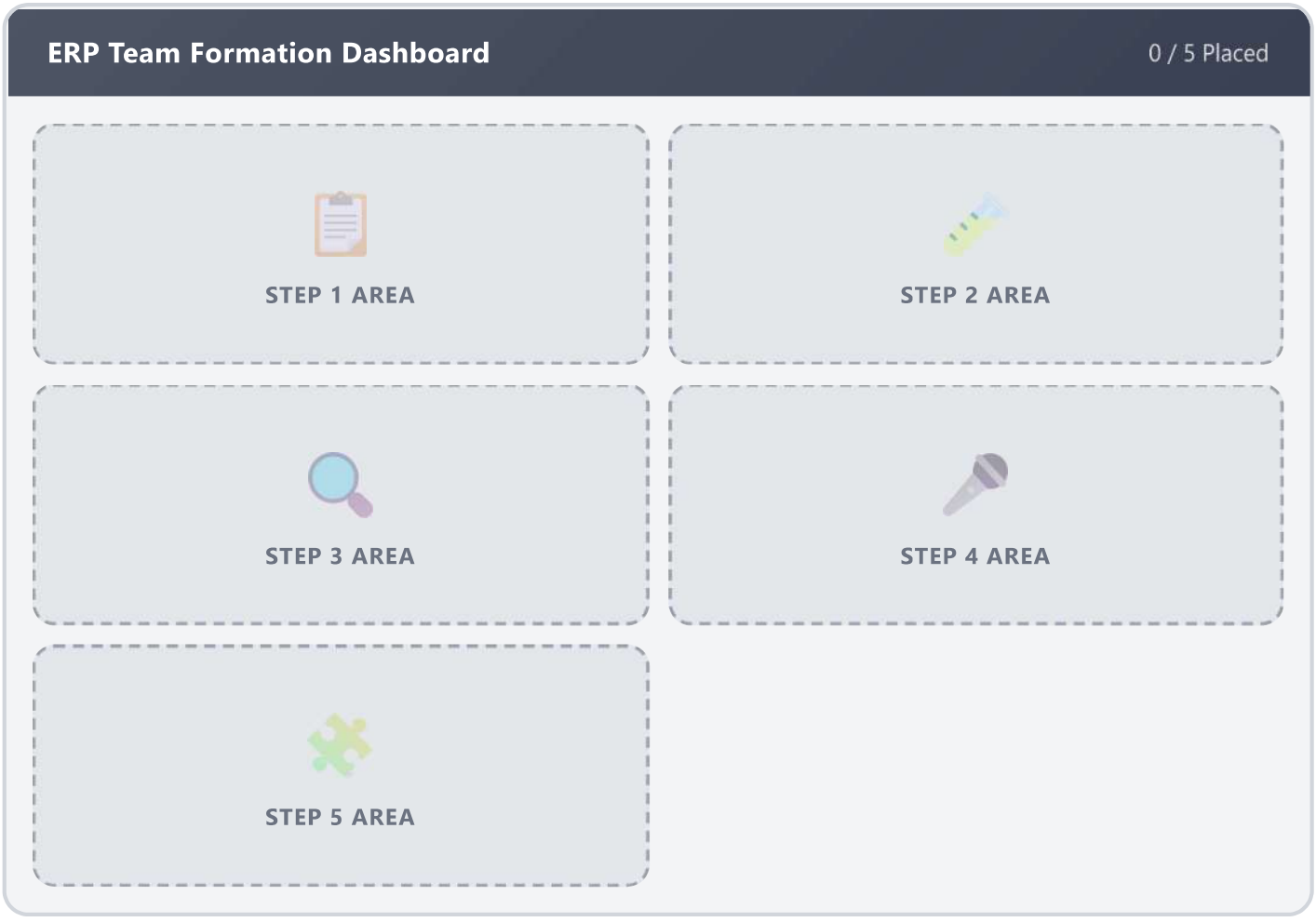
Two-thirds of mid-size businesses run old versions of their on-premise ERP software. Why? Because each upgrade risks losing critical customizations and integrations, making the pain of re-implementation too great. Cloud-based solutions like NetSuite solve this — customizations and integrations automatically update with system upgrades, so you always have the most advanced capability without the upgrade pain.

Team Formation & Consultant Intervention

Forming the right project team and engaging the right consultants are **make-or-break decisions** in ERP implementation. You need team members with required competencies who can work well in a project environment (which differs significantly from daily operations), and you may need external ERP consultants to fill knowledge gaps.

Interactive Dashboard: Team Formation Steps

Click a component from the pool below, then click the matching gray zone to reveal the five best-practice steps for selecting ERP project team members.



AVAILABLE COMPONENTS

Click a component, then click the matching gray zone on the dashboard above.

ERP Consultant: Role, Skills & Rewards

The ERP consultant needs an **equally broad base of understanding** of all elements that comprise the system. Although ERP consulting is a specialized field, there exists within it a need for knowledge of most aspects of a business enterprise.

- **Generalist Path:** Possess at a high level a solid understanding of all key elements — enables comprehensive solution design and implementation services
- **Specialist Path:** Offer deep expertise in one area (accounting, HR, logistics, sales & marketing) — powerful when collaborating with complementary specialists
- **Technical Training:** Obtained from top software providers — Oracle, SAP, and Microsoft collaborate with educational institutions to offer ERP training
- **Experience:** Most successful consultants have worked with a major ERP vendor or in a customer organization where one of these applications was adopted

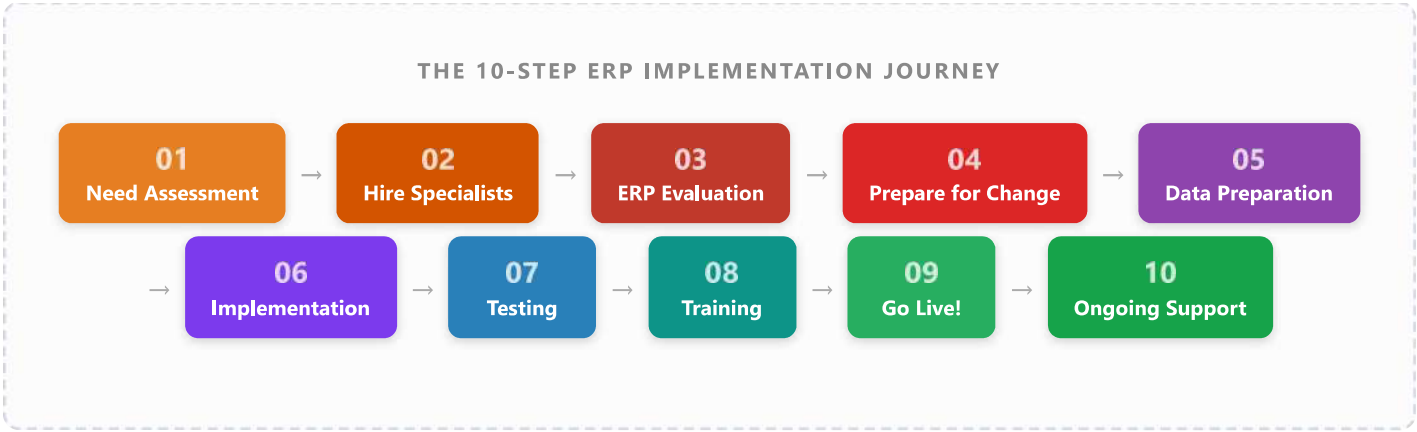
- **High Demand:** ERP consultants are highly sought after — demand exceeds supply significantly
- **Premium Rates:** Consulting rates are significantly higher than average; SAP or PeopleSoft projects are plentiful with companies willing to pay a premium
- **Compensation:** Can start in six figures, rising to seven figures or more for larger consulting firms
- **Project Earnings:** $120K net per project as a rookie working solo; up to $250K with experience — sometimes more depending on scope and negotiation
- **Less Business Development:** High demand means more time doing the job rather than drumming up business
- **Global Exposure:** Organizations benefiting from ERP often have global operations — extensive travel may be required (advantage or disadvantage depending on personal situation)

Benefits of Scientific Team Selection

- Provides a **scientific and objective assessment method** supplementing other sources of candidate data
- Identifies **potential development needs** of individuals moving from operational to project environments
- Identifies potential issues associated with building an **effective project team culture**
- Good decisions on team members help significantly to **deliver projects on time and on budget** while optimizing business benefits

Ten Steps to Successful ERP Implementation

The ERP implementation process follows a structured sequence of ten critical steps. Each step builds upon the previous one, and skipping or rushing any step can jeopardize the entire project. Click each step below to learn the details.



The first step is to identify your company's needs. Start by finding and documenting the **critical business processes, inflection points, and key performance indicators (KPIs)**. This helps pinpoint the right ERP solution and the appropriate specialists needed for the transition.

Key Insight: This is about *business processes* that support the enterprise. While IT considerations are important, IT shouldn't be the driver of the analysis. The business must lead.

Many organizations lack the **internal expertise and experience** for ERP implementation. Consider hiring or contracting with an experienced professional to guide you through the process. This prevents headaches and wasted time, and gives you the space to focus on other tasks.

Selection of the right ERP solution is one of the **most critical steps**. The right program depends on your industry, business needs, and system preferences. Key consideration: on-premise vs cloud computing ERP. Many businesses have jumped to the cloud, but it's not always the best option — it depends on your specific requirements and organizational capacity. Your ERP consultant can assist with this decision.

Implementing a new ERP system can be a **major change**, especially if you've never used one before. Stay transparent with employees about the change and engage them when possible. ERP transitions are **as much about changing culture as changing technology**.

Don't assume all current data can be converted into the new system. **Analyze your current data** and decide which pieces need to be converted. After data is entered into the ERP database (most allow entry via spreadsheet), clean it up — review and weed out unnecessary data like old customers or those no longer in business. **Data integrity is critical** to a successful implementation.

After completing the first five steps, it's time to **launch the new system**. Your ERP consultants have mapped out a specific plan for installing, configuring, and optimizing the system for your working environment. Plans are bound to change (at least slightly) during this process — always keep your major needs and end goals in mind.

While ERP systems have come a long way since their introduction in the 90s, **do not assume everything will execute smoothly**. System testing and **user acceptance testing (UAT)** are key phases of the implementation process. This is where you verify that the system works as intended in your real-world environment.

After the system is configured, **train your employees** on how to use the new program. Consider finding a group that specializes in onsite training to ensure everyone is up to speed. Your IT team should receive an **extra session of training** to handle any technical problems that may arise in the future.

Once the system has been configured, tested, and employees trained, it's time to **go live**. You can now safely activate your ERP system. It wouldn't hurt to do another round of testing after launching, just in case — this is called "post-go-live testing" and catches edge cases that only appear under real operational load.

You can't just install your ERP system and never touch it again. Advanced ERP implementations require **time and attention beyond the initial installation**. This includes upgrades to your system and general maintenance. For cloud-based solutions, the provider handles most of this — but for on-premise systems, this is an ongoing internal responsibility that many organizations underestimate.



Timeline Comparison

The average NetSuite (cloud) customer implements ERP within **3 months**. The on-premise ERP implementation industry average is **1.5 years (18 months)**. That's a 6x difference — primarily because cloud eliminates hardware setup, reduces configuration complexity, and the provider handles upgrades.

ERP Project Management & Monitoring

ERP systems have fundamentally changed the work of IT organizations. The sheer size and complexity of ERP implementations makes managing these projects difficult. There are really **two basic sides** to ERP management: **people** and **technology**. An ERP package touches the entire organization and can affect nearly every employee.

Top 10 ERP Project Management Headaches

According to surveys of experienced ERP project managers from corporate IT departments and Big 5 consulting companies:

1. Project Size
2. Staffing (Includes Turnover)
3. Risk Management
4. Unreasonable Deadlines
5. Funding
6. Organizational Politics
7. Scope Creep
8. Unexpected Gaps
9. Interfaces
10. Resistance to Change

Key Project Management Areas

Perhaps the **single most decisive element** of ERP success or failure is the knowledge, skills, abilities, and experience of the project manager. Key characteristics include:

- Must understand **both the business and the technology**
- Must understand the impact of ERP on the business and work with managers for smooth transition from "as is" to "to be" environment
- Must be **flexible** enough to roll with changes and not lose it when unpleasant surprises pop up
- Must work with **nearly every individual** in the organization — from technical IT staff to the mailroom
- Must possess the ability to **learn extremely fast** — understanding business issues in unfamiliar areas
- Must be **highly disciplined** — clearly envision the end game and hold the organization to the road leading there
- Must be willing to make **tough decisions** that will upset some and please others — a thick skin is an asset

Scope management procedures must be created and enforced to prevent **"never ending project" syndrome**. Constant scope changes cause confusion among team members.

- **Define what's in and out:** The primary focus is on defining and controlling what is and is not included
- **Business processes:** Decide which business processes will be included — this determines which ERP functionality to implement
- **Phased rollout:** If multiple business units exist, decide which divisions to include in each phase
- **Technology decisions:** Determine which technologies will be replaced, upgraded, or will exchange data through interfaces
- **Change control:** Clearly define and enforce change control procedures — tight procedures may cause tension but the project can't succeed if hitting a constantly moving target

Essential: Ensure a project charter or mission statement exists. Nail down requirements and have them documented and signed by users and senior management.

No software fits every organization perfectly. Despite what ERP vendors claim, there will always be gaps — small or extremely large and problematic.

- **Political minefield:** Each time a gap costs additional dollars, someone will ask "Why did we spend all that money if the software doesn't do what we need?"
- **Executive pushback:** Sponsors may look bad and push the project manager to "make the gaps go away" — leading to user frustration
- **Solution:** Be extremely thorough in package selection. Ensure everyone at every level knows what the software can and can't do
- **Gap document:** Start creating it early — it's useful for stakeholder management and provides clear direction on what needs to be done
- **Review process:** Detail the change scope, cost, benefit, and adapted project plan for each gap

It's **absolutely critical to get the right people involved early**. Leaving out the wrong person has both project-related and political implications.

- **Avoid tunnel vision:** Don't just work with immediate rollout users — examine all interfaces and involve every communicating business area
- **Staffing challenge:** Good technology staff with deep ERP experience are extremely hard to find
- **Retention:** Identify critical but high-turnover-risk staff early. Develop recognition programs and consider flextime (studies show significant productivity increases)
- **Preventing brain drain:** Pair consultants with corporate employees in both technology and business areas. Work side-by-side throughout implementation for constant knowledge transfer
- **Real example:** One large midwestern food producer solved the knowledge drain problem by pairing consultants with corporate staff — ensuring nearly constant information flow

Scheduling ERP projects is like **"herding cats"** — lots of people, lots of subprojects, and many potentially conflicting issues.

- **Plan thoroughly:** Develop a clear, concise, and thorough project plan before starting — flexible enough for inevitable bumps
- **Account for people:** Identify time issues with assigned people, account for vacations, sick days, and other leave
- **Critical path analysis:** Determine which resources absolutely must be present at certain times (e.g., the DBA must be there when the database is built)
- **Compassionate micromanagement:** Create very specific work assignments for developers. Schedule technical and management reviews at least once a week
- **Phase reviews:** Do a project review at the end of each phase and the project as a whole
- **Success criteria:** Define them clearly early — if they're obsolete at the end, the project can't be evaluated as successful

Managing Chaos

Managing an ERP project is unlike any other effort because of the huge number of variables, people, and risks. The complete replacement of an organization's information systems has a tremendous impact on the people, the company, its suppliers, and even its customers. The project manager must guide with a firm, practiced hand that both encourages innovation and ensures everything moves in the right direction.

Measuring Benefits of ERP

A lot of tools help with the "before" part of ERP evaluation — TCO and ROI calculators, for example. But measuring the "after" part is often neglected — yet it's critical to determine whether the job is complete or there's still more to do to wring expected efficiencies out of the implementation.

Conceptual Explanation

The "after" part of ERP evaluation is **different enough from the "before" part** that it needs its own set of evaluation methods. The following six methods provide a comprehensive framework — from immediately visible results to long-term strategic advantages — each capturing a different dimension of ERP success that a single metric would miss.

Interactive Dashboard: Six Ways to Measure ERP Success

Click a component from the pool below, then click the matching gray zone on the dashboard to reveal each measurement method.

ERP Benefit Measurement Dashboard 0 / 6 Placed

METHOD 1 | METHOD 2 | METHOD 3

METHOD 4 | METHOD 5 | METHOD 6

AVAILABLE COMPONENTS

Click a component, then click the matching gray zone on the dashboard above.

Visible Immediate → **Measurable** Hard Numbers → **Tangible** Real but Hard to Quantify → **Strategic** Long-Term Value

Important Distinction

Measurable benefits are always tangible, but not all tangible benefits are readily measurable. For example, improved communication is quite obvious and real to employees and managers, but harder to put a number on. Nonetheless, you should find a way to include ALL tangible benefits in your post-implementation evaluation — measurable or not.

ERP Lifecycle, XRP & Features

Information Technology / Management Information Systems — S.Y. B.Com (A&F)

ERP Implementation Lifecycle

The ERP implementation follows a structured lifecycle. Click on each phase below to learn more about the specific activities involved.

Phase 1: Pre-Evaluation Screening

The organization evaluates whether it requires an ERP system. Management reviews existing operational bottlenecks (e.g., delayed monthly financial closes, inventory mismatches) and eliminates software vendors that do not match the company's business model.

Phase 2: Package Evaluation & Selection

The organization analyzes commercial packages (e.g., SAP S/4HANA, Oracle Cloud ERP, TallyPrime Enterprise).

- **Evaluation Criteria:** Total Cost of Ownership (TCO), vendor stability, module scalability, and functional fit.
- **Rule of Thumb:** A commercial ERP package generally fits **80%–85%** of standard business processes out of the box.

Phase 3: Project Planning

The implementation team defines timelines, budgets, resource allocation, and committee roles. Key deliverables include project milestones, risk management frameworks, and Steering Committee establishment.

Phase 4: Gap Analysis

This phase compares how the business currently operates (**"As-Is" state**) against how the selected ERP package processes data (**"To-Be" state**). Gap analysis determines the delta that requires process adjustments

or custom code.

Phase 5: BPR & Customization

- **BPR (Business Process Re-engineering):** Restructuring business workflows to align with ERP best practices.
- **Customization:** Writing custom modules or scripts to handle unique business rules.

Phase 6: System Configuration & Data Migration

- **Configuration:** Setting system parameters, chart of accounts, tax structures (GST/TDS), and role-based permissions without modifying source code.
- **Data Migration:** Cleaning, formatting, and transferring historical ledger entries, customer data, and vendor profiles from legacy systems into the new ERP database.

Phase 7: Testing & End-User Training

- **Testing:** Conducting unit testing, integration testing, stress testing, and User Acceptance Testing (UAT).
- **Training:** Training accounts, inventory, and sales staff on transaction screens and approval workflows prior to launch.

Phase 8: Going Live (Cutover)

The organization transitions from legacy systems to the live ERP system.

Cutover Strategies:

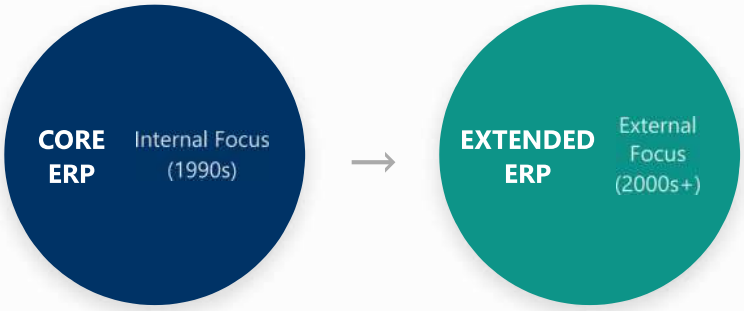
- **Direct Cutover:** Instant switch.
- **Parallel Running:** Operating both systems concurrently for 1–2 accounting periods.
- **Phased Rollout:** Rolling out by module or location.

Phase 9: Post-Implementation Support

Post-launch activities focus on bug fixes, system optimizations, patch installations, and periodic system audits.

Concept of Extended ERP (XRP / ERP II)

EVOLUTION: CORE VS. EXTENDED



Core ERP: Focuses on internal enterprise functions (Finance, HR, Production, Internal Inventory). Connects internal departments around a shared database.

Extended ERP (XRP): Extends system access beyond internal operations to connect business partners, suppliers, logistics providers, and customers in real time via web services and APIs.

Interactive Component Map

Match the function on the left with the correct Extended ERP component on the right to reveal the details.

XRP Component Dashboard 0 / 4 Placed

SUPPLY CHAIN MGMT

CUSTOMER RELATIONSHIP MGMT

SUPPLIER RELATIONSHIP MGMT

BUSINESS INTELLIGENCE

Features of Commercial ERP Software

Commercial ERP solutions (e.g., SAP, Oracle, Microsoft Dynamics 365, TallyPrime) provide structured functional frameworks tailored for enterprise resource tracking.



Key Features Explained

1. Centralized Database (Single Source of Truth)

Eliminates redundant data entry across departments. An invoice generated by the sales department automatically updates revenue ledgers, inventory balances, and accounts receivable in real time.

3. Role-Based Security and Internal Controls

Supports Segregation of Duties (SoD) to prevent fraud. For example, the employee who creates a purchase order cannot approve payment for that purchase order within the system.

2 **Modular Architecture**
Structured into distinct functional units (FI, CO, MM, HCM). Organizations can activate modules as business requirements expand.

4 **Comprehensive Audit Trail**
Logs transaction details (user ID, timestamp, original value, modified value), supporting statutory compliance and external audit procedures.

5 **Multi-Currency and Multi-Company Support**
Automatically handles foreign currency translations, exchange gain/loss calculations, and consolidated financial statements across international subsidiaries.

6 **Process Automation & Workflow Management**
Automates recurring accounting tasks such as monthly depreciation calculations, bank reconciliations, and payment approval routing.